

Security Models and Cryptographic Protocols in a Quantum World

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ABSTRACT

The emergence of quantum computing has provided new paradigms for cryptography. On the one hand, it poses significant new threats to existing classically cryptographic systems, requiring the community to define new security models that capture what a quantum adversary can do. On the other hand, it gives us new tools to design cryptographic protocols, with weaker assumptions than in the classical world, or even protocols that are impossible classically.

In this survey, we first give an overview of new security definitions for classical cryptography, considering quantum adversaries who can either only use local quantum computation (post-quantum security), or even send quantum messages and in particular have access to oracle in superposition (quantum security). We explore these new notions through the examples of commitments, zero-knowledge proofs, encryption, and signatures. Then, we present what is arguably the most famous application of quantum cryptography: quantum key distribution (QKD) protocols that take advantage

of unique properties of quantum mechanics to provide secure communication unconditionally. We also explore cryptography beyond QKD, focusing on unclonable cryptography: a family of cryptographic functionalities, built with quantum states, and designed to be resistant to counterfeit by leveraging the “no-cloning” theorem. We examine in particular quantum money, but also the recent notions of unclonable encryption and copy-protection, including related variants.

By presenting a comprehensive survey of these topics, this paper aims to provide a thorough understanding of the current landscape and future potential of quantum cryptography.

1

Introduction

In 1981, physicist Richard Feynman proposed in his lecture [1] the following question: Is it possible to simulate physics with a new kind of computer – a “quantum computer” which would “not [be] a Turing machine, but a machine of a different kind”? In 1985, Deutsch formally suggested the notion of a quantum Turing machine, and this model has since been refined in Deutsch and Jozsa [3] and Bernstein and Vazirani [4]. Such a system is now known as a *quantum computer*. As its name suggests, quantum computers operate according to the theory of quantum mechanics – a mathematical theory that allows us to calculate and predict “probabilities” in the quantum world.

In quantum mechanics, the basic quantities that describe a system are not classical probabilities but *amplitudes*, which may be negative or even complex numbers. While the measurable probabilities obtained from these amplitudes are always nonnegative (since they are computed as squared magnitudes), the relative signs and phases of amplitudes allow for interference effects that have no classical analogue. These interference effects challenge our classical intuition, as they enable phenomena such as superposition and quantum interference, which are essential to the power of quantum computation.

Beyond these amplitude-based effects, quantum mechanics exhibits other deeply nonclassical features that cannot be explained within the framework of classical probability theory. Together, these principles, which we briefly explain below, represent fundamental departures from classical intuition and form the conceptual foundation of quantum information processing.

- *Superposition*: this is a quantum phenomenon where a particle exists in multiple states or positions simultaneously until it is measured. This means that, rather than having a single, definite state, the particle can be in a linear combination of all possible states at once, only collapsing into one specific state upon observation.
- *Interference*: amplitudes can “add or cancel each other out”. This unique feature allows quantum systems to reinforce certain computational paths while canceling out others. It is a fundamental reason why quantum algorithms, such as Shor’s algorithm (Shor [5]), can achieve exponential speedups over classical algorithms for specific problems by harnessing this interference to find solutions more efficiently.
- *Quantum Uncertainty*: this is also known as the destructiveness of measurement, which refers to the principle that measuring one property of a particle, such as its position or momentum, fundamentally limits the accuracy with which another property can be known. This is because the act of observation itself disrupts the particle’s state, making it impossible to precisely measure both properties simultaneously.
- *Entanglement*: particles become so correlated that the state of one particle instantly influences the state of the other, no matter how far apart they are. Quantum correlation is much stronger than any classical correlation, in the sense that there exist quantum correlations that cannot be prepared by two non-interacting parties classically.

Still today, we have no real evidence that quantum computers, if built, will offer truly significant advantage for the applications of

computing. However, there is one cryptography-related exception: Shor’s factoring algorithm [5] provably shows that quantum computers can solve the integer factoring and discrete logarithm problems in quantum polynomial time. If we go back a bit further, the relationship between quantum information and cryptography is almost half-a-century old: in a remarkable 1968 manuscript (published more than a decade later [6]) that first discussed the idea of *conjugate coding*, Wiesner also proposed a scheme for *quantum money*, as the first ever application of quantum cryptography, and is also credited for the invention of oblivious transfer – a key concept in modern cryptography that was re-discovered years later by Rabin [7]. These works demonstrate that quantum information can challenge classical cryptography as we know it today and simultaneously pave the way for a new cryptographic landscape that cannot be achieved using classical information alone. This paper aims to provide an overview of recent advances in the development of quantum information for cryptography, as well as the limitations and challenges that cryptographers face in light of quantum information, with a particular focus on the famous quantum no-cloning theorem and its role in the interplay between quantum information and cryptography.

Overview. The ambivalence of the consequences of quantum information in the specific field of quantum cryptography has led to the development of two very active research domains. The so-called *post-quantum cryptography* studies the (negative) consequences of the emergence of fault-tolerant quantum computers. These computers are assumed to be controlled by an adversary who can use them to carry out attacks against traditional cryptography (referred to as *classical* in this context). The availability of quantum information to cryptography presents unique challenges and limitations, some of which are inherently trivial in the classical setting. Conversely, *quantum cryptography* seeks to leverage the new possibilities offered by quantum mechanics to develop cryptographic functionalities that go beyond classical limitations. The most well-known application of quantum cryptography is the unconditionally secure quantum key distribution, known as QKD [8].

Section 2 provides a brief introduction to the mathematical formal-

ism of quantum information and quantum computation, as well as the notation that will be used later in the paper. Section 3 describes the crucial property of quantum *no-cloning*, which informally states that it is physically impossible, in general, to clone an unknown quantum state. We then explain the seminal idea of *conjugate coding*, proposed by Wiesner, which serves as the main source of inspiration for many of the quantum constructions described later. We conclude with the counterintuitive phenomenon of *entanglement*, particularly the property of *monogamy of entanglement*, which will prove crucial in the context of unclonable cryptography.

Section 4 is devoted to the limitations and challenges in defining *quantum security* for *classical* cryptographic primitives and protocols. It is widely believed that ensuring the resistance of computational hardness assumptions against quantum computers is sufficient to guarantee the security of a cryptographic protocol. Post-quantum security thus usually builds solely on quantum-hard assumptions. However, contrary to this intuition, many issues arise. Modeling the adversary quantumly not only gives them more tools to attack a protocol but also affects how the security of the protocol is proven. Note, however, that this does not mean an attack has been found on the scheme, but rather that traditional security proof techniques no longer apply. In classical provable security, one typically needs to simulate the view of the adversary, which quantumly requires simulating quantum processes instead of classical ones. The adversary should be unable to distinguish between a real honest execution and a simulated view with no access to the secret information held by the honest parties. Such a quantum-simulated view is, unsurprisingly, usually incompatible with existing classical simulation techniques, such as rewinding the simulator or programming a random oracle, which cannot be directly applied in a quantum setting due to several technical obstacles. These obstacles can be traced to the quantum no-cloning theorem and the destructiveness of quantum measurements.

We first describe in this section the changes regarding post-quantum security, where the adversary has access only to quantum computation and no quantum communication. In such a setting, when switching from the *random oracle* model (ROM), first formalized by Bellare

and Rogaway [9] to its quantum version (QROM), the most important consideration is how we can transpose the modeling and proof techniques. This models the fact that an attacker could implement on their own the random oracle (which is instantiated by a secure, public hash function with no secret information) and query this oracle in superposition to take advantage of it. We then consider the case of *commitments*. Although the definition of computational hiding straightforwardly follows from the classical definition, this is not the case for computational binding. We explain why the classical definition turns out to be inadequate in the quantum setting and describe the new notion of computational collapse-binding. Finally, we consider the *rewinding* technique, which allows the simulator to copy the internal state and grants them the right to rewind to an earlier step if they made a wrong guess about the adversary’s choice, for example. The problem — general quantum states cannot be copied — was identified quite early by Graaf [10], and several quantum rewinding techniques have been developed.

In the second part of this section, we consider *quantum security*, where the adversary has access to quantum keyed oracles, which would allow it to carry out *superposition attacks*. In such a setting, even defining security notions for basic primitives such as encryption and signatures becomes a non-trivial task. We describe in this section the case of IND-CCA2 security definition for encryption schemes, where manipulations involving “storing (message, ciphertext) tuples” and “checking whether the ciphertext used in a decryption query by the adversary is not the challenge ciphertext” become notoriously hard to achieve in a quantum setting. We also discuss the definitions of security for authentication and digital signatures, where the familiar notions of unforgeability are difficult to translate into meaningful quantum notions. In this case, the difficulty comes from the notion of “predicting an unqueried value”.

Finally, in Sections 5 and 6, we describe some of the applications made possible by quantum information. Section 5 is mainly dedicated to *quantum key distribution* (QKD), probably the most widely known quantum protocol. It is mainly based on the notion of conjugate coding, presented in Section 3.2. This protocol allows two users, Alice and Bob, to share a common secret key without an eavesdropper being

able to intercept it without being detected. Alice and Bob have access to an authenticated (but not secure) classical communication channel and a quantum communication channel. We also present everlasting security and device-independent cryptography. QKD is arguably the most famous protocol in quantum cryptography, whose security is based on the no-cloning principle. Furthermore, its simplicity stands out, and presenting it as an introduction to quantum cryptography in this paper provides a good first intuition for why quantum mechanics can allow for better security (i.e., quantum advantages). At the end of the section, we give an overview of quantum cryptography beyond QKD. Among these research areas, we dedicate Section 6 to *unclonable cryptography*.

Unclonable cryptography naturally extends the application of the no-cloning principle in cryptography with more functionality. Again, protocols in unclonable cryptography inherently leverage the no-cloning principle, which fundamentally belongs to the quantum world, and not to the classical one. In Section 6, we first present a toolkit for constructing unclonable cryptography. We recall BB84 states, as well as the monogamy-of-entanglement and certified deletion properties of this family of states. We then present coset states, along with their direct product hardness and monogamy-of-entanglement properties. Coset states allow for verifiability, but BB84 states are simpler and can be implemented with current technology.

Next, we introduce some unclonable objects following a hierarchy inspired by [11]. The first level is the *authenticity* level, where objects do not hold any data. This level includes *quantum money* and *quantum lightning*. Quantum money is a functionality introduced by [6] in the same paper in which he introduced conjugate coding. A digital money scheme consists of units of money, which are quantum states. Thanks to the no-cloning theorem, these quantum coins cannot be copied. In this part, we present the quantum money functionality and review some of the important protocols that have been proposed for it. Quantum lightning is a new functionality [12] which can be seen as a quantum money with the collision-resistance property. It provides a way to prepare quantum states called quantum bolts associated with a serial number, and to verify that such a bolt has been prepared honestly. Unlike quantum money, this functionality ensures that no one, even the

one who prepared the bolt, can produce two valid bolts with the same serial number.

The second level is the *information level*, where objects that hold information are made unclonable. The main example is unclonable encryption, which can be seen as a classical encryption scheme with quantum ciphertexts. These ciphertexts must be unclonable, meaning that no adversary can, given a ciphertext of a message m , split this ciphertext into two ciphertexts that can both be decrypted to m . Unclonable symmetric encryption is inspired by Wiesner’s conjugate coding and has been defined by [13]. It was extended to asymmetric encryption by [14]. In this survey, we present a formal definition of unclonable encryption and an example of construction. We also define the property called certified deletion for encryption schemes, describe the protocol presented in [15], and discuss the recent advances in the area. This notion means that if a user places encrypted private information on a server, they can request a certificate of deletion for the ciphertext, ensuring that the server has actually destroyed it, and thus no longer has any information about the plaintext.

Finally, the third level is the *functionality level*, where we make program-like objects unclonable. This includes copy-protection schemes and single decryptors. Copy-protection, introduced in [16], allows a software vendor to sell a copy of their software to a client without the client being able to copy it. This task, impossible classically, is made possible quantumly by encoding the program into a quantum state. Being able to copy the program would thus imply breaking the no-cloning theorem. Unfortunately, constructing such an encoding is a very difficult task, and copy-protection constructions have been found for only a few families of functions. In this section, we provide a history of this functionality by presenting several interesting results in this field. We introduce the families of point functions and compute-and-compare functions and present the copy-protection protocols given by [17]. We also present the notion of single decryptors, which can be seen as copy-protection schemes where the protected function is the decryption algorithm of some encryption scheme (with the decryption key being a quantum state that cannot be copied). We conclude the section with some discussions on related notions such as secure software

leasing and revocable cryptography.

Further Reading. Since quantum cryptography is a flourishing area of research, we have chosen to give an introduction to a limited numbers of topics. The reader is encouraged to consult other references, such as the initial review on “Quantum cryptography beyond quantum key distribution” by Broadbent and Schaffner [18], its follow-up survey by Bozzio, Crépeau, Wallden, and Walther [19], and a short review on unclonable cryptography by Sattath [20]. Here, we briefly mention some other active research areas that are not covered in this survey:

- **Classical verification of quantum computation (CVQC) and certified randomness.** CVQC is a protocol that allows a classical client to verifiably delegate a BQP computation to a potentially cheating quantum server. Since the major breakthrough of Mahadev [21], classical verification of quantum computation has remained a central primitive of study in quantum cryptography Zhang [22], Bartusek, Kalai, Lombardi, Ma, Malavolta, Vaikuntanathan, Vidick, and Yang [23], and Natarajan and Zhang [24]. This can also be extended with the notion of blindness, which allows a classical client to outsource a quantum computation of its choice to a quantum server without leaking anything about the actual description of the computation. In another major breakthrough, Mahadev [25] gave the first construction of this primitive by building quantum fully-homomorphic encryption from lattice-based hardness assumptions. We refer the reader to the survey [26] and a course by Vidick¹.
- **Quantum pseudorandomness and quantum black-box reductions.** Pseudorandomness is ubiquitous in theoretical computer science and has found applications in many areas, ranging from cryptography, to computational complexity, to the study of randomized algorithms, and many more. The notion of quantum pseudorandomness (Ji, Liu, and Song [27]) has emerged as the natural quantum analogue of pseudorandomness in the classical world.

¹Course FSMP, Fall’20: Interactions with Quantum Devices.
<http://users.cms.caltech.edu/vidick/teaching/fsmp/fsmp.pdf>

The work of Kretschmer [28] has shown that such fully quantum cryptographic primitives can exist in a world in which no classical cryptography exists, including one-way functions. Quantum pseudorandomness has applications in many areas of quantum information, ranging from entanglement theory (Aaronson, Bouland, Fefferman, Ghosh, Vazirani, Zhang, and Zhou [29], Bouland, Fefferman, Ghosh, Metger, Vazirani, Zhang, and Zhou [30]), quantum learning theory (Zhao, Lewis, Kannan, Quek, Huang, and Caro [31]), the foundations of quantum cryptography (Morimae and Yamakawa [32], Ananth, Qian, and Yuen [33]) and many more. While these recent results are remarkable, it is still unclear how they exactly relate to the classical ones. Even more, these (inherently quantum) assumptions might not necessarily be captured by classical complexity theory. In this line of research, there are also examples of quantum/classical separations, but known results are usually shown for classical cryptographic primitives in the quantum setting, leaving a wide unexplored area of the complexity of quantum cryptographic primitives such as quantum money and unclonable cryptography. In particular, these recent results raise new questions on several levels, culminating in a fundamental question of minimal assumption for quantum cryptography.²

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²For more example and relation, we refer the reader to the graph: <https://sattath.github.io/qcrypto-graph/>.

2

Basics of Quantum Information and Quantum Computation

This section contains the rudiments of quantum information that are used in the following chapters. We assume only basic knowledge of linear algebra and refer the interested reader to textbook references on quantum information, in particular Nielsen and Chuang [34], for more details.

2.1 State Space

In quantum information processing, the fundamental unit of information is called a *qubit*, which is described mathematically by a vector of length one in a two-dimensional complex vector space, that is, the two-dimensional Hilbert space. We use the standard *ket-bra* notation to denote vectors representing qubits (and, more generally, quantum states), enclosing vectors in a *ket*, yielding $|\phi\rangle$. We can write any state of one qubit as $|\phi\rangle := \alpha|0\rangle + \beta|1\rangle$, where the states $|0\rangle$ and $|1\rangle$ form a basis for the underlying two-dimensional vector space, and where α and β are complex numbers (so that they can be negative as we mentioned earlier) such that $|\alpha|^2 + |\beta|^2 = 1$. If neither α nor β equals zero, we say that $|\phi\rangle$ is in a *superposition* (linear combination) of both $|0\rangle$ and $|1\rangle$. The quantum state of two (or more) qubits can be

described by the tensor product of two quantum states of one (or more) qubit(s). For example, a basis for two qubits is the tuple of states $(|0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle)$, which is usually abbreviated as $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$. Extending this concept to multiple qubits, we see that an n -qubit system can be in any superposition of the n -bit basis states $|0\dots 0\rangle, |0\dots 1\rangle, \dots, |1\dots 1\rangle$. Hence, an n -qubit quantum state $|\phi\rangle$ is described by 2^n complex coefficients as $|\phi\rangle := \sum_x \alpha_x |x\rangle$ where $x \in \{0, 1\}^n$ and $\sum_x |\alpha_x|^2 = 1$.

When we refer to a quantum register in the following, we simply mean a collection of qubits that we wish to view as a single unit and to which we give some name. Names of registers will always be uppercase letters, such as X, Y and Z . For example, in case of bi-partite quantum states shared among Alice and Bob, subscripts can be used to indicate which player holds which qubits: the 2-qubit state $|0\rangle_A \otimes |0\rangle_B = |00\rangle_{AB}$ means that Alice and Bob both hold a qubit in state $|0\rangle$.

2.2 Unitary Evolution and Circuits

Basic evolutions of a quantum system are described by linear operations that preserve the norm. Formally, these operations can be expressed as *unitary* complex matrices: We say that a complex matrix U is unitary if $UU^\dagger = \mathcal{I}$, where $U^\dagger$ is the complex-conjugate transpose of U and $\mathcal{I}$ is the identity matrix. Note in particular that this definition directly yields that any unitary matrix U is invertible, with inverse $U^\dagger$.

In this paper, we describe quantum algorithms within the quantum circuit model, in which unitary operations, called *gates*, are arranged to form circuits. It is also convenient to allow quantum circuits to include two simple, non-unitary gates: *ancillary gates*, which take no input and output a single qubit in the state $|0\rangle$, and *trace-out gates*, which take one input qubit and produce no output, effectively discarding the qubit. In addition to these two gates, quantum circuits may include any universal set of unitary gates, that we will define later. A quantum circuit may therefore have a different number of input and output qubits. We say that a circuit is of type (n, m) if it has n input qubits and m output qubits. More generally, any admissible map from n qubits to m qubits is said to be of type (n, m) . The *size* of a type (n, m) quantum circuit

is defined as the total number of gates in the circuit, plus $n + m$. When Q denotes such a circuit, we identify Q with the admissible map from n qubits to m qubits that is induced by executing Q , and we denote its size by $|Q|$.

Quantum Algorithms

A *quantum algorithm* is a uniform family of quantum circuits, i.e., such that there exists a (classical) Turing machine which, given the security parameter expressed in unary 1^n as input, runs in time at most polynomial in n , and outputs a (classical) description of the n -th member of the quantum circuit family. QPT stands for “quantum polynomial time”, and a QPT algorithm is a uniform family of quantum circuits of size polynomial in the security parameter.

Recall that the classical complexity class P contains all decision problems that can be solved by a deterministic (classical) Turing machine in polynomial time. As quantum algorithms are probabilistic by nature, there is no quantum analogue for the class P. However, there is an analogue for BPP (which is the class of decision problems solvable by a *probabilistic* Turing machine in polynomial time with an error probability bounded away from, say, $\frac{1}{3}$ for all instances), which is the complexity class BQP. Formally, BQP is the set of all languages $\mathcal{L}$ for which there exists a QPT algorithm $\mathcal{M}$ such that:

1. $\forall x \in \mathcal{L}, \Pr[\mathcal{M}(x) = 1] \geq \frac{2}{3}$; and
2. $\forall x \notin \mathcal{L}, \Pr[\mathcal{M}(x) = 0] \geq \frac{2}{3}$.

Alternatively, a language $\mathcal{L}$ is in BQP if and only if there exists a polynomial quantum Turing machine that accepts $\mathcal{L}$ with an error probability of at most $\frac{1}{3}$ for all instances.

Oracle Quantum Algorithms

Let $\mathcal{X}, \mathcal{Y}$ be any sets. For any function $f : \mathcal{X} \rightarrow \mathcal{Y}$, define U_f as the unitary transformation that maps $|x, y\rangle$ to $|x, y \oplus f(x)\rangle$. By definition, the inverse transformation of U_f is itself.

Let O_f be an oracle that computes the unitary transformation U_f . Oracle quantum algorithms are quantum algorithms whose circuits, in addition to having the gates as described above, have the ability to query (perhaps in superposition) the oracle O_f which may act on many qubits. Each oracle call is counted as a single gate towards the size complexity of a circuit. The notion of time- and space-uniformity for oracle algorithms is the same as with non-oracle algorithms: there is a polynomial-time/polynomial-space Turing machine – which does not have access to the oracle – that outputs the description of the circuits.

We also distinguish between classical and quantum queries. We say that an oracle algorithm $\mathcal{A}^{O_f}$ makes quantum queries if it can query O_f in superposition. We say that $\mathcal{A}^{O_f}$ makes classical queries if, before every oracle call, the input qubits to the oracle are measured in the standard basis. In this case, the algorithm would be querying the oracle on a probabilistic mixture of inputs.

Finally, one can also define oracle algorithm $\mathcal{A}^{O_{f_1}, O_{f_2}, \dots, O_{f_k}}$ with access to a collection of oracles $\{O_{f_1}, O_{f_2}, \dots, O_{f_k}\}$, which is polynomially equivalent to an oracle algorithm $\mathcal{B}^O$ with access to a single oracle $O(k; x) = O_{f_k}(x)$. We also call this oracle O a *keyed oracle*.

2.2.1 Quantum Gates

We refer to the following well-known unitary gates:

- *Pauli gates:* $X : |a\rangle \mapsto |1 - a\rangle$, $Z : |a\rangle \mapsto (-1)^a |a\rangle$ and $Y := iXZ$, for each $a \in \{0, 1\}$.
- *Hadamard gate:* $H : |a\rangle \mapsto \frac{1}{\sqrt{2}} |0\rangle + \frac{(-1)^a}{\sqrt{2}} |1\rangle$, for each $a \in \{0, 1\}$. For example, applying a Hadamard gate to only the first qubit in the state $|00\rangle$ would be represented as $(H \otimes \mathcal{I}) |00\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$.
- *Rotation gates:* $R_\phi : |a\rangle \mapsto e^{ia\phi} |a\rangle$, for each $a \in \{0, 1\}$. We obtain the T gate when $\phi = \frac{\pi}{4}$, the phase gate P when $\phi = \frac{\pi}{2}$.
- *Controlled gates:* for any k -qubit unitary quantum gate U , we define the controlled- U as: $\text{Ctrl-}U : |a\rangle |x\rangle \mapsto |a\rangle U^a |x\rangle$, for each $a \in \{0, 1\}$ and $x \in \{0, 1\}^k$. In particular, we write the controlled-NOT gate as $\text{CNOT} : |a\rangle |b\rangle \mapsto |a\rangle |b \oplus a\rangle$.

- *Toffoli gates:* CCNOT : $|a\rangle |b\rangle |c\rangle \mapsto |a\rangle |b\rangle |c \oplus (a \cdot b)\rangle$ for each tuple $(a, b, c) \in \{0, 1\}^3$.

Two examples (among other known examples) of universal gate sets are the following:

- Hadamard, phase and Toffoli gates: $\{H, P, \text{CCNOT}\}$.
- Hadamard and controlled-phase gates: $\{H, \text{Ctrl-P}\}$.

2.3 Measurement

A quantum measurement involves a collection of operators $\{M_i\}_i$ acting on the state space of the system to be measured and satisfying: $\sum_i M_i^\dagger M_i = \mathcal{I}$. The label i indicates a potential measurement outcome. Given a state $|\psi\rangle$ to be measured, the probability of obtaining outcome i is $p(i) := \langle \psi | M_i^\dagger M_i | \psi \rangle$ and the state of the system after the measurement will be $\frac{M_i |\psi\rangle}{\sqrt{p(i)}}$. If we are only interested in the probabilities of the different outcomes and not in the post-measurement state, then we can denote $E_i := M_i^\dagger M_i$ and we refer to the set $\{E_i\}_i$ as a *positive-operator valued measure* (POVM). When performing a measurement in an orthonormal basis $\mathcal{B} := \{|i\rangle\}_i$, we are essentially choosing $M_i := |i\rangle\langle i|$, where $\langle i|$ is defined as the complex conjugate $(|i\rangle)^\dagger$ of $|i\rangle$. The most well-known bases are the computational basis (also called the standard basis, or the rectilinear basis) $\mathcal{B} := \{|i\rangle\}_{i \in \{0,1\}^*}$, and the Hadamard basis (also called the diagonal basis), in which we apply Hadamard to the state before measuring it in the computational basis.

As a simple example of quantum measurements, consider the states $|\psi_1\rangle = |0\rangle$ and $|\psi_2\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Then measuring the state $|\psi_1\rangle$ in the computational basis yields the outcome 1 with probability 1 and the state remains $|0\rangle$, while measuring $|\psi_2\rangle$ also in the computational basis yields the outcome 0 or 1, each with probability 1/2, and the post-measurement state is $|0\rangle$ if we observed outcome 0 and $|1\rangle$ if we observed 1.

As a final note, in general, quantum measurement is an irreversible process, and we usually refer to this property as *the destructiveness* of quantum measurement.

2.4 Mixed States

Consider a scenario where we are given a state sampled from a distribution of quantum states described as $\{(p_i, |\psi_i\rangle)\}_{i \in I}$. That is, we are given $|\psi_i\rangle$ with probability p_i . The *mixed state* formulation provides a convenient way of manipulating such a state.

We define the mixed state corresponding to this scenario as $\rho = \sum_{i \in I} p_i |\psi_i\rangle\langle\psi_i|$, where $\langle\psi|$ is defined, similarly as above, as the complex conjugate $(|\psi\rangle)^\dagger$ of $|\psi\rangle$ for any state $|\psi\rangle$ — we call this transposed vector a *bra*. Note in particular that this mixed state is a matrix, as opposed to the *pure states* $|\psi\rangle$ defined in the previous sections, that we defined as vectors.

Now, recall that we defined the unitary evolution of a pure state $|\psi\rangle$ as the action of a matrix U on the state, yielding the state $U|\psi\rangle$. The same unitary evolution applied to the state obtained from our scenario then yields $U|\psi_i\rangle$ with probabilistic p_i , which corresponds to following mixed state:

$$\sum_{i \in I} p_i U|\psi_i\rangle\langle\psi_i|U^\dagger = U\rho U^\dagger$$

Measuring a mixed state with a measurement $\{M_j\}_j$ yields j with probability $\text{Tr}(M_j M_j^\dagger \rho)$, and the corresponding post-measurement state is described as the mixed state

$$\frac{M_j \rho M_j^\dagger}{\text{Tr}(M_j M_j^\dagger \rho)}$$

2.5 Computational Indistinguishability in the Quantum Setting

Definition 2.1 (Norms). Let A be a linear operator on a Hilbert space with singular values $\lambda_1, \dots, \lambda_n \geq 0$. Then, the trace norm is defined as

$$\|A\|_1 = \sum_i \lambda_i,$$

and the Schatten ∞ -norm is defined as

$$\|A\|_\infty = \max_i \lambda_i.$$

Definition 2.2 (Trace distance). For two quantum states ρ, σ , the trace distance between them is

$$\text{TD}(\rho, \sigma) := \frac{1}{2} \|\rho - \sigma\|_1.$$

Definition 2.3 (Computationally indistinguishable states). A distinguisher $\mathcal{D}$ is a (not necessarily polynomially-time-bounded) quantum algorithm, that receives as input a quantum state and outputs a classical bit (i.e. a distribution over $\{0, 1\}$). The distinguishability is the trace distance between $\mathcal{D}(\psi)$ and $\mathcal{D}(\psi')$. For two quantum states ψ, ψ' which are computationally indistinguishable up to δ (i.e., no QPT distinguisher has advantage exceeding δ in distinguishing ψ from ψ'), we write:

$$\psi \approx_\delta \psi'.$$

We can also define computational indistinguishability with respect to *non-uniform* QPT algorithms with *quantum advice*, denoted by $A := \{A_\lambda, \phi_\lambda\}_{\lambda \in \mathbb{N}}$, where each A_λ is the classical description of a polynomial-size quantum circuit, and ϕ_λ is some (not necessarily efficiently computable) non-uniform quantum advice of size polynomial in the security parameter λ . We usually consider computational indistinguishability with respect to non-uniform QPT adversaries with quantum advice, unless stated explicitly otherwise. All asymptotic statements are understood to be in the limits $\delta \rightarrow 0$ and $\lambda \rightarrow \infty$.

3

Fundamental Properties of Quantum Mechanics

We describe in this chapter several crucial specificities of quantum mechanics, which are the roots of the differences between classical and quantum cryptography, be they on the negative (quantum attacks) or positive (protocols beyond classical impossibility results) sides.

3.1 Quantum No-Cloning

One of the most fundamental properties of quantum information is the so-called *no-cloning theorem* [35]. Informally, this theorem states that it is not physically possible, in general, to *clone* an unknown quantum system, that is, there is no physical process that takes as input an arbitrary single quantum state, and outputs two identical copies of its input.

Theorem 3.1. There exists no quantum operation C such that

$$C(|\phi\rangle \otimes |0\rangle) = |\phi\rangle \otimes |\phi\rangle$$

for all inputs $|\phi\rangle$.

The fact that we cannot clone arbitrary quantum states shows that quantum information is really different from classical information. For

instance, this means that we cannot determine α and β from a single copy of $|\phi\rangle := \alpha|0\rangle + \beta|1\rangle$. Indeed, while this has some nice features – for example, an inbuilt copy protection mechanism, which we describe later in Section 6 – it also presents some definitional obstacles, which usually appear in the context of security definition for encryption or zero-knowledge (see Section 4).

3.2 Conjugate Coding

Conjugate Coding has been introduced by Wiesner in the 60's as a technique to protect classical information. However, as the field of quantum information was at this time not as developed as it is now, this technique did not get a lot of attention and we had to wait until 1983 for Wiesner's paper [6] to be published.

The idea of conjugate coding is to encode classical information into quantum states, using two conjugate bases. As in Chapter 2, we consider the *computational basis* $\{|0\rangle, |1\rangle\}$ and the *Hadamard basis* $\{|+\rangle, |-\rangle\}$, where

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \text{ and } |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Definition 3.1 (Conjugate Coding). The encoding of a bit $x \in \{0, 1\}$ in a basis $\theta \in \{0, 1\}$ (0 for computational or 1 for Hadamard) is defined as $H^\theta |x\rangle$ (see Table 3.1).

Basis θ	0	0	1	1
Bit x	0	1	0	1
Encoding $H^\theta x\rangle$	$ 0\rangle$	$ 1\rangle$	$ +\rangle$	$ -\rangle$

Table 3.1: Conjugate coding

It is important to notice that, if a state is prepared in one of these two bases and measured in the other one, the outcome is 0 or 1 with the same probability $1/2$ and the post-measurement state is different from the original one. The best strategy for a malicious user who owns the encoding $H^\theta |x\rangle$ of a bit x and who wants to learn x but does not know θ is to make a random guess on the basis and to measure $H^\theta |x\rangle$

in this basis. It yields the correct bit x with probability $3/4$ but, in the case where the wrong basis is chosen, it perturbs the state. These phenomena are, as we see later, used in several protocols to ensure that no one tried to learn sensitive information.

3.3 Quantum Entanglement and Nonlocality

A crucial and rather counter-intuitive feature of quantum mechanics is *quantum entanglement*, a physical phenomenon that occurs when quantum particles behave in such a way that the quantum state of each particle cannot be described individually. In other words, entangled states are those that cannot be written as a tensor product of separate states. The most famous one is the EPR-pair:

$$|\Phi^+\rangle_{AB} := \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB}).$$

When Alice measures her qubit (in system A) and obtains the outcome a , the overall state of the entangled system *collapses* so that Bob's qubit (in system B) is now in the corresponding state $|a\rangle_B$. Consequently, when Bob later measures his qubit in the same basis, he obtains the same result $b = a$. It is important to emphasize that this correlation does not imply that any information travels between Alice and Bob at the time of measurement. Rather, quantum mechanics predicts that their measurement outcomes will be perfectly correlated whenever they later compare their results, even if the measurements occurred at space-like separation – so far apart that no signal moving at or below the speed of light could have connected them.¹

It turns out that entanglement allows correlations between two particles – two qubits, or two parties – that cannot be replicated classically. In this case, physicists say that the correlations violate a Bell inequality [36]. The modern way to understand Bell inequality is by means of so-called *non-local* games. The most well-known example of such an inequality was proposed by Clauser, Horne, Shimony, and Holt [37]. It can be described as a non-local game among two players

¹This nonlocal correlation, famously referred to by Einstein as “spooky action at a distance”, reflects the fundamental nonclassical nature of entanglement.

Alice and Bob. In this CHSH game, Alice and Bob can initially discuss in order to establish a joint strategy, i.e., Alice and Bob are allowed to share classical randomness. Once the game starts, they are separated and cannot communicate. They receive as input uniformly random bits x and y and have to output bits a and b respectively. They win the game if and only if $a \oplus b = x \wedge y$ (imagine a third party, called a referee who chooses x and y , receives a and b and checks whether the relationship $a \oplus b = x \wedge y$ holds). One can verify (by trying out all possible strategies for Alice and Bob) that classically maximum winning probability that can be achieved is $\frac{3}{4}$. In other words, we have the Bell inequality $\Pr[\text{classical players win CHSH game}] \leq \frac{3}{4}$. However, if Alice and Bob share a maximally entangled state (e.g., an EPR pair), they can perform a quantum measurement which allows them to win the CHSH game with probability $\cos^2(\frac{\pi}{8}) \approx 0.85$ which is strictly larger than $\frac{3}{4}$, hence violating the Bell inequality.

3.4 The Monogamy of Entanglement

Entanglement has another interesting property called “monogamy”, which states that if two systems are maximally entangled with each other, then they cannot have any entanglement with any other system: equivalently, they must be in tensor product with the remainder of the universe. For example, let’s consider the “maximally entangled” EPR-pair $|\Phi^+\rangle_{AB}$. It is an entangled state between two systems A and B , but what about a third system, call it C for Charlie? Is it possible to create a joint state $|\Psi\rangle_{ABC}$ in which the strong correlations of the EPR pair are shared simultaneously between all three systems? The monogamy property says that it is impossible for C to share any correlations with either A or B . We note that one way to qualify monogamy is through the use of non-local games, such as the CHSH game that we discussed in Section 3.3.

We will see later in Section 6.1 that the concept of non-local games and monogamy of entanglement are formally used as the main tool to define and prove security for unclonable cryptographic objects.

4

Quantum Security of Classical Cryptosystem

Modern cryptography is based on the *provable security* paradigm. This is a methodological approach to assessing the security of a cryptosystem, where rigorous mathematical models and proofs are required in order to show that the security of the cryptosystem can be formally validated. To achieve this goal, one needs to first specify the attacker's capabilities and the security goals that a given cryptographic scheme must meet. Next, a security proof – or reduction – is constructed to show that if an efficient adversary can break the cryptographic scheme within the defined security model, then an efficient algorithm must also exist for solving some computational problems. Ultimately, the underlying computational problems are inherent to the difficulty of solving certain mathematical problems (such as integer factorization) which, for theoretical or historical reasons, are widely considered to be very hard to solve even with the help of the most powerful supercomputers known today. These computational problems are also called *hardness assumptions*. Finally, one needs to provide a reduction which shows how to transform an adversary that breaks the security goals of that cryptosystem into an adversary against the hardness assumptions which the scheme is based on. Theoretically, it is always possible for an adver-

sary with enough computational resources to solve the mathematical problem, thus breaking the security of that cryptosystem. However, from a practical standpoint, this would require either an unreasonable amount of time, or an unreasonable amount of computational resources (such as storage, memory, power, etc.), or both.

The advantage of having a provably secure cryptosystem is that, as long as the security model used is sound and the underlying hardness assumptions hold, one can stay assured that the cryptosystem cannot be broken. Arguably, this approach has turned cryptography to a well established science, with clear security definitions and objectives.

However, the effectiveness of provable security strongly relies on the hardness assumptions used, which are not guaranteed. Cryptographic hardness assumptions are based on the observation that algorithmical advances on solving the underlying mathematical problem would imply (unlikely) breakthrough results of scientific importance. Shor's celebrated quantum polynomial-time algorithm for solving factorization and discrete logarithms, both problems believed to be classically intractable, is a breakthrough result in this sense. These problems are at the base of many modern cryptosystems such as RSA by Rivest, Shamir, and Adleman [38] and elliptic curve cryptography by Miller [39] and Koblitz [40], which are widely adopted today. Indeed, the security of classical cryptosystems relies on computational assumptions which were usually made with respect to classical adversaries; if quantum adversaries are to be taken into account, the landscape of security is bound to change dramatically.

This led to the development of quantum-resistant cryptography, that is, cryptography that can run on classical computers but should withstand attackers equipped with quantum computing power. The first task, unsurprisingly, is to find new mathematical problems which are supposed to be “hard” even for quantum computers, so that new quantum-resistant cryptosystems can be constructed by relying on such new quantum hardness assumptions. Some notable examples of these assumptions are: finding the shortest vector on lattices, inverting hash functions, decoding certain types of linear codes, and few others (see Bernstein [41]). Another task is to define new security models, incorporating quantum computation capabilities of the adversary. In

other words, this task aims to formalize mathematically what it means by saying that “*the cryptosystem should be secure against a quantum adversary*”. This task is the main focus of this section.

Before going into details, we briefly discuss two different quantum-resistant security notions for cryptosystems: *post-quantum security* and *quantum security*.

- **Post-quantum Security:** A cryptosystem is said to have *post-quantum security* if it is secure against adversaries with quantum computers. This notion represents the *first level of upgrade* from the standard classical security model, where adversaries are modeled as classical probabilistic algorithms. The distinction between post-quantum and classical security lies in the computational model of the adversary: rather than being limited to classical computation (as in the standard Turing machine model), a post-quantum adversary may perform arbitrary quantum computations *locally*. However, the communication between parties, and between the system and the adversary, remains entirely classical. That is, they can only exchange classical messages.

As in the classical setting, the adversary is typically modeled as an oracle algorithm. In the post-quantum setting, this becomes an *oracle quantum algorithm* as defined in Section 2.2. Usually, the adversary is assumed to have only classical oracle access; however, when the security model allows the adversary to evaluate certain functions locally (for example, a publicly known function modeled as an oracle), quantum oracle access is considered instead. An example of this will be illustrated in Section 4.2.1.

This is the model most often considered in quantum-resistant cryptography, as its attacks are particularly relevant due to their impact on current data communication technology. This security model is also called QS1 by Gagliardoni [42] or Q1 by Kaplan, Leurent, Leverrier, and Naya-Plasencia [43].

- **Quantum Security:** A cryptosystem is said to have quantum security if it is secure against adversaries with quantum computers that have *quantum* oracle access in its security definition. A scheme

with quantum security not only has post-quantum security, but it also remains secure in arbitrary intermediate security notions between quantum security and post-quantum security. Therefore, in this sense, quantum security is theoretically the ultimate security goal that cryptosystems can achieve. This stronger model is sometimes called “superposition-based security” by Damgård, Funder, Nielsen, and Salvail [44], QS2 by Gagliardini [42], or Q2 by Kaplan, Leurent, Leverrier, and Naya-Plasencia [43].

4.1 Importance of Studying Quantum Security

While the notion of post-quantum security is more realistic, we argue why it is also worth studying the notion of quantum security. Some of the reasons why this notion is important to consider are also explained in detail by Gagliardini [42].

- **Composability:** to ensure that classical cryptosystems retain their security even if executed on a quantum computer, possibly in complex environments or protocols where composition should be taken into account.
- **Security proof concerns:** security reductions might require more than just post-quantum security of underlying primitives. An example is the need of quantum-secure pseudorandom functions in order to simulate a quantum random oracle, which is a concept that can be considered solely in post-quantum security model. Another example is the case of code obfuscation (see Sahai and Waters [45]), in which the obfuscated program will be run on the adversary’s quantum computers; for example if the adversary is given a public encryption key which is generated by hardcoding a symmetric key into an obfuscated encryption program.
- **Device-independent security:** to ensure security against *exotic quantum attacks*, which include attacks “tricking” a classical device into being quantum (e.g., the “frozen smart-card attack” presented by Gagliardini, Hülsing, and Schaffner [46] – in this setting, a quantum hacker can trick the encryption chip into quantum

behavior, which allows the adversary to query the target device (i.e., encryptor) in superposition); or cases where a quantum computer is used to run a classical algorithm, but an adversary manages to have control over the measurement devices of the victim.

A closely related research topic in the quantum security model is quantum cryptanalysis. In fact, several cryptosystems proven to be secure in the post-quantum security model are broken when considering superposition attacks. Notable examples include the works of Kaplan, Leurent, Leverrier, and Naya-Plasencia [47] who break a series of symmetric-key systems in polynomial time using quantum period finding algorithm, key-recovery attacks against Even-Mansour constructions by Kuwakado and Morii [48], distinguishers against 3-round Feistel constructions by Kuwakado and Morii [49], key recovery attacks against FX constructions by Leander and May [50], and many more (see Schrottenloher [51]).

4.2 Post-quantum security

As we said, the two main tasks we need to consider when designing post-quantum secure cryptosystems are: (i) the underlying hardness assumptions; and (ii) security notions and security reductions. Let's focus on the latter, then several natural questions come to mind:

- Would it be possible to re-use classical security definitions *as-is*, except a minor modification that we lift the adversary's power from being classical and polynomially bounded, to being polynomially bounded, but quantum?
- Given that the underlying assumptions are quantum-hard, would the same security reductions used in the classical setting go through without making any changes?

While the answer for these questions are “yes” in some cases, for example, encryption and authentication/digital signatures in the standard plain model, it is not the case for many other cryptographic schemes. The following points are among those to be discussed in this section:

- We will see that new security models are required to capture the power of quantum adversaries. Even more, classical intuitions of security might be inadequate in the quantum setting. We will look into these problems through the cases of the (quantum) random oracle model in Section 4.2.1 and the computational binding property of bit commitments in Section 4.2.2. New security notions that are inherently different from the classical setting will be discussed, followed by examples of constructions.
- Proving post-quantum security is considerably harder than proving classical security. We will demonstrate this through an example of proving zero-knowledge property of the graph isomorphism proof system by Goldreich, Micali, and Wigderson [52]. Classically, this is done by using a basic *rewinding* argument. However, quantumly, this is not possible in general. Indeed, for some time the Goldreich-Micali-Wigderson graph isomorphism proof system was not known to be zero-knowledge against quantum attacks, leading some to question whether the zero-knowledge property could ever be established in the quantum setting. Fortunately, we now know that this is possible, albeit new proof techniques are required. A fact known as *quantum rewinding lemma* will be introduced and illustrated for the Goldreich-Micali-Wigderson graph isomorphism proof system in Section 4.2.3.

4.2.1 The Quantum Random Oracle Model

The (Classical) Random Oracle Model. Let us first briefly recall what the random oracle is, and why we should care about it. For that, we first need to discuss *hash functions*. Essentially, these are functions that take a (potentially) long input, and “hash” it down to a fixed-size value. It turns out that throughout the field, we use hash functions a lot in cryptography: it appears in the context of digital signatures, encryption, key derivation problem, to name a few. For it to be useful, a *cryptographic* hash function H has to satisfy some of the following properties:

- **Pre-image Resistance.** We expect it to be “one-way”, meaning

that it is hard to “invert” the function. More precisely, given a hash value y , it should be computationally infeasible to find any message m such that $y = H(m)$.

- **Second Pre-image Resistance.** Given an input m_1 , it should be computationally infeasible to find a different input m_2 such that $H(m_1) = H(m_2)$.
- **Collision Resistance.** It should be computationally infeasible to find a pair of two different messages (m_1, m_2) such that $H(m_1) = H(m_2)$. Such a pair is called a cryptographic hash collision.

Sometimes we demand even more from our hash functions. Notably, we want them to behave like a *random function*: every single output value is an independent, uniformly random string. Sometimes, this property is crucial for the security reduction to go through. And we also want to use them in practice: it must be practically implementable. But the problem is: the set of output strings are perfectly random. Finding a way to represent them in a compact representation of the function (that is, in a string of length polynomial of the output length) would be equivalent to compressing random data, and a simple counting argument says this is not possible: there are 2^{n2^m} many functions that map a binary string of length m to a binary string of length n , and with $\text{poly}(n)$ -length strings, one can express at most $2^{\text{poly}(n)}$ many functions, where $\text{poly}(\lambda)$ is some polynomial.

And this is where we are: cryptographers introduce *random oracles*, a useful *heuristic* that we can use to model the use of perfect hash functions in a variety of applications. Indeed, this has become a well-known and popular heuristic that is used to justify numerous cryptographic constructions. A construction or a security proof in the random oracle model is as follows: the interaction and computation of two or more parties of the scheme is the same as in the plain model, except that no party is allowed to compute the hash function. Hashing is done by a new magical party called the “oracle”. If any party needs to hash something, they send the message (over an imaginary secure channel) to the oracle, who computes the hash and sends it back to them. Formally, any party (including the adversary) is given oracle access to a random

hash function $H : \{0, 1\}^m \rightarrow \{0, 1\}^n$, and it can only learn a value $H(x)$ by querying the oracle H at the classical string x . This formalization essentially places two restrictions on the random oracle model:

- First, it must implement a genuine random function.
- Secondly, the responses given by the oracle must be consistent. In other words, querying a given string through the random oracle multiple times will produce the same result, regardless of which party asks for it.

One last thing we want to say is how we actually implement a random oracle in security proofs. Rather than starting with an entire truth table of the function all filled in, we begin our analysis with an empty truth table and generate it as we go. This is also known as the “lazy sampling” technique.

1. During the game, the oracle maintains a truth table of the function. At the beginning of the game, the oracle’s truth table is empty.
2. Whenever some party asks the oracle to hash a message, the oracle first checks to see if that input value is already stored in the table.
3. If not, it generates a random output string, stores the input message and the new output string in its table, and returns the output string to the caller.
4. If the oracle does find the input value in the table, it returns output value that is already stored there.

It is clear that the actual formalization of the random oracle model and the lazy sampling technique are perfectly indistinguishable from the point of view of a classical adversary. In particular, the classical lazy sampling technique trivially satisfies the two restrictions that we mentioned above. However, guaranteeing these restrictions in the quantum setting poses many technical challenges as we will quickly see below.

Random Oracles in a Quantum World. Here is a fact about the random oracle model: everyone has access to the random oracle. When replacing the oracle with a hash function (like SHA3), the algorithms for computing this function are public, which means that we have to take into account adversaries who use the hash function H as a subroutine, computing its value $H(m)$ on their favorite inputs m at various points during their attack. This is clear in the purely classical setting. What is maybe less obvious is that the adversary with a quantum computer in their basement can do something more insidious: they can download the circuit for H off the Internet (it is public after all), rewrite it as a reversible classical circuit, and then run that circuit on their quantum computer. They are now under no obligation to execute this circuit on classical inputs; in particular, they can evaluate it in superposition, and prepare states like $\sum_m |m\rangle |H(m)\rangle$. Note that this is a completely generic strategy that uses the hash function in a black-box way, that is, it does not depend on the structure of the hash function in any way. So, what does this mean? First, this kind of attack is totally fit in the post-quantum security model. It means that one cannot claim post-quantum security for a scheme involving hash functions without also claiming security against adversaries with quantum query access to the hash. Second, it means that one should work in the *quantum* random oracle model rather than the random oracle model, when analyzing post-quantum security of the scheme.

What is the quantum random oracle model? Essentially, we allow the adversary to query the random oracle in “superposition”, that is, the adversary can submit a quantum state to the oracle H and receives back the evaluated state. This quantum query is the following operation:

$$\sum_{x \in \{0,1\}^m, y \in \{0,1\}^n} \alpha_{x,y} |x, y\rangle_{XY} \mapsto \sum_{x \in \{0,1\}^m, y \in \{0,1\}^n} \alpha_{x,y} |x, y \oplus H(x)\rangle_{XY}.$$

We call the registers X the input registers, and the registers Y the responses (or the output) registers.

Proving Security in the Quantum Random Oracle Model. Unlike the classical case, proving security in the quantum random oracle model can be extremely difficult. In the classical random oracle model, one can

copy down the adversary's queries as a means to learning what points the adversary is interested in. Many classical security proofs crucially use this information in order to construct a new adversary which solves some hard underlying problem, reaching a contradiction. In the quantum setting, such copying is impossible by no-cloning (Theorem 3.1). One can try to record some information about the query, but this amounts to a measurement of the adversary's query state which can be detected by the adversary. A mischievous adversary may refuse to continue if it detects such a measurement, rendering the adversary useless for solving the underlying problem.

To illustrate what might go wrong, let consider the natural approach of “lazy-sampling” that is analogous to the classical setting: the oracle performs a map that may look like $|x, y\rangle \mapsto |x, y \oplus u\rangle \otimes |x, u\rangle$ for each query, where u is chosen uniformly at random, and everything to the right of the tensor is the oracle's internal state.

But now consider the following query by an adversary. It sets up the uniform superposition $\sum_{x,y} |x, y\rangle$ and queries. In the case where H is a single fixed classical function, then the state after the query is

$$\sum_{x,y} |x, y \oplus H(x)\rangle = \sum_{x,y} |x, y\rangle.$$

We can see that the state is unchanged by making the query. In contrast, when implementing the lazy-sampling, the query would result in

$$\sum_{x,y} |x, y \oplus u\rangle \otimes |x, u\rangle.$$

Here, the adversary's state (the left side of the tensor) is entangled with the oracle's state (the right side of the tensor). It is straightforward to detect this entanglement by applying the Quantum Fourier Transform to the adversary's input registers, and measuring the result in the computational basis. The Quantum Fourier Transform performs the following operation efficiently:

$$|x\rangle \mapsto \frac{1}{\sqrt{q^n}} \sum_{y \in \{0,1\}^n} \omega_q^{x \cdot y} |y\rangle,$$

where $\omega_q := \exp(\frac{2\pi i}{q})$, and $x \cdot y$ denotes the dot product. In the former, this process will result in a 0, since applying Quantum Fourier Transform

(or simply Hadamard gates if the input is in $\mathbb{Z}_2$) to $\sum_x |x\rangle$ maps it to $|0\rangle$. In the latter, this process will result in a random string $x \in \{0, 1\}^m \neq 0$ with overwhelming probability, since in this case, we are measuring a uniform superposition state. Thus, the two cases are easily distinguishable, undermining the applicability of lazy sampling.

To remedy the problem of simulating random oracles in the quantum setting, new quantum-sound proof techniques have been developed. The first ones are stateless simulations. Here, the simulator (who is trying to simulate the random oracle in the security proof) commits to a function to implement the oracle in the very beginning, and then sticks with this implementation throughout the entire security game. Using this strategy, it is easy to verify that one of the requirements of the random oracle model is satisfied: namely, the consistency of the responses given by the oracle. We also need to make sure that the simulator implements a genuine random function. For this requirement, we have the following results. If the number of the queries to the oracle is bounded, say, at most q queries, then Zhandry [53] shows that $2q$ -wise independent functions is good enough.

Theorem 4.1. For any sets X and Y , we can efficiently construct a random oracle from X to Y capable of handling q quantum queries, where q is a polynomial in the security parameter. More specifically, the behavior of any quantum algorithm making at most q queries to a $2q$ -wise independent function is identical to its behavior when the queries are made to a random function.

Besides simulability, security proofs in random oracle model usually require more:

- The reduction can observe the input points at which the adversary makes queries to the random oracle. We will refer to this ability of the challenger as *observability*.
- The challenger can control the response of the random oracle at these input points, often embedding instances of some hard problem in the response. We will refer to this ability of the challenger as *reprogrammability*.

Again, it is clear that classical lazy-sampling allows us to achieve these two abilities trivially. Fortunately, it is possible to do some sort of on-the-fly simulation analogous to the classical lazy-sampling simulation if care is taken to implement the oracle correctly, as shown in the breakthrough work by Zhandry [54]. We shall not go into the details of the technique, but we give an informal and slightly inaccurate description of the idea here. Concretely, Zhandry simulates the random oracle as a stateful oracle which stores a quantum superposition of databases, where a database is just a list of input/output pairs (x, y) . A database intuitively represents a partial specification of the oracle: if a pair (x, y) is in the database, it means the oracle on input x is set to y , whereas if there is no pair that begins with x , it means the oracle is un-specified at x . Since the oracle actually stores a superposition of databases, a point x can be in superposition of being specified and unspecified. Originally, the database starts out empty. On query x , very roughly, if x is not found in the database, a pair (x, y) is added, where y is in *uniform superposition* over all possible y values (instead of being a classical value as in the classical setting). Recall that the query can be quantum, so this addition to the database is happening in superposition. Then once x is guaranteed to be specified, the query is answered (again in superposition). Now, an important difference from the classical setting is this: in order to maintain perfect indistinguishability from a truly random oracle, a particular test is performed on the database after answering the query. This test determines whether the adversary maintains any knowledge of the oracle at input x . If not, the pair (x, y) is removed from the database.

This technique has been found in many applications, allowing us to translate more classical results to the quantum setting, including the famous Fiat-Shamir transformation by Liu and Zhandry [55].¹

Finally, to conclude this section, we mention one of the most common tools that is used for the remaining task of *reprogrammability*: the One-Way To Hiding (OW2H) lemma, first introduced by Unruh [57]. Informally, the lemma states that given two oracles $G, H : \{0, 1\}^m \rightarrow$

¹An independent and concurrent work of Don, Fehr, Majenz, and Schaffner [56] achieves the same result albeit from using a different technique.

$\{0, 1\}^n$ whose outputs differ with respect to a set of inputs $S \subseteq \{0, 1\}^m$, and an algorithm A that has quantum access to either G or H , the probability that A can distinguish between G and H is essentially bounded by the square root of the probability that measuring a random quantum oracle query made by A results in a classical string x in S .

As a side note, the original version of the OW2H lemma (introduced by Unruh [57]) handled only random oracles G and H , and found widespread use in proving security of cryptosystems in the quantum random oracle model. This lemma has been generalized to the setting in which not only it allows G and H to have an arbitrary output distribution but also allows the distinguisher's auxiliary input to be arbitrarily correlated with G, H and the difference set S . Due to its relative simplicity and that it does not result in a significant loss in security proofs, the generalized OW2H lemma can be useful in many applications, even in the standard model. Because of this, we will state here the generalized version of the OW2H lemma, introduced by Ambainis, Hamburg, and Unruh [58].

Theorem 4.2. Let $\mathcal{S} \subseteq \{0, 1\}^m$ be a random subset. Let $G, H : \{0, 1\}^m \rightarrow \{0, 1\}^n$ be oracles satisfying $G(x) = H(x)$ for every $x \notin \mathcal{S}$. Let z be a random bit string. ($\mathcal{S}, G, H, z$ may have arbitrary joint distribution.) Let A be a quantum oracle algorithm making at most q quantum queries to its corresponding oracle (either G or H). Let B^H be an oracle algorithm that on input z does the following: picks $i \leftarrow \{1, \dots, q\}$ randomly, runs $A^H(z)$ until (just before) the i -th query, measures all query input registers in the computational basis, and outputs the set $\mathcal{T} = \{t_1, \dots, t_{|\mathcal{T}|}\}$ of measurement outcomes (if A makes less than i queries, the measurement outcomes are taken to be $\perp \notin \{0, 1\}^m$). Let

$$\begin{aligned} P_{\text{left}} &= \Pr[1 \leftarrow A^H(z)] \\ P_{\text{right}} &= \Pr[1 \leftarrow B^H(z)] \\ P_{\text{guess}} &= \Pr[\mathcal{S} \cap \mathcal{T} \neq \emptyset : \mathcal{T} \leftarrow B^H(z)] \end{aligned}$$

Then $|P_{\text{left}} - P_{\text{right}}| \leq 2q\sqrt{P_{\text{guess}}}$. The same result also holds with B^G instead of B^H in the definition of P_{guess} .

Let us explain the lemma: P_{left} denotes the probability that $b = 1$

when b is the return value of running adversary A with oracle access (quantum queries) to H and input z . Analogously for P_{right} . P_{guess} is the probability of extracting a “reprogrammed” position. So $|P_{\text{left}} - P_{\text{right}}|$ is the probability with which the adversary distinguishes H and G , and the lemma bounds that probability. Combining the OW2H lemma for reprogrammability and the simulatability techniques described above (by either using the compressed oracle technique, or $2q$ -wise independent hash functions) allows us to translate many important results in the classical setting to the quantum setting, see Ambainis, Hamburg, and Unruh [58].

4.2.2 Computational Binding Quantum Commitments

A commitment protocol is a protocol with two parties, Alice and Bob (or sender and recipient). It consists of two phases, the *commit* and the *open* phase. In the commit phase, Alice runs with some input $b \in \{0, 1\}$ and Bob has no input. No output is made. In the open phase, both Alice and Bob have no input. Bob outputs a bit $b' \in \{0, 1\}$ or aborts. Intuitively, we require that Bob will output the bit b' that Alice committed herself to in the first phase, that is, on one hand Alice cannot change her mind about the bit (binding property). On the other hand, we do not want Bob to learn the bit b before the open phase (hiding property).

Formally, a commitment scheme $(\text{Commit}, \text{Vf})$ consists of a randomized algorithm Commit and a deterministic algorithm Vf . $(c, u) \leftarrow \text{Commit}(1^\lambda, m)$ returns a commitment c and the opening information u for the message m and security parameter λ . To open, the committer sends (m, u) to the recipient who checks whether $\text{Vf}(1^\lambda, m, c, u) = 1$. Both Commit and Vf have classical input and output. For simplicity, we take the message space of Commit as $\{0, 1\}$. We will focus on non-interactive classical commitments, that is, the commit and open phase consists of a single classical message. However, the adversary who tries to break the binding or hiding property will be a quantum-polynomial-time algorithm.

The basic properties of a commitment scheme are the following:

Definition 4.1 (Completeness). $(\text{Commit}, \text{Vf})$ has perfect completeness iff $\forall m \in \{0, 1\}, \Pr[\text{Vf}(1^\lambda, m, \text{Commit}(1^\lambda, m))] = 1$.

Definition 4.2 (Statistical Hiding). $(\text{Commit}, \text{Vf})$ is statistically hiding iff for all $\lambda \in \mathbb{N}$, for all (possibly unbounded) non-uniform adversary $\mathcal{A}$, we have that $\text{TD}(\mathcal{A}(\text{Commit}(1^\lambda, 0)), \mathcal{A}(\text{Commit}(1^\lambda, 1))) \leq \text{negl}(\lambda)$ for a negligible function $\text{negl}(\cdot)$.

One can see that the definition of computational hiding is straightforward, we just take the classical definition but consider quantum-polynomial-time adversaries instead of classical ones, in the same way as in the definition of semantic security for encryption in the post-quantum setting.

Definition 4.3 (Computational Hiding - Informal). No quantum-polynomial-time algorithm V^* can distinguish the commitment of 0 and the commitment of 1, except with negligible probability.

One might be tempted to assume that the definition of the *computational* binding property in the quantum setting is also straightforward; we can define it in the same way as of the hiding property, by considering quantum-polynomial-time adversaries:

Definition 4.4 (Classical-style Computational Binding - Informal). No quantum-polynomial-time algorithm C^* can output, except with negligible probability, a commitment c (i.e., the message sent during the commit phase) as well as two openings u, u' that open c to two different messages m, m' .

Unfortunately, this definition turns out to be inadequate in the quantum setting. Consider the usual way of building a computationally binding commitment from a collision resistant hash function H : to commit to a message $m \in \{0, 1\}$, choose a random $r \in \{0, 1\}^n$ and output $H(m, r)$. Classically, this is computationally binding by the collision resistance of H : if an adversary can open the commitment to two different values, this immediately yields a collision for H . Unruh [59], based on the work of Ambainis, Rosmanis, and Unruh [60], shows in the quantum setting, collision resistance – even against quantum adversaries – is not enough. Indeed, Unruh shows that for certain hash functions H it may be possible for the adversary to produce a commitment, and only afterward decide on the committed value.

Essentially, the adversary, given access to some oracles, constructs a uniform superposition of inputs, then applies the function H in superposition, and measures the output register to obtain a value y . The state collapses to the superposition $|\psi_y\rangle$ of all pre-images $x = (m, r)$ of y . Then, using particular properties of H , the adversary is able to perturb $|\psi_y\rangle$ so that it becomes a different superposition of pre-images of y . One simply de-commits to any message by first modifying the superposition and then measuring. However, this does not violate the collision-resistance of H . Notice that, while it is easy to sample states $|\psi_y\rangle$, it is impossible to sample two copies of the same $|\psi_y\rangle$. Indeed, given two copies of $|\psi_y\rangle$, simply measure both copies. Since these are superpositions over many inputs, each state will likely yield a different x . The two x 's obtained are both pre-images of the same y , and therefore constitute a collision for H . Now, since the adversary cannot copy $|\psi_y\rangle$, the adversary can only ever perform the above procedure once and obtain only a single de-commitment (and effectively destroy the state $|\psi_y\rangle$ during this process). Thus, the adversary cannot compute two openings simultaneously, but he can open the commitment to an arbitrary message once. This is in clear contradiction to the intuition of the binding property.

Remark 4.1. As a general rule of thumb, in a quantum world, being able to perform two tasks with good probability, does not imply being able to perform both tasks simultaneously. In the above example, the adversary can open the commitment to either the message 0 or 1 with high probability, but he cannot do so simultaneously.

To overcome this potential limitation, we need a new definition for computational binding property, which is called the *collapsing* property [59]. To explain the definition of collapse-binding commitments, for a moment let's consider a perfectly-binding commitment. That is, when an adversary A outputs a commitment c , there is only one possible message m_c that A can open c to. Hence, if the adversary A outputs a superposition of messages that he can open c to, that superposition will necessarily be in the (classical) state $|m_c\rangle$. We can characterize perfectly-binding commitments by requiring: when an adversary outputs a superposition of messages that he can open the commitment c to, that

superposition will necessarily be a single computational basis vector (i.e., no non-trivial superposition).

This idea is depicted in Figure 4.1. Consider the circuit (a). Here,

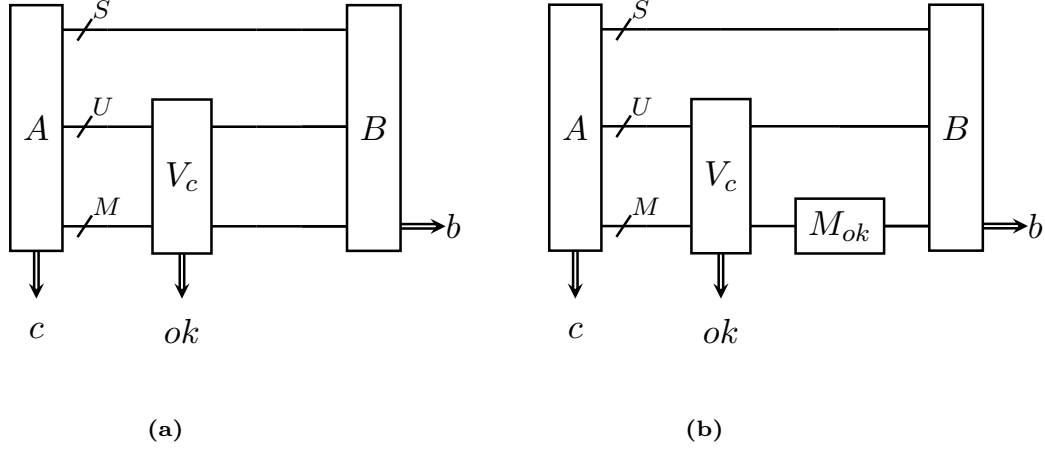


Figure 4.1: Games from the definition of collapse-binding commitments.

the adversary is characterized by two quantum algorithms A and B . The first part A of the adversary outputs a commitment c (which is a classical message). Furthermore, he outputs three quantum registers S, U, M . S contains his *internal* state. M is supposed to contain a superposition of messages, U a superposition of corresponding opening informations. Then we apply the measurement V_c which checks whether U, M contain matching opening information/message. More formally, V_c measures whether U, M is a superposition of states $|u, m\rangle$ such that u is valid opening information for message m and commitment c . Let $ok = 1$ if the measurement succeeds. Then we feed the registers S, U, M back to the second part B of the adversary. B outputs a classical bit b .

Now consider the circuit in Figure 4.1(b). Here we added a measurement M_{ok} on the registers M after V_c . M_{ok} is a complete measurement in the computational basis, but is executed only if $ok = 1$. Since M_{ok} disturbs the state of M iff that state is not a computational basis vector, we can rephrase the definition of perfectly-binding commitments: A commitment is perfectly-binding iff, for all computationally unbounded adversaries (A, B) , $\Pr[b = 1]$ is equal in Figure 4.1(a) and Figure 4.1(b) where b is the output of B .

Now we are ready to weaken this characterization to get a computa-

tional binding property. Basically, we require that the same holds for quantum-polynomial-time adversaries.

Definition 4.5 (Computational Collapse-Binding - Informal). A commitment is collapse-binding iff, for all quantum-polynomial-time adversaries (A, B) , $\Pr[b = 1]$ in Figure 4.1(a) is negligibly close to $\Pr[b = 1]$ in Figure 4.1(b).

Collapse-binding commitments are useful only if they exist. Perfectly-binding commitments are easily seen to be collapse-binding, but then we cannot have statistically hiding or short commitments. Indeed, a notion of *collapsing* hash function can be defined similarly. Recall our example of building commitments from collision-resistant hash functions above. Roughly, these are hash functions for which $|x\rangle$ and $|\psi_y\rangle$ are computationally indistinguishable. Using such hash functions to build commitments, one obtains collapse-binding commitments, for which the attack described above is impossible. Finally, Unruh [59] shows that a random oracle is collapse-binding. Further note that the definition of the collapse-binding property (4.5) given above was formulated specifically for non-interactive commitments where only classical messages are exchanged and where the verification in the opening phase is deterministic, but the definition can be generalized to the interactive setting as well. Interestingly, the existence of non-collapse binding commitments implies the existence of some unclonable cryptographic objects that we will discuss in Chapter 6.

4.2.3 The Quantum Rewinding Lemma

In this section, we will analyze the failure of basic techniques used in classical security proofs to carry over directly to the quantum setting. In particular, we take the case of the *rewinding* technique in the context of zero-knowledge proof systems into consideration. We first briefly recall the definition of a zero-knowledge proof system here, and then analyze the security proof of the Goldreich, Micali, and Wigderson [52] in both classical and quantum settings.

A zero-knowledge proof is, intuitively speaking, a protocol in which a prover P is able to convince a verifier V of the truth of a statement

x in such a way that the verifier learns nothing (except, of course, the fact that x is true).

More formally, we fix a relation R . If $(x, w) \in R$, we say that w is a witness for the statement x . We defined the language $\mathcal{L}_R$ of true statements as follows:

$$\mathcal{L}_R := \{x : \exists w \text{ s.t. } (x, w) \in R\}.$$

In other words, $x \in \mathcal{L}_R$ iff there is a witness for x . We also employ the following notation: For two machines A, B , $\langle A(a), B(b) \rangle$ denotes the output of B after an interaction of A and B where A gets input a and B gets input b .

Definition 4.6 (Proof Systems). We call a pair (P, V) of interactive machines a *proof system* for the relation R with *soundness-error* ϵ iff the following two conditions are fulfilled:

- **Completeness:** For any $(x, w) \in R$, we have that

$$\Pr[\langle P(x, w), V(x) \rangle = 1] = 1.$$

- **Soundness:** For any (potentially computationally unbounded) machine P^* , and for any $x \notin \mathcal{L}_R$, we have that

$$\Pr[\langle P^*(x), V(x) \rangle = 1] \leq \epsilon.$$

Next, we define what it means that a proof system is zero-knowledge.

Definition 4.7 (Classical Zero-knowledge). A pair (P, V) of interactive machines is *zero-knowledge* if for any polynomial-time machine V^* there exists a polynomial-time machine S (called the simulator) and a negligible $\text{negl}(\cdot)$ such that for all $(x, w) \in R$ and all $z \in \{0, 1\}^*$, we have

$$\langle P(x, w), V^*(x, z) \rangle \approx_{\text{negl}(|x|)} S(x, z). \quad (4.1)$$

A well-known example for a zero-knowledge proof is the Goldreich-Micali-Wigderson for the Graph Isomorphism problem stated as follows.

Definition 4.8 (Graph Isomorphism). The relations R_{GI} is defined as follows: $(x, w) \in R_{GI}$ iff $x = (G_1, G_2)$ and $w = \sigma$ where G_1, G_2 are graphs and $\sigma : G_1 \rightarrow G_2$ is a graph isomorphism.

Definition 4.9 (Graph Isomorphism Proof System). Let GIP denote the following protocol between two polynomial-time machines P and V . Both P and V get as input a pair G_0, G_1 of simple, undirected n -vertex graphs. The prover P also gets as input a permutation $\sigma \in S_n$, for S_n denoting the symmetric group on indices $\{1, \dots, n\}$, that satisfies $\sigma(G_1) = G_0$ if G_0 and G_1 are isomorphic.

- **Prover's Step 1:** Choose a random permutation $\pi \in S_n$ uniformly and send the graph $H = \sigma(G_0)$ to the verifier.
- **Verifier's Step 1:** Choose $a \in \{0, 1\}$ uniformly at random and send a to the prover. (Roughly speaking, the verifier is challenging the prover to exhibit an isomorphism between G_a and H).
- **Prover's Step 2:** Set $\tau = \pi \circ \sigma^a$ and send τ to the verifier. (Note that if it is the case that $\sigma(G_1) = G_0$, it must hold that $\tau(G_a) = H$).
- **Verifier's Step 2:** Accept if $\tau(G_a) = H$, reject otherwise.

We then have that

Theorem 4.3. GIP is a zero-knowledge proof system.

We now present the definitions of zero-knowledge proofs for the quantum case. For two quantum machines A, B , $\langle A(a), B(b) \rangle$ denotes the quantum state of B after an interaction of A and B where A gets input a and B gets input b . Here a and b may be classical values or density operators. The definition of being a proof system (i.e., completeness and soundness) is identically the same as in the classical case (4.6), except that P^* is allowed to be a quantum machine. Next, we present the definition of quantum zero-knowledge.

Definition 4.10 (Quantum Zero-knowledge). A pair (P, V) of interactive machines is *quantum zero-knowledge* if for any quantum-polynomial-time machine V^* there exists a quantum-polynomial-time machine S (called the simulator) and a negligible $\text{negl}(\cdot)$ such that for all $(x, w) \in R$ and all ρ , we have

$$\text{TD}(\langle P(x, w), V^*(x, \rho) \rangle, S(x, \rho)) \leq \text{negl}(|x|). \quad (4.2)$$

Note that the fundamental difference between the quantum case and the classical case is that, in the former, the “auxiliary input” ρ that V^* and the simulator S get is a quantum state.

To show that GIP is QZK, we need to construct a suitable simulator S . However, it turns out that the construction from the classical case does not directly carry over. The reason is that the simulator in the classical case uses *rewinding*: It tries to produce a simulation, and, if it fails, it tries again (until it guesses correctly the verifier’s challenge). In the quantum case, trying again does not necessarily work, because the first try may have destroyed the input state ρ , so in the second try the adversary might abort, causing the simulation to fail.

To remedy the problem, we would need a new technique, called the *quantum rewinding lemma*, which allows some interesting interactive proof systems to be proven to possess the zero-knowledge property in the quantum setting. This lemma was first introduced in Watrous [61], and the version we present here is given in Bitansky and Shmueli [62].

Lemma 4.4 (Quantum Rewinding Lemma). *There is a quantum algorithm R that gets as input:*

- *A general quantum circuit Q with n input qubits that outputs a classical bit b and an additional m qubits.*
- *An n -qubit state $|\psi\rangle$.*
- *A number $t \in \mathbb{N}$.*

R executes in time $t \cdot \text{poly}(|Q|)$ and outputs a distribution over m -qubit states $D_\psi = R(Q, |\psi\rangle, t)$ with the following guarantees.

For an n -qubit state $|\psi\rangle$, denote by Q_ψ the conditional distribution of the output distribution $Q(|\psi\rangle)$, conditioned on $b = 0$, and denote by $p(\psi)$ the probability that $b = 0$. If there exist $p_0, q \in \{0, 1\}, \epsilon \in (0, \frac{1}{2})$ such that:

- *Amplification executes for enough time: $t \geq \frac{\log(1/\epsilon)}{4 \cdot p_0(1-p_0)}$,*
- *There is some minimal probability that $b = 0$: For every n -qubit state $|\psi\rangle$, $p_0 \leq p(\psi)$,*

- $p(\psi)$ is input-independent, up to ϵ distance: For every n -qubit state $|\psi\rangle$, $|p(\psi) - q| < \epsilon$, and
- q is closer to $\frac{1}{2}$: $p_0(1 - p_0) \leq q(1 - q)$,

then for every n -qubit state $|\psi\rangle$,

$$\text{TD}(Q_\psi, D_\psi) \leq 4\sqrt{\epsilon} \frac{\log(1/\epsilon)}{p_0(1 - p_0)}.$$

How do we use this quantum rewinding lemma to show the quantum zero-knowledge property of GIP protocol? The idea is that, first we construct a polynomial-time simulator S_1 , that tries to produce a simulation and either produces a perfect simulation or aborts, and that aborts with probability exactly $\frac{1}{2}$, independent of the auxiliary input ρ . The construction of this simulator is analogous to the classical case and the proof that it produces a perfect simulation with probability $\frac{1}{2}$ also follows very closely the lines of the proof in the classical case. We then purify S_1 resulting in Q , which can be used to construct the algorithm R mentioned in the lemma. The final simulator S runs R and outputs its state.

Note further that the above lemma (Lemma 4.4) also covers the case where S_1 produces a quantum state that has negligible trace distance from $\langle P(x, w), V^*(x, \rho) \rangle$, and even protocols that are only computationally quantum zero-knowledge.

As a final word, note that in general, rewinding is not possible in the quantum setting, and Watrous' quantum rewinding lemma only works in some specific cases. One particular requirement is that the success probability of the algorithm Q must be independent of the auxiliary input $|\psi\rangle$.

Further reading. For quantum zero-knowledge, see [61] and the survey given by Vidick and Watrous [63]. In the discussion above, we see how to recover computational zero-knowledge in the quantum setting using Watrous' quantum rewinding lemma. What about the computational soundness? Again, many classical reductions are via a rewinding argument. In particular, the classical reduction needs to perform the following (abstract) task. Assuming that a malicious prover P^* , up to

round i -th, answers a random challenge r with success probability ϵ , the reduction rewinds the malicious prover and obtains some number k of different accepting transcripts $(\tau, r_1, z_1), \dots, (\tau, r_k, z_k)$, with the same prefix τ (which is the transcript of the protocol up to round i -th). Here, not only the reduction needs to rewind undetectably, but also needs to be able to *record* (or extract) the prover’s response. In the classical setting, this is an elementary task. The works of Unruh [64] and Chiesa, Ma, Spooner, and Zhandry [65] allow one to recover this type of rewinding in the quantum setting, provided the protocol satisfies a special “collapsing” condition (which is defined similarly to the notion of collapse-binding for commitments in Section 4.2.2). Roughly speaking, this condition says that the measurement performed by the reduction in the rewinding loop to obtain the response is indistinguishable (to the adversary) from a binary measurement of whether the obtained response is valid or not. Notably, the work of Chiesa, Ma, Spooner, and Zhandry [65] introduces a new technique, the so-called state repair technique, to recover the success probability of a state after it is disturbed by a (binary) projective measurement, which allows them to bypass the rewinding barriers faced by the previous technique of Unruh [64]. This technique was developed in subsequent works to recover post-quantum security of many rewinding-based protocols (e.g., Lombardi, Ma, and Spooner [66] and Lai, Malavolta, and Spooner [67]).

4.3 Quantum Security

As previously discussed, post-quantum security focuses on resisting attacks where the attacker has a quantum computer and can only evaluate publicly available functions in superposition. However, if post-quantum security is cryptographically fully relevant (i.e., when universal quantum computers become available), it also makes sense to consider quantum security against superposition attacks as relevant. One reason is that, once quantum computers are available, they should be used in a hybrid computation model together with classical computers,² and thus quantum computers might also be accessible to honest parties. Now, in

²After all, it is hard to justify that we build quantum computers only to break the current classical cryptography.

this setting, the honest party can be tricked into evaluating queries on a quantum computer such that an attacker can now, for example, evaluate encryption queries in superposition. In a sense, quantum security is the most conservative security notion, where one places minimal assumptions on the underlying hardware: In the world where end users run quantum computers, the use of “classical computers” boils down to the assumption on the hardware to ensure that there are no quantum effects in the classical computers running the algorithms. This is similar to the line of research on device-independent quantum key distribution (QKD) [68], where one does not want to make any assumption on the hardware implementing QKD.

However, extending security notions to the “fully” quantum setting is far less straightforward. In contrast to the post-quantum setting, it is hard to define security notions for even simple primitives such as encryption or digital signatures. The reasons are mainly due to fundamental properties of quantum mechanics including the destructiveness of quantum measurement, and the quantum no-cloning theorem (Theorem 3.1).

4.3.1 Defining Security for Encryption Against Quantum Adversaries

Classical Security Notions The standard security notion for encryption is defined using the indistinguishability game which is modeled as a game between a challenger and an adversary A . Classically, there are several variants of this model which are all equivalent: Find-then-Guess, Real-or-Random, and Left-or-Right. In the Find-Then-Guess style, the game starts with a first learning phase (with access to some oracles), followed by a challenge phase where A sends a challenge query (two messages x_0 and x_1 to be encrypted) and receives a challenge ciphertext (encryption of x_b). Afterward, a second learning phase follows, and finally, A outputs a solution (its guess for the bit b). The security reduction consists of constructing a new adversary which simulates A and solves some hard underlying problem. The learning phases define the type of attacks: chosen-plaintext attacks (CPA) if the adversary has access to an encryption oracle in both learning phases, and chosen-

ciphertext attacks (CCA) in case it also has access to a decryption oracle in the learning phases (non-adaptive or CCA1 if it is restricted to the first learning phase, and adaptive or CCA2 otherwise).

Indistinguishability against adaptive chosen-ciphertext attack (IND-CCA2) is usually considered the most desirable security notion for encryption. In the CCA2 games, the adversary is restricted not to ask for decryption of the challenge ciphertext, otherwise, this would lead to a trivial guess of the bit b . It is the role of the challenger to ensure that the adversary obeys this rule, which intrinsically requires the ability to copy, store and compare classical strings. For the rest of this section, we will focus on the problem of defining chosen-ciphertext security for classical encryption.

Quantum Indistinguishability Security Notions Boneh and Zhandry [69] propose the first definition of IND-CCA for both symmetric and public-key encryption schemes against quantum adversaries allowed to make quantum encryption and decryption queries. However, they also show that the natural translation of the classical Find-then-Guess paradigm to the quantum setting is unachievable, even for IND-CPA security.

Definition 4.11 (Chosen-plaintext Security with Quantum Challenge Queries). A symmetric key encryption scheme $\mathcal{E} = (\text{Enc}, \text{Dec})$ is indistinguishable under a fully quantum chosen plaintext attack (IND-fqCPA secure) if no efficient adversary A can win the following game, except with probability at most $\frac{1}{2} + \text{negl}(\lambda)$, where $\text{negl}(\cdot)$ is a negligible function in the security parameter λ :

- **Key Gen:** The challenger picks a random key k and a random bit b .
- **Encryption Queries:** A is allowed to make chosen message queries on superpositions of message pairs. For each such query, the challenger chooses randomness r , and encrypts the appropriate message in each pair using r as randomness:

$$\sum_{x_0, x_1, y} \alpha_{x_0, x_1, y} |x_0, x_1, y\rangle \rightarrow \sum_{x_0, x_1, y} \alpha_{x_0, x_1, y} |x_0, x_1, y \oplus \text{Enc}(k, x_b; r)\rangle$$

- **Guess:** A produces a bit b' , and wins if $b' = b$.

We note that in the above definition, as well as all the other definitions in this section, the adversary now has *quantum* access to the keyed oracle. In particular, the keyed oracle – the encryption oracle that implements $\text{Enc}(k, \cdot)$ – is evaluated in superposition, which is not possible in post-quantum security.

Remark 4.2. If an oracle O implements a classical randomized algorithm, there are several choices for how the randomness is used in each query if the oracle is queried in superposition. One option is to choose fresh randomness for each message in the superposition. Another option is to choose a single randomness value for each query, and generate output in the superposition with that randomness. We note that there is a simple transformation that converts an oracle requiring independent randomness for every message into a scheme that is secure when a single randomness value is used for an entire query: for each query, choose a fresh random key k for a quantum pseudorandom function (QPRF) ([70]). This will be the single per-query randomness value. Each input x in the superposition will be answered using randomness obtained by applying the QPRF to x using the key k . From the adversary's point of view, this is indistinguishable from choosing independent randomness for each message. Indeed, Theorem 4.1 shows that we can replace the QPRF with a function drawn from a pairwise independent function family, which allows us to achieve perfect simulability. For this reason, requiring global randomness per query does not change the oracle from the adversary's point of view, but greatly simplifies its implementation. Thus, we choose to implement all randomized oracles this way.

Theorem 4.5. No symmetric-key encryption scheme can achieve IND-fqCPA security.

Proof. The adversary A chooses as challenge messages the states $|0^m\rangle$ and $H|0^m\rangle$ (where H denotes the m -fold tensor Hadamard transform), i.e., he prepares the register in the state $\sum_x \frac{1}{2^{m/2}} |0^m, x, 0^n\rangle_{X_1 X_2 Y}$, where Y is the ciphertext registers. When the challenger performs the encryption, we can have two cases:

- If $b = 0$, that is, the first message state is chosen, the state is transformed into:

$$\sum_x \frac{1}{2^{m/2}} |0^m, x, \text{Enc}(0^m)\rangle = |0^m\rangle \otimes H |0^m\rangle \otimes |\text{Enc}(0^m)\rangle.$$

- If $b = 1$, that is, the second message is chosen, the state is transformed into:

$$\sum_x \frac{1}{2^{m/2}} |0^m, x, \text{Enc}(x)\rangle = |0^m\rangle \otimes \sum_x \frac{1}{2^{m/2}} |x, \text{Enc}(x)\rangle.$$

Notice that in the second case we have a fully entangled state between the second and the third registers X_2, Y . At this point, A does the following:

- measures (traces out) the Y registers;
- apply H to the X_2 registers;
- measures the X_2 registers;
- outputs $b' = 1$ iff the outcome of this last measurement is 0^m , else outputs 0.

We now analyze the success probability of A . If $b = 0$, then the X_2 register is left untouched: By applying again the Hadamard transformation it will be reset to the state $|0^m\rangle$, and a measurement on this state will yield 0^m with probability 1. If $b = 1$ instead, tracing out one half of a fully entangled state results in a complete mixture in the X_2 registers. Applying a Hadamard transform and measuring in the computational basis necessarily gives a fully random outcome, and hence outcome 0^m only with probability $\frac{1}{2^m}$. Thus A can distinguish the two cases with probability significantly larger than the trivial probability $\frac{1}{2}$. $\square$

Remark 4.3. This impossibility shows that some seemingly natural notions of quantum indistinguishability are too strong and might go beyond what a meaningful notion of indistinguishability should achieve. Recall the basic security requirement of an encryption scheme: *the ciphertext does not leak any information about the message, except its*

length. A scheme is broken if it leaks at least one bit of information to the adversary. However, any scheme would be broken in the sense of IND-fqCPA notion even the adversary has no information whatsoever. The attack works only because of properties of quantum mechanics. To put it another way, there are encryption schemes that are insecure in the sense of IND-fqCPA but may be arguably secure for practical use. This highlights the difficulty when defining security notions in the quantum setting.

Chosen-ciphertext Security with Classical Challenge Queries To overcome this impossibility, we can resort to considering quantum queries during the learning phases only, and classical queries during the challenge phase. This approach has another advantage: since challenge queries are classical, when we move to CCA security, we can check if a ciphertext was the result of a challenge query and reject decryption queries for these ciphertexts. Below, we give the definition for symmetric-key encryption, but the same notion can be easily defined for public-key encryption.

Definition 4.12. A symmetric key encryption scheme $\mathcal{E} = (\text{Enc}, \text{Dec})$ is indistinguishable under a fully quantum chosen ciphertext attack (IND-qCCA secure) if no efficient adversary A can win the following game, except with probability at most $\frac{1}{2} + \text{negl}(\lambda)$, where $\text{negl}(\cdot)$ is a negligible function in the security parameter λ :

- **Key Gen:** The challenger picks a random key k and a random bit b . It also creates a list C which will store challenge ciphertexts.
- A is allowed to make three types of queries:
 - **Encryption Queries:** For each such query, the challenger chooses randomness r , and encrypts the appropriate message in each pair using r as randomness:

$$\sum_{x,y} \alpha_{x,y} |x, y\rangle \rightarrow \sum_{x,y} \alpha_{x,y} |x, y \oplus \text{Enc}(k, x; r)\rangle.$$
 - **Challenge Queries:** A sends two classical messages x_0 and x_1 , to which the challenger responds with $c^* = \text{Enc}(x_b)$. The challenger also adds c^* to the list C .

- **Decryption Queries:** For each such query, the challenger decrypts all ciphertexts in the superposition, except those that were the result of a challenge query:

$$\sum_{y,x} \alpha_{y,x} |y, x\rangle \rightarrow \sum_{y,x} \alpha_{y,x} |y, x \oplus f(y)\rangle ,$$

where $f(y) = \perp$ if $y \in C$, and $f(y) = \text{Dec}(k, y)$ otherwise.

- **Guess:** A produces a bit b' , and wins if $b' = b$.

Chosen-ciphertext Security with Quantum Challenge Queries We make some observations on the former definition. In addition to looking artificial, the inconsistency between the learning phases and the challenge phase may lead to a cryptographic construction that fulfills this security notion (IND-qCPA or IND-qCCA) while being subject to an attack. For instance, in Anand, Targhi, Tabia, and Unruh [71], the authors verify IND-qCPA security of XTS mode of operation (with quantum learning queries and classical challenge queries). They design a block cipher such that an encryption scheme in XTS mode, instantiated with that block cipher, can be attacked during the learning phase using quantum learning queries. However, this attack cannot be used to violate the IND-qCPA security definition. The explanation for this inconsistency is that this attack cannot be implemented in the challenge phase due to the classical restriction imposed on the adversary. This example supports the claim that the inconsistency between the learning phases and the challenge phase can be problematic and should be overcome.

Now, apart from the impossibility mentioned above, let us highlight the barriers when one tries to translate classical indistinguishability notions to the quantum setting, when the adversary can send superposition queries to the challenger. The main obstacle is to define how the challenger should reply to the quantum decryption queries after the adversary has made the quantum challenge queries. When the challenge queries are classical, they can be recorded and later the challenger can return $\perp$ if the adversary submits one of them as a decryption query. Although it is trivial and inherent to record the challenge ciphertext in

the classical setting, it is highly non-trivial to record ciphertexts in the quantum world, due to a number of technical obstacles, all of which can be traced to quantum no-cloning and the destructiveness of quantum measurements.

Since we now consider the adversary’s challenge queries as quantum states, it may be tempting to think that the approaches from the literature on quantum encryption (that is, the problem of encrypting *quantum* data) would work here. The notorious “recording barrier” that we discussed above has arisen previously in the literature on quantum encryption. In particular, the notions of quantum ciphertext indistinguishability under adaptive chosen-ciphertext attack and quantum authenticated encryption — as developed by Broadbent and Jeffery [72] and Alagic, Gagliardini, and Majenz [73] — requires circumventing similar obstacles. Unfortunately, we will explain below that this approach unlikely works for classical encryption schemes.

On a high level, in the security game for a quantum authenticated encryption, an adversary A has negligible probability in distinguishing between two experiments: in the real one, it has access to encryption and decryption oracles with no restrictions, whereas in the random one, the challenge encryption oracle replaces A ’s queried plaintexts by random ones (half of a maximally-entangled state), and the decryption oracle answers with the originally queried plaintexts if the adversary asked for decryption of a challenge ciphertext (which can be done by first decrypting the ciphertext and applying a measurement on the entangled state), otherwise it answers normally. However, in the context of classical encryption, the standard oracle for encryption is a map $|x, y\rangle \mapsto |x, y \oplus f(x)\rangle$ where f is the encryption algorithm. The adversary can then use the same strategy to detect the random experiment’s simulation: it prepares a maximally-entangled state $|\phi^+\rangle_{XX'}$ and uses half of it (the registers X) as the challenge plaintext, and keeps X' . After receiving the challenge ciphertext, it measures the plaintext registers and X' , and trivially distinguishes whether it is in the random experiment. We note that this attack cannot be performed without relaying, that is the plaintext registers X need to be available to A after the challenge encryption. However, non-relaying is indistinguishable from being traced out the plaintext registers (from A ’s perspective).

This inherently reduces to a definition with classical challenge queries.

We further note that Theorem 4.5 only applies to the Find-then-Guess paradigm (and it also extends to the Left-or-Right paradigm, essentially by the same attack), but it leaves open the possibility to define security notions with quantum challenge queries for the Real-or-Random paradigm. In fact, chosen-ciphertext security with quantum queries has been defined for both public-key and symmetric-key encryption by Chevalier, Ebrahimi, and Vu [74] in the Real-or-Random paradigm, using the compressed oracle technique by Zhandry [54]. We also refer the reader to the works of Gagliardoni, Hülsing, and Schaffner [46] and Carstens, Ebrahimi, Tabia, and Unruh [75], where the authors studied and characterized several security notions for symmetric-key encryption against quantum *chosen-plaintext* attacks, which motivate the notions given by Chevalier, Ebrahimi, and Vu [74]. Furthermore, Chevalier, Ebrahimi, and Vu [74] show that any encryption scheme satisfying quantum indistinguishability with classical challenge queries can be lifted to their stronger indistinguishability with quantum challenge queries *unconditionally*, in a black-box way. It is thus easier to work it with the IND-q(CPA/CCA) security notion (Definition 4.12) and then invoke their lifting theorem to achieve full quantum security.

4.3.2 Defining Security for Authentication Against Quantum Adversaries

Another fundamental operation in classical cryptography is messages authentication: a sender Alice wants to send a message m over an insecure channel to a receiver Bob, with the guarantee that the message was not tampered with in transit. Similar to the case of encryption, formulating and designing authentication of *classical* messages in the presence of adversaries with quantum query access to the signing function has been a longstanding challenge, as the familiar classical notions of unforgeability do not directly translate into meaningful notions in the quantum setting. In the context of authentication, a particular difficulty is how to fairly capture the notion of “predicting an unqueried value” when the adversary can query in quantum superposition.

For simplicity, we focus on the symmetric-key setting, that is, we

consider message authentication codes (MACs), but the discussion in this section applies to classical digital signature schemes identically. We first review classical definition for MACs, and then discuss several approaches for this problem.

Definition 4.13 (EUF-CMA security). A MAC system is said to be existentially unforgeable under a classical chosen message attack if no PPT attacker can win the following game: a random key k is chosen from the key space and the attacker is presented with a signing oracle $S(k, \cdot)$. Queries to the signing oracle are called chosen message queries. Let $L = \{(m_i, t_i)\}_{i=1}^q$ be the set of message-tag pairs that the attacker obtains by interacting with the signing oracle. The attacker wins the game if it can produce an existential forgery, namely a valid message-tag pair (m^*, t^*) satisfying $(m^*, t^* \notin L)$. The MAC system is said to be secure if no PPT adversary can win this game with non-negligible probability in the security parameter λ .

When modeling the entire security game above as a quantum experiment and allow the adversary to issue quantum chosen message queries, that is, the adversary can submit a superposition of messages from the message space and in response receive a superposition of MAC tags on those messages, a quantum chosen message query performs the following transformation on a given superposition of messages:

$$\sum_{m,t} \alpha_{m,t} |m, t\rangle \rightarrow \sum_{m,t} \alpha_{m,t} |m, t \oplus S(k, m)\rangle,$$

where $S(k, m)$ is a tag on the message m with secret key k .

The first definition for authentication in the quantum setting was proposed by Boneh and Zhandry [76]. Roughly speaking, they define a MAC to be unforgeable if, after making q queries to the MAC, no QPT adversary can produce $q + 1$ valid input-output pairs except with negligible probability. We will refer to this notion as “+1 security”.

Definition 4.14 (+1 security). We define a quantum MAC game as follows.

- **Quantum chosen message queries.** The adversary maintains its own quantum state and issue quantum queries to the signing oracle. Let $\sum_{m,x,y} \alpha_{m,x,y} |m, x, y\rangle$ be the adversary’s state just prior

to issuing a signing query. The MAC signing oracle transforms this state as follows:

1. it chooses a random string r that will be used by the MAC signing algorithm; if the signing algorithm is deterministic, skip this step,
2. it signs each “slot” in the given superposition by running $S(k, m; r)$, that is running algorithm S with randomness r (again, the randomness is ignored if the signing algorithm is deterministic). More precisely, the signing oracle performs the following transformation:

$$\sum_{m,x,y} \alpha_{m,x,y} |m, x, y\rangle \rightarrow \sum_{m,x,y} \alpha_{m,x,y} |m, x \oplus S(k, m; r), y\rangle.$$

- **Existential forgery.** After issuing q quantum chosen message queries the adversary wins the game if it can generate $q + 1$ valid classical message-tag pairs.

A MAC system is existentially unforgeable under a quantum chosen message attack (EUF-qCMA) if no QPT adversary can win the quantum MAC game with non-negligible advantage in the security parameter λ .

One interesting property of this definition is that the classical Carter-Wegman construction for *one-time* MACs (that is, the adversary can issue only *one* chosen message query) is not secure anymore (even though, it is secure *unconditionally* in the classical setting).

Theorem 4.6. One-time MACs constructed from pairwise-independent hash function is not secure in the quantum setting according to Definition 4.14.

The proof of this theorem follows from the quantum polynomial interpolation algorithm by Boneh and Zhandry [76]: given oracle access to the polynomial, the algorithm reconstructs its coefficients. In particular, Boneh and Zhandry [76] shows that for any polynomial of degree d , d quantum queries are sufficient to interpolate the polynomial, but classical algorithms require $d + 1$ classical queries. Pairwise-independent

hash functions can be constructed from random linear polynomials, which can be interpolated using a single quantum query.

Fortunately, Boneh and Zhandry [76] also show that while pairwise independence is not enough, 4-wise independence is.

Theorem 4.7. Let A be any quantum algorithm that makes c classical queries and q quantum queries to an oracle H . If H is drawn from a $(c + 2q)$ -wise independent function, then the output distribution of A is identical to the case where H is truly random, which effectively yields a q -time +1-secure MAC construction.

The theorem follows readily from the fact that $2q$ -wise independent functions is perfectly indistinguishable from random oracle (c.f Theorem 4.1), when the adversary makes at most q quantum queries.

On the other hand, similarly to the classical setting, pseudo-random functions (PRFs) are sufficient to achieve +1-secure MACs, namely $S(k, m) = PRF(k, m)$, where PRF is a quantum-secure pseudo-random function. We give its security definition below.

Definition 4.15 (Zhandry [70]). A quantum-secure pseudorandom function is a family of efficient classical functions $PRF : \{0, 1\}^\lambda \times \{0, 1\}^m \rightarrow \{0, 1\}^n$ such that the following holds. For any polynomially bounded $m = m(\lambda)$ and $n = n(\lambda)$, and any QPT adversary A , A cannot distinguish $PRF(k, \cdot)$ for a random $k \leftarrow_R \{0, 1\}^\lambda$ from a truly random function $H : \{0, 1\}^m \rightarrow \{0, 1\}^n$. That is, there exists a negligible $\text{negl}(\cdot)$ such that

$$\left| \Pr \left[A^{PRF(k, \cdot)}(1^\lambda) = 1 \mid k \leftarrow_R \{0, 1\}^\lambda \right] - \Pr \left[A^{H(\cdot)}(1^\lambda) = 1 \right] \right| \leq \text{negl}(\lambda),$$

where the adversary A has *superposition* access to either the oracle $PRF(k, \cdot)$ or $H(\cdot)$.

Theorem 4.8. If $PRF : K \times X \rightarrow Y$ is a quantum-secure pseudo-random function and $1/|Y|$ is negligible, then $S(k, m) = PRF(k, m)$ is a EUF-qCMA-secure MAC.

One can easily see that EUF-qCMA security of the construction S in Theorem 4.8 is reducible to the quantum security of the pseudorandom

function PRF . In fact, an attacker for EUF-qCMA security of S can even be used to construct an attacker breaking the one-wayness of PRF .

Next, one may wonder is this definition good enough? Let's recall the problem of defining computational binding of commitment that we discussed in Section 4.2.2. In particular, quantum algorithms must sometimes “consume” (i.e., fully measure) a state to extract some useful information (e.g., output a valid opening information for the commitment without actually “committing”). For the +1-security notion, there might be an adversary that makes one query but then must consume the post-query states completely in order to make a single, but convincing, forgery. We will see that indeed, this definition has some unsatisfying properties.

- +1-security does not rule out attacks in which the adversary is given *one* signed superposition, and is able to produce *one* valid message/tag pair where the message is not part of the original superposition. For example, consider the case where the sender only signs messages that start with the email address of some intended recipient, say, `bob@gmail.com`. Suppose the adversary tricks the sender into signing a superposition of messages that all begin with `bob@gmail.com`, but then manipulates the signed superposition into a different superposition that includes valid signed messages that do not start with `bob@gmail.com`. Clearly, this is an undesirable outcome.
- Suppose an adversary tricks the sender into signing a uniform superposition on messages, and then produces a classical signed message (m, σ) . From the sender's perspective, each message has weight $1/|M|$, where M is the message space. However, the sender cannot prevent the adversary from “mauling” the signed superposition state and bias the output of the measurement to obtain, say, (m^*, σ^*) with probability much higher than $1/|M|$. Again, +1-security definition does not preclude such a biasing, since the adversary only ever obtains a single signed message. Thus, this definition does not capture natural non-malleability properties one would hope for from an authentication scheme.

To address these above limitations, several approaches have been proposed. At a high level, the situation can be divided into two settings: one-time setting in which the adversary can only make one chosen message query, and many-time setting in which the adversary can make more than one chosen message query, or even unbounded number of queries (as long as the number of queries is polynomial in the security parameter). Below, we discuss another solution for the latter setting, proposed by Alagic, Majenz, Russell, and Song [77] and then conclude the section with a solution for the former setting, which is the best possible one can hope for classical authentication protocols.

Blind Unforgeability In this approach we examine the behavior of adversaries in the following experiment. Let $\varepsilon : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ be an efficiently computable function, and B_ε be a subset of the message space $\mathcal{M}$, that is selected by placing each $m \in \mathcal{M}$ independently with probability $\varepsilon(\lambda)$ for the security parameter λ . Define the blind signing oracle $B_\varepsilon S$ as follows.

$$B_\varepsilon S(k, m) = \begin{cases} \perp & \text{if } m \in B_\varepsilon \\ S(k, m) & \text{otherwise} \end{cases},$$

where $S(k, \cdot)$ is the signing algorithm. Given an adversary A and A -selected parameter ε , a signing oracle $B_\varepsilon S$, the “blind forgery experiment” is:

1. Generate a signing key k and random blinding B_ε .
2. Grant quantum oracle access to the oracle $B_\varepsilon S(k, \cdot)$ to the adversary A and obtain an output m, σ from $A^{B_\varepsilon S(k, \cdot)}(1^\lambda)$.
3. Output 1 if m, σ is a valid forgery and $m \in B_\varepsilon$, otherwise output 0.

Definition 4.16 (Alagic, Majenz, Russell, and Song [77]). A MAC is blind-unforgeable if for every adversary (A, ε) , the probability of winning the blind forgery experiment is negligible.

Interestingly, even though Alagic, Majenz, Russell, and Song [77] show that there exists a construction that $+1$ -secure but blind-forgeable, it is still open whether blind-unforgeability implies $+1$ -security.

Security Definition for One-time MACs To address the above limitations, [78] gave new security notions for authentication in the quantum setting, strictly restricted to the one-time setting. More generally, their definition is inspired by the simulation paradigm in classical cryptography. In this framework, one first defines a class $\mathfrak{A}$ of *ideal* adversaries. Intuitively, ideal adversaries are those that cannot be avoided in a real execution of an ideal authentication protocol, such as those that discard messages, or ones that carry out actions explicitly allowed by the protocol. An authentication protocol is secure with respect to the class $\mathfrak{A}$ if the behavior of any adversary (not necessarily ideal) in the protocol can be approximately simulated by an ideal adversary in $\mathfrak{A}$.

Following this idea, the most important part is to define the capability of quantum ideal adversaries. For classical authentication protocols, what a quantum adversary can do trivially (without being detected) is: perform some (partial) measurement of the message in the computational basis, and controlled on the outcome of the partial measurement, perform some quantum operation on his own private qubits. Indeed, these are trivially possible on *any* classical protocol. The new security definition is defined according to this class of ideal adversaries as follows.

Definition 4.17 (Garg, Yuen, and Zhandry [78]). Let A be a QPT adversary, and consider the following experiment:

- The challenger samples a random key k .
- A makes a single quantum query to the function $m \mapsto S(k, m)$.
- A outputs a superposition $|\phi\rangle = \sum_{m, \sigma, \text{aux}} \alpha_{m, \sigma, \text{aux}} |m, \sigma, \text{aux}\rangle$ of message/tag pairs as well as auxiliary information.
- The challenger runs $S(k, \cdot)$ in superposition on the m, σ registers. The result is XORed into the σ registers. The challenger then measures the σ registers.
 - If the output is 0, the challenger outputs the post-measurement state $|\phi'\rangle$.
 - Otherwise, it outputs $\perp$.

Call this output $\text{GYZ-Expt}(\lambda, A)$.

We call A *m-respecting* if, after the oracle query, A cannot directly modify the m registers, but is allowed to operate on its auxiliary registers, potentially based on the contents of m . Alternatively, A can replace all registers with a special symbol $\perp$. A message authentication code scheme is one-time GYZ-secure if, for any QPT adversaries A , there exists an m -respecting adversary B such that the following two distributions on quantum states are quantum-polynomial-time indistinguishable:

$$\text{GYZ-Expt}(\lambda, A) \text{ and } \text{GYZ-Expt}(\lambda, B).$$

Unfortunately, it is not clear how to extend Definition 4.17 to two or more queries. We leave open the problem of giving a unified definition for quantum-secure authentication in the quantum setting, supporting many chosen message queries.

5

Introduction to Quantum Cryptography

Although quantum computers bring significant new threats to the current cryptography models, they also bring new possibilities for defining cryptosystems and even new cryptographic objects that do not exist in the classical world. In this part, we present several subfields of quantum cryptography. More precisely, we introduce *quantum key distribution* (QKD), probably the most widely known quantum protocol. Through the example of QKD, we present *everlasting security* and *device-independent cryptography*. We also show that quantum cryptography is not only made of QKD by presenting other subfields, namely the very broad *quantum secure multi-party computation* and *unclonable cryptography*.

5.1 Quantum Key Distribution

Quantum key distribution is highly based on conjugate coding. We first remind what is conjugate coding: the following is taken verbatim from Section 3.2.

The idea of conjugate coding (summarized in Table 5.1) is to encode classical information into quantum states, using two conjugate bases. We will denote these bases as the *computational basis* $\{|0\rangle, |1\rangle\}$ and the *Hadamard basis* $\{|+\rangle, |-\rangle\}$.

Basis θ	0	0	1	1
Bit x	0	1	0	1
Encoding $H^\theta x\rangle$	$ 0\rangle$	$ 1\rangle$	$ +\rangle$	$ -\rangle$

Table 5.1: Conjugate coding

As a first use case for conjugate coding, we present a *quantum key distribution* protocol — often called the *BB84 protocol* — that has been introduced by Bennett and Brassard [8].

Consider Alice who wants to share a key with Bob, and Eve who wants to intercept the key without being detected. They have access to an authenticated (but not secure) classical communication channel and to a quantum communication channel.

A first protocol without Eve. For ease of presentation, we first introduce the protocol without Eve, so the purpose of Alice and Bob is just to share a key. In particular, they know that no one is trying to learn the key.

- **Encoding:** Alice first samples n bases $(\theta_i)_i$ (computational or Hadamard) and n bits $(x_i)_i$ at random and encodes them into a quantum state $|\psi\rangle$, where every one-qubit register $|\psi\rangle_i$ of $|\psi\rangle$ is the encoding of x_i with the basis θ_i using conjugate coding.
- **Quantum communication:** Then, Alice sends $|\psi\rangle$ to Bob.
- **Measurement:** At this point, Bob is not able to learn which bits are encoded in the qubits since he does not know which bases have been sampled by Alice. He samples n bases $(\gamma_i)_i$ (computational or Hadamard) at random and measures every register $|\psi\rangle_i$ in the basis γ_i . We note the measurement outcomes $(y_i)_i$.

Note that, for all register $|\psi\rangle_i$, if Bob chooses the same basis as Alice (*i.e.* if $\theta_i = \gamma_i$), then the measurement outputs the correct bit (*i.e.* $y_i = x_i$), otherwise, he gets the correct bit only with probability $1/2$.

- **Classical communication:** In this step of the protocol, Alice and Bob publicly reveal the bases they used. Thus, they now know which bits Bob correctly guessed. They discard all bits for which Alice's basis θ_i is different from Bob's basis γ_i .

Note that, at this point, if no one tampered with the qubits, then Alice and Bob have the same remaining bits. Also note that, on average, half of the initial bits are discarded. Thus, if we assume that is no malicious user is trying to learn the key, we can stop the protocol at this step and Alice and Bob obtain a shared key of length $n/2$ on average. We provide an example of an execution of such a simplified BB84 protocol in Table 5.2.

Alice's bits	0	1	0	0	0	1	0	1
Alice's bases	H	H	H	C	H	H	C	C
Alice sends $ \psi\rangle$	$ +\rangle$	$ -\rangle$	$ +\rangle$	$ 0\rangle$	$ +\rangle$	$ -\rangle$	$ 0\rangle$	$ 1\rangle$
Bob's bases	H	C	C	C	C	C	C	C
Bob's measurements	0	0	0	0	1	1	0	1
Bases reveal	✓	✗	✗	✓	✗	✗	✓	✓
Shared key	0	.	.	0	.	.	0	1

Table 5.2: Example of a run of the BB84 protocol (without Eve) with 8 qubits. Computational and Hadamard bases are respectively denoted as C and H. The ✓ (resp. the ✗) symbol corresponds to the event where Alice and Bob's basis are the same (resp. different).

A second protocol considering Eve. Now, assume that a malicious user Eve has access to both the quantum and the classical channel and that she is trying to learn the key without being detected. The beginning of the protocol is the same, but we add one more step in order to detect any attempts from Eve to learn the key.

- **Tampering detection:** Alice samples a subset $I \subset \{1, \dots, n\}$ of $n/4$ indices at random such that every index in I corresponds to a bit that *has not* been discarded in the previous step. (*i.e.* $i \in I \implies \theta_i = \gamma_i$). Then, she sends the set $S_A = \{(i, x_i)\}_{i \in I}$ to

Bob. Bob sends $S_B = \{(i, y_i)\}_{i \in I}$ to Alice. Alice and Bob finally compare these bits.

Note that, as said above, if no one tampered with $|\psi\rangle$, then the $(x_i)_{i \in I}$ must be the same as the $(y_i)_{i \in I}$. Hence, if for one index $i \in I$, $x_i \neq y_i$, then it means that Eve made an attempt to learn the key, and tampered with $|\psi\rangle$ by doing so.

If this case happens, Alice and Bob abort the protocol. Otherwise, they discard the bits at indices $i \in I$, and they output the remaining bits as the shared key. We provide an example of an execution of this protocol in Table 5.3.

Alice's bits	0	1	0	0	0	1	0	1
Alice's bases	H	H	H	C	H	H	C	C
Alice sends $ \psi\rangle$	$ +\rangle$	$ -\rangle$	$ +\rangle$	$ 0\rangle$	$ +\rangle$	$ -\rangle$	$ 0\rangle$	$ 1\rangle$
Eve's bases	C	H	C	H	C	C	H	C
Eve's measurements	1	1	0	1	1	1	0	1
Post-measurement state	$ 1\rangle$	$ -\rangle$	$ 0\rangle$	$ -\rangle$	$ 1\rangle$	$ 1\rangle$	$ +\rangle$	$ 1\rangle$
Bob's bases	H	C	C	C	C	C	C	C
Bob's measurements	1	1	0	0	0	1	0	1
Bases reveal	✓	✗	✗	✓	✗	✗	✓	✓
Alice's bits	0	.	.	0	.	.	0	1
Bob's bits	1	.	.	0	.	.	0	1
S_A with $I = \{1, 8\}$	0	.	.	.	.	.	.	1
S_B with $I = \{1, 8\}$	1	.	.	.	.	.	.	1
$S_A - S_B$ comparison	✗	.	.	.	.	.	.	✓

Table 5.3: Example of a run of the BB84 protocol (with Eve) with 8 qubits. Computational and Hadamard bases are respectively denoted as C and H. The ✓ (resp. the ✗) symbol corresponds to the event where Alice and Bob's basis are the same (resp. different). In this example, the final comparison yields an error for the first bit. As a result, Alice and Bob abort the protocol

The security of this protocol has been an open problem for years, and various proofs have finally been written, including the ones of Shor and Preskill [79] and of Renner [80]. Hence, this “BB84” protocol is an information-theoretically secure key-agreement protocol while all

key-agreement protocols we know in classical cryptography protocols provably rely on computational assumptions.

5.2 Information Reconciliation and Privacy Amplification

Remark that for a real world usage, we have to consider noise in the quantum communication, and thus use some techniques, namely *information reconciliation* and *privacy amplification* in order to make sure that the remaining bits are indeed the same and that the hypothetical malicious user has no information *at all* on the shared key.

Introducing noise. First, we present information reconciliation. In the description of the QKD protocol above, we assumed that there is no noise during the quantum communication step. Then, if no tampering is detected during the *tampering detection* step, Alice and Bob will have the same key at the end of the protocol with high probability. However, when it comes to real implementations of this protocol, it is highly improbable that there is no noise at all. Then, the noise will perturb the quantum state sent during the *quantum communication* step and Alice and Bob might detect tampering even if there is no one trying to learn the key. Thus, we need to take noise into account. The first thing we need to add in the protocol is a threshold τ used this way: Alice and Bob, during the *tampering detection* step, count the number of bits x_i different from y_i (for all $i \in I$). If this number is greater than the threshold τ , they abort the protocol. Otherwise, they continue. Note that τ is defined according to the expected noise during the quantum communication. Informally, if no one is trying to learn the key, the number of $x_i \neq y_i$ will be lower than τ with high probability, but if a malicious user Eve is trying to learn the key, then the perturbations she induces, in addition to the noise, will increase the number of $x_i \neq y_i$ enough to make it exceed τ .

Information reconciliation. Now that the tampering problem is solved, we need to solve another important problem: because of the noise, the key k_A of Alice at the end of the protocol will probably be different from the key k_B of Bob. More precisely, $k_{A,i}$ will be different from

$k_{B,i}$ for a few number of i , depending on how much noise there is. Then, before the protocol ends, Alice and Bob perform a so-called *information reconciliation* protocol (see Brassard and Salvail [81] for more information). Broadly, in such a protocol, they exchange classical data related to their respective keys and use these data to correct the possible errors between their two keys, respectively getting $\widehat{k}_A$ and $\widehat{k}_B$ such that $\widehat{k}_A = \widehat{k}_B$. One way of doing information reconciliation is to use *syndrome decoding* from the error correcting codes field.

Privacy amplification. Another problem Alice and Bob could have is that Eve could have gained partial information on the resulting key. In order to make the key completely unknown to Eve, Alice and Bob, at the end of the protocol, perform what is called a *privacy amplification* procedure. The idea of such a procedure is, assuming Alice and Bob share the exact same key and Eve has bounded information on this key, to extract a new shared key for Alice and Bob from their initial key. This new key, while being smaller than the initial one, will be completely unknown to Eve. Renner [80] showed how *privacy amplification* can be done with a family of *two-universal hash functions*.

5.3 Everlasting Security

A criticism that can be made to this protocol is that it requires Alice and Bob to share an authenticated channel, otherwise it becomes vulnerable to man-in-the-middle attacks. There are mainly two ways of constructing such a channel, namely using public or private key cryptography.

For public key cryptography, the existing protocols rely on computational assumptions, and so the protocol becomes computationally secure instead of information-theoretically secure. However, we present here the notion of *everlasting security* (sometimes also referred as *long-term security*), which lies between computational security and unconditional security. This notion has been introduced by Müller-Quade and Unruh [82] and Unruh [83]. Informally, we say that a protocol is everlastingly secure if it relies on computational assumptions - that is, on the power of the adversary and on the hardness of some problems - only during the protocol execution, and, when the protocol ends, it remains secure even

if we remove these assumptions. As an example of a protocol that is not everlastingly secure, consider the Diffie-Hellman key exchange protocol Diffie and Hellman [84]. This protocol allows two users to exchange a private key, and is based on the hardness of the discrete logarithm problem. Assuming that this assumption holds, no adversary can obtain information on the exchanged key. However, an adversary could store the transcript of the protocol — all public communication between the users — and wait to get a computer with enough computational power break the discrete logarithm problem, and thus obtain the private key. This exactly means that the Diffie-Hellman protocol is not everlastingly secure: if it was, the key would remain secret after the execution of the protocol (that is after the two users both obtain the key) regardless of the amount computational power any adversary can obtain in the future. This example illustrates how hard it is to construct everlastingly secure cryptographic protocol in the classical world, and we show in the following how quantum communication can help overcoming these difficulties.

The introduction of these notions was motivated by the fact that, while the hardness of some problems used in cryptography is commonly assumed but is still an open problem, it seems fairly reasonable to assume that no one will break any of these problems within a short amount of time. The same goes with the computational power of the adversary: while it is hard to predict the future computational power of an adversary, it is reasonable to make an assumption on this power up to a near future.

With this new notion in mind, we can imagine an everlastingly secure key distribution protocol. For the initial step, Alice and Bob perform a first QKD with a classical authenticated channel constructed from computational assumptions, yielding a shared key. Every time they need to perform a new execution of QKD, they use a piece of the key they got from the previous QKD execution to construct the classical authenticated channel. Since this previous key is only known to both of them, they can construct this channel using private key cryptography. Thus, the protocol only relies on computational assumptions for the first channel in the first step.

Instead of using public key cryptography to construct the authen-

enticated channel, we can use private key cryptography, but it requires Alice and Bob to share an initial key before the protocol starts, which is the very purpose of the QKD. Then in this case, it can be more relevant to see this protocol as a *key expansion* protocol instead of a *key distribution* one.

5.4 Device-Independent Quantum Key Distribution (DIQKD)

Device-independent security aims to decrease the assumptions made on the devices we use. Indeed, when we prove that a quantum protocol has the required properties, we usually assume that the devices used by the different parties in the protocol work as intended.

This is the case, for instance, of the QKD protocol presented above. But if in this protocol, the devices Alice and Bob use to prepare and measure do not work as expected, this could lead to dysfunction in the protocol. Imagine for instance that the measurement device of Bob measures in the wrong basis once every five measurements in average. Then it would increase the number of different bits in the *tampering detection* step, and thus could make Alice and Bob to always abort the protocol, even if Eve is not listening. Or worse, some malfunction in Alice's device could lead her to leak more information than expected to Eve without her noticing.

Following the line of research of Ekert [85] and Bennett, Brassard, and Mermin [86], Mayers and Yao [87] had the idea of *self-testing* devices: quantum devices which would be provided with test protocols such that if a device passes the tests, then we can conclude that the device work as intended. Device independent protocols and especially DIQKD generates a lot of interest from several researchers. Among all the work done in this field, we can cite in particular Barrett, Hardy, and Kent [88]; Pironio, Acín, Brunner, Gisin, Massar, and Scarani [89]; Hänggi, Renner, and Wolf [90]; Vazirani and Vidick [91]; Metger, Dulek, Coladangelo, and Arnon-Friedman [92].

We briefly present a device-independent QKD protocol, using the protocol that we presented above as a base. First, we need to introduce a modification in the protocol above. We replace the three first steps, namely *encoding*, *quantum communication* and *measurement* by the

following steps:

- **States preparation and quantum communication:** Alice prepares n EPR states $|\psi_i\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Then, she sends the second register of every one of these states to Bob and keeps the first one for her.
- **Measurement:** Alice samples n bases $(\theta_i)_i$ (computational or Hadamard) and measures every register $|\psi_i\rangle_A$ in the basis θ_i . We note the measurement outcomes $(x_i)_i$. Bob samples n bases $(\gamma_i)_i$ (computational or Hadamard) and measures every register $|\psi_i\rangle_B$ in the basis γ_i . We note the measurement outcomes $(y_i)_i$.

The next steps, namely *classical communication* and *tampering detection* are the same as above. Note that the first protocol we described above and this one with the two new steps are equivalent.

With this modified protocol, Alice and Bob now have, at the beginning of the protocol, several EPR states. Yet, we know that two players playing the CHSH game we presented in Section 3.3, have significantly more chances to win it if they share an EPR state than if they do not share entanglement. More precisely, the higher probability of success they can get without entanglement is $3/4$ while the best strategy with shared EPR states yields a probability of success of $\cos^2(\frac{\pi}{8}) \approx 0.85$. Even better, Reichardt, Unger, and Vazirani [93] showed that if Alice and Bob win with this $\cos^2(\frac{\pi}{8})$ probability, then they must share a quantum state close to an EPR state. With this game and this result in mind, we can now think of a device-independent QKD protocol. Roughly, Alice and Bob play the CHSH game with a random subset of their EPR states and share the results. If they win with a probability significantly smaller than $\cos^2(\frac{\pi}{8})$, they abort the protocol. Otherwise, they consider the EPR they share as valid, and use the remaining ones for the rest of the protocol.

5.5 Beyond Quantum Key Distribution

Quantum key distribution is the most well known functionality in quantum cryptography, so much that quantum cryptography is sometimes

reduced to QKD only. But as we will see, quantum cryptography is much more than just QKD. In this section, we briefly present multi-party quantum computation (MPQC) and unclonable cryptography. We cover unclonable cryptography in more details in Chapter 6, but we only briefly cover MPQC. We refer to the survey of Broadbent and Schaffner [94] and to the *quantum protocol zoo* (Kashefi and Singh) - a project whose purpose it to provide a wiki of quantum cryptography protocols - for more information.

Multi-party quantum computation. A multi-party computation (MPC) protocol for a function f is a protocol between n parties, where n parties jointly compute the function f on their inputs $x_1, \dots, x_n$. The protocol has correctness if their output is indeed the evaluation of the function $f(x_1, \dots, x_n)$, and security if no party $\mathcal{P}_i$ learns more information than $f(x_1, \dots, x_n)$. This field has been introduced by Yao [96] and by Goldreich, Micali, and Wigderson [97] in the late eighties. The latter provided a first universal protocol for multi-party computation based on computational assumptions.

Crépeau, Gottesman, and Smith [98] extended this functionality to the quantum world by writing the first definition of the so-called *multi-party quantum computation (MPQC)*. Similarly to its classical counterpart, a MPQC protocol allows n parties to jointly compute the result of some unitary operator U applied on their joint inputs $\rho_1 \otimes \dots \otimes \rho_n$. In the same paper, they constructed an information theoretically secure protocol for verifiable quantum secret sharing and extended it to an information theoretically secure MPQC scheme.

Delegated computation. Another similar field is *delegated computation*, where one or several clients send their inputs to a powerful server which computes a function for them. This functionality makes a lot of sense in quantum cryptography, since a world in which anyone has their own quantum computer is unlikely to happen in a near future. What we can imagine, however, is a world where only a few number of people (the servers) possess a quantum computer, and perform quantum computations on the behalf of the rest of the users (the clients). Ideally of course, we would like the computation instructed by the clients to be

verifiable, and to be unlearnable by the server - in which case we say that the protocol is *blind*. Child [99] defined this functionality in the case where the clients are able to perform simple quantum computation only (think of preparing and sending single qubits) and proposed a blind protocol, and Broadbent, Fitzsimons, and Kashefi [100] proposed a blind and verifiable universal protocol. Mahadev [21] considered a setting where the clients are completely classical, and provided a blind and verbatim protocol in this case, based on computational assumptions.

Quantum-enhanced multi-party computation. We can also mention the area of quantum-enhanced secure multi-party computation (QMPC), in which the parties aim to compute a classical function on their classical inputs, but they can perform quantum computations. Interestingly, this new ability yields different results than in classical MPC. One famous example of such separation is the relation between two central two-party primitives: *oblivious transfer* (Wiesner [6] and Rabin [7]) and *bit commitment* (Blum [101]). Oblivious transfer is a primitive in which a first party Alice holds two messages, and a second party Bob asks her one, and only one message. The security of this primitive requires that Alice should not learn which message has been requested by Bob, while the latter should not learn anything about the message he did not request. Bit commitment on the other hand, asks one player Alice to commit to a bit, in such a way that she can later open the commitment and reveal the value of the bit to the second player Bob. Such a protocol must be *hiding*, meaning that Bob cannot learn anything on the bit from Alice's commitment, and *binding*, meaning that, if Alice committed to some bit b , she cannot convince Bob that she committed to the other bit $1 - b$.

Oblivious transfer in particular is arguably one of the most fundamental primitives in MPC, as secure MPC can be built based on this primitive. In classical MPC, it is a folklore result that oblivious transfer implies bit commitment, but not the other way around. In the quantum world however, these two primitives are in fact equivalent: Bennett, Brassard, Crépeau, and Skubiszewska [102] showed how to construct oblivious transfer from bit commitment. More recently, Grilo, Lin, Song, and Vaikuntanathan [103] and Bartusek, Coladangelo, Khurana, and Ma

[104] showed that, unlike what is believed in classical MPC, the same oblivious transfer primitive can be constructed from one-way functions only.

Unclonable cryptography. Quantum mechanics can not only improve classical cryptography, but also provides ways to create functionalities that simply do not exist in classical cryptography. Most of these new functionalities use the no-cloning theorem to build unclonable objects, hence, this field is called *unclonable cryptography*. In the following, we present this field in more details.

6

Unclonable Cryptography

In this section, we present a few examples of cryptographic primitives which belong to unclonable cryptography, and hence cannot be constructed in classical settings.

Several unclonable objects have been investigated, and we can establish the following hierarchy inspired by Broadbent, Jeffery, Lord, Podder, and Sundaram [11]. The first level is the *authenticity* level, where objects do not hold any data. This level includes quantum money and quantum lightning. The second one is the *information* level, where we make objects that hold information unclonable. This includes unclonable encryption. Finally, the third level is the *functionality* level, where we make program-like objects unclonable. This includes single-decryptors and copy-protection schemes.

In the following, we cover the following primitives: quantum money, quantum lightning, unclonable encryption, single-decryptors and copy-protection. These unclonable primitives and their properties share some similarities, in fact, they are all based on the no-cloning theorem and their security games are usually stated in the form of two non-communicating adversaries. Ananth, Kaleoglu, and Liu [105] proposed a unified way to define them. As we think it brings more clarity

for the reader, we do not use this unified representation, but rather define each primitive one by one in the following.

6.1 Toolkit for Constructing Unclonable Cryptography

In this section, we present useful tools when it comes to construct unclonable primitives. Specifically, we first formally present the monogamy-of-entanglement property of BB84 states, already mentioned in Section 3.4, and the certified deletion property of this family of states. Then, we present *coset states*, and their direct product hardness and monogamy-of-entanglement properties. The main feature of coset states as we will see shortly, is their verifiability. On the other hand, BB84 states are simple and can be implemented with the current technology.

As presented in the next sections, unclonable cryptography is about encoding a cryptographic object (like a decryption key, a ciphertext, *etc.*) in a quantum state in an unclonable way, that is in a way that prevent any adversary to copy such a state to obtain two, or more, states that all feature the encoded object's property (in the case of a key for instance, the property would be being able to decrypt ciphertexts). The families of states and theorems presented in this section are useful to perform such encodings. Indeed, they all capture a specific type of unclonable property: none of these theorems would be true if one could copy a quantum state from these families.

6.1.1 Monogamy-Of-Entanglement and Certified Deletion of BB84 States

Monogamy of entanglement describes the property that two maximally entangled parties cannot be entangled with any other party. We capture this property through a game - played by three players, Alice, Bob, and Charlie - that follows a template that we extensively use in the context of unclonable cryptography. A player Alice is given a random BB84 state, and is asked to split it. That is, she performs an arbitrary quantum operation on it, and returns a two-registers quantum state (intuitively, think that Alice tries to produce a copy of the original state, or something as close as possible). She then sends the first register to

Bob, and the second one to Charlie. Bob and Charlie are then given the basis of this BB84 state, and are both asked to return the value of the state. Tomamichel, Fehr, Kaniewski, and Wehner [106] prove that Bob and Charlie cannot both answer correctly. Later, Culf and Vidick [107] prove a stronger result, that is this game is still hard to win if Bob only needs to correctly guess the values of qubits that have been encoded in the rectilinear basis, and the same goes for Charlie for the qubits encoded in the diagonal basis. As this strong version of monogamy-of-entanglement implies the regular one, we simply denote the strong version as monogamy-of-entanglement in the following.

Game 1: Monogamy-of-Entanglement Game of BB84 States

Parameters: a security parameter λ .

Parties: a challenger and three adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$, such that $\mathcal{B}$ and $\mathcal{C}$ cannot communicate.

1. The challenger samples two bitstring $x, \theta \in \{0, 1\}^\lambda$, such that the size of $I = \{i \in \{1, \dots, \lambda\} : \theta_i = 1\}$ is $\lambda/2$, and prepares the corresponding BB84 state $|x^\theta\rangle = H^\theta |x\rangle$.
2. Then, the challenger sends $|x^\theta\rangle$ to $\mathcal{A}$.
3. $\mathcal{A}$ prepares two (possibly entangled) quantum states σ_B and σ_C , then sends σ_B to $\mathcal{B}$ and σ_C to $\mathcal{C}$.
4. The challenger sends θ to both $\mathcal{B}$ and $\mathcal{C}$.
5. $\mathcal{B}$ returns x_1 and $\mathcal{C}$ returns x_2 ; they win if and only if $(x_1)_i = x_i$ for all $i \in I$, and $(x_2)_i = x_i$ for all $i \in \bar{I}$.

We denote the binary random variable indicating whether the adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ win the game by $\text{MoE-BB84}_\lambda(\mathcal{A}, \mathcal{B}, \mathcal{C})$.

Theorem 6.1 (Monogamy-of-Entanglement of BB84 States). No triple of adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ wins the monogamy-of-entanglement game of BB84 states (game 1) with non-negligible probability in λ . Formally, for all triple of adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$,

$$\Pr(\text{MoE-BB84}_\lambda(\mathcal{A}, \mathcal{B}, \mathcal{C}) = 1) = \text{negl}(\lambda)$$

Proof. We now give a sketch of the proof for this monogamy-of-entanglement game, following the formal proof given in Culf and Vidick [107]. We start by reducing this game to the corresponding *extended non-local game*, with only two players, which simplifies the analysis. This technique comes from a work of Johnston, Mittal, Russo, and Watrous [108]. In all of the following part, given a security parameter λ , we denote by $\mathcal{S}_{\lambda/2}^\lambda$ the set of bitstrings $\theta \in \{0, 1\}^\lambda$ such that the size of $\{i : \theta_i = 1\}$ is $\lambda/2$.

Lemma 6.2 (Reduction to Extended Non-Local Game). *Consider the following extended non-local game, between a challenger and two players $\mathcal{B}$ and $\mathcal{C}$.*

- $\mathcal{B}$ and $\mathcal{C}$ prepare a tripartite quantum state ρ_{ABC} ; they send the first register ρ_A to the challenger, $\mathcal{B}$ keeps ρ_B , and $\mathcal{C}$ keeps ρ_C . After this point, the players are not allowed to communicate.
- The challenger samples λ BB84 bases $\theta \in \mathcal{S}_{\lambda/2}^\lambda$ at random, then measures each qubit $\rho_{A,i}$ of ρ_A in the corresponding basis θ_i ¹; let x_i denote the outcome. Finally, the challenger sends θ to both players.
- $\mathcal{B}$ and $\mathcal{C}$ respectively return bitstrings x_1 and x_2
- They win if and only if $(x_1)_i = x_i$ for all i such that $\theta_i = 0$, and $(x_2)_i = x_i$ for all i such that $\theta_i = 1$.

Assume there is a triple of adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ winning the monogamy-of-entanglement game of BB84 (game 1) with probability p , then there is a pair of adversaries $(\mathcal{B}, \mathcal{C})$ who wins the extended non-local game with the same probability p .

We now analyze the extended non-local game. A pair of adversaries for this game can be expressed as a tripartite quantum state ρ_{ABC} as well as two families of projective measurements, $\{B^\theta\}$ and $\{C^\theta\}$, both indexed by bases $\theta \in \mathcal{S}_{\lambda/2}^\lambda$. Define then the projector

¹By measuring $\rho_{A,i}$ in the basis θ_i , we mean measuring in the computational basis if $\theta_i = 0$, or in the Hadamard one otherwise.

$\Pi_\theta = \sum_{x \in \{0,1\}^\lambda} |x\rangle\langle x|^\theta \otimes B_{x|_0}^\theta \otimes C_{x|_1}^\theta$ — where $x|_b = \{x_i \mid \theta_i = b\}$. The winning probability of this strategy is then

$$p_{win} = \frac{1}{|\mathcal{S}_{\lambda/2}^\lambda|} \sum_{\theta \in \mathcal{S}_{\lambda/2}^\lambda} \text{Tr}(\Pi_\theta \rho_{ABC})$$

We then use the following lemma adapted from Tomamichel, Fehr, Kaniewski, and Wehner [106].

Lemma 6.3. *Let $A_1, A_2, \dots, A_n$ be positive semi-definite operators on a Hilbert space. Let $\{\pi_k\}_{k \in \{1, \dots, n\}}$ be a mutually orthogonal family of permutations, that is a family of permutations such that for all x , $\pi_k(x) = \pi_{k'}(x) \implies k = k'$. Then,*

$$\left\| \sum_{i=1}^n A_i \right\|_\infty \leq \sum_{k=1}^n \max_{i \in \{1, \dots, n\}} \left\| \sqrt{A_i} \sqrt{A_{\pi_k(i)}} \right\|_\infty$$

where $\|\cdot\|_\infty$ denotes the Schatten ∞ -norm defined in Definition 2.1. Furthermore, when the A_i 's are projectors, we have

$$\left\| \sum_{i=1}^n A_i \right\|_\infty \leq \sum_{k=1}^n \max_{i \in \{1, \dots, n\}} \|A_i A_{\pi_k(i)}\|_\infty$$

which follows directly from projectors' property $A_i^2 = A_i$.

The following upper-bound follows.

$$\begin{aligned} p_{win} &= \frac{1}{|\mathcal{S}_{\lambda/2}^\lambda|} \sum_{\theta \in \mathcal{S}_{\lambda/2}^\lambda} \text{Tr}(\Pi_\theta \rho_{ABC}) \\ &= \frac{1}{|\mathcal{S}_{\lambda/2}^\lambda|} \text{Tr} \left(\sum_{\theta \in \mathcal{S}_{\lambda/2}^\lambda} \Pi_\theta \rho_{ABC} \right) \\ &\leq \frac{1}{|\mathcal{S}_{\lambda/2}^\lambda|} \left\| \sum_{\theta \in \mathcal{S}_{\lambda/2}^\lambda} \Pi_\theta \right\|_\infty \\ &\leq \frac{1}{|\mathcal{S}_{\lambda/2}^\lambda|} \sum_{1 \leq k \leq |\mathcal{S}_{\lambda/2}^\lambda|} \max_{\theta} \left\| \Pi_\theta \Pi_{\pi_k(\theta)} \right\|_\infty \end{aligned}$$

where the first equality follows from linearity of the trace, the first inequality from definition of the Schatten ∞ -norm, the second one from lemma 6.3 above, and where $\{\pi_k\}_{k \in \{1, \dots, |\mathcal{S}_{\lambda/2}^\lambda|\}}$ is some mutually orthogonal family of permutations to be defined later.

The following lemma allows us to upper-bound the quantity to be maximized in the equation above by a quantity that decreases exponentially in the distance between θ and θ' .

Lemma 6.4. *Let $\theta, \theta' \in \{0, 1\}^\lambda$, and denote by $d(\theta, \theta')$ the number of indices i on which $\theta_i \neq \theta'_i$. Then, using the notations above,*

$$\|\Pi_\theta \Pi_{\theta'}\|_\infty^2 \leq 2^{-d(\theta, \theta')/4}$$

We can then rewrite the equation above as

$$p_{win} \leq \frac{1}{|\mathcal{S}_{\lambda/2}^\lambda|} \sum_{k=1}^{|\mathcal{S}_{\lambda/2}^\lambda|} \max_{\theta} \left(2^{-d(\theta, \pi_k(\theta))/2} \right)$$

Finally, we use the two last lemmas from Culf and Vidick [107] to complete the proof.

Lemma 6.5. *Let n be an even integer. Then there is a family of mutually orthogonal permutations $\{\pi_1, \dots, \pi_{|\mathcal{S}_{\lambda/2}^\lambda|}\}$ of $\mathcal{S}_{\lambda/2}^\lambda$ such that the following holds. For each $i \in \{0, \dots, n/2\}$ there are exactly $\binom{n/2}{i}^2$ permutations π_j such that the number of indices on which θ and $\pi_j(\theta)$ are both 1 is $\frac{n}{2} - i$ (i.e. θ and θ' differ on $2i$ indices).*

Lemma 6.6. *For all even integer $n \geq 2$,*

$$\frac{1}{\binom{n}{n/2}} \sum_{i=0}^{n/2} \binom{n/2}{i}^2 \sqrt{2^{-i}} \leq \sqrt{e} \left(\cos \frac{\pi}{8} \right)^n$$

Plugging the first lemma in the equation above gives

$$p_{win} \leq \frac{1}{|\mathcal{S}_{\lambda/2}^\lambda|} \sum_{i=0}^{\lambda/2} \binom{\lambda/2}{i}^2 2^{-i/2}$$

As $|\mathcal{S}_{\lambda/2}^\lambda| = \binom{\lambda}{\lambda/2}$, we directly obtain

$$p_{win} \leq \sqrt{e} \left(\cos \frac{\pi}{8} \right)^\lambda$$

which is negligible in λ . □

Certified deletion of BB84 states. We now present the certified deletion property of BB84 states. This property was formalized by Behera, Sattath, and Shinar [109], based on a first idea of Broadbent and Islam [110] and a subsequent work of Ben-David and Sattath [111]. Informally, this property captures the idea that measuring an unknown BB84 state in the Hadamard basis information theoretically destroys information on the bits encoded in the computational basis. This property can be used to construct so-called certificates of deletion, as we will see in Section 6.3.2. Consider a scenario where Alice is given a state $|x^\theta\rangle$, and some secret hidden by the bits of x encoded in the computational basis. Alice can generate a certificate of deletion by measuring the whole $|x^\theta\rangle$ in the Hadamard basis, and send it to an authority (typically the one who generated the state and the secret) who can then check its validity. If the certificate is valid, it ensures that Alice loses the information on the bits of x encoded in the computational basis, and thus all information on the secret.

Game 2: Certified Deletion Game of BB84 States

Parameters: a security parameter λ .

Parties: a challenger and an adversary $\mathcal{A}$.

1. The challenger samples two bitstrings $x, \theta \in \{0, 1\}^\lambda$, such that the size of $I := \{i \in \{1, \dots, \lambda\} : \theta_i = 1\}$ is $\lambda/2$, and prepares the corresponding BB84 state $|x^\theta\rangle = H^\theta |x\rangle$.
2. The challenger sends $|x^\theta\rangle$ to $\mathcal{A}$.
3. $\mathcal{A}$ sends a bitstring $x_1 \in \{0, 1\}^\lambda$ back to the challenger.
4. The challenger sends θ to $\mathcal{A}$.
5. $\mathcal{A}$ returns a bitstring $x_2 \in \{0, 1\}^\lambda$ back to the challenger.
6. $\mathcal{A}$ wins if and only if $(x_1)_i = x_i$ for all $i \in I$, and $(x_2)_i = x_i$ for all $i \in \bar{I}$.

We denote the binary random variable indicating whether the adversary $\mathcal{A}$ wins the game by $\text{CD-BB84}_\lambda(\mathcal{A})$.

Theorem 6.7 (Certified Deletion of BB84 States). No adversary $\mathcal{A}$ wins the certified deletion game of BB84 states (game 2) with non-negligible probability. Formally, for all adversary $\mathcal{A}$,

$$\Pr(\text{CD-BB84}_\lambda(\mathcal{A}) = 1) = \text{negl}(\lambda)$$

Bartusek and Khurana [15] proved a stronger form of this certified deletion property, where the adversary tries to guess, in the second part of the game, the parity of the bits x_i encoded in the rectilinear basis instead of trying to guess all the bits.

Game 3: Certified Deletion Game of BB84 States

Parameters: a security parameter λ .

Parties: a challenger and an adversary $\mathcal{A}$.

1. The challenger samples two bitstrings $x, \theta \in \{0, 1\}^\lambda$, such that the size of $I := \{i \in \{1, \dots, \lambda\} : \theta_i = 1\}$ is $\lambda/2$, and prepares the corresponding BB84 state $|x^\theta\rangle = H^\theta |x\rangle$.
2. The challenger sends $|x^\theta\rangle$ to $\mathcal{A}$.
3. $\mathcal{A}$ sends a bitstring $x^* \in \{0, 1\}^\lambda$ back to the challenger.
4. The challenger sends θ to $\mathcal{A}$.
5. $\mathcal{A}$ returns a bit $b^* \in \{0, 1\}$ back to the challenger.
6. $\mathcal{A}$ wins if and only if $(x^*)_i = x_i$ for all $i \in I$, and $b^* = \bigoplus_{i \in \bar{I}} x_i$.

We denote the binary random variable indicating whether the adversary $\mathcal{A}$ wins the game by $\text{CD2-BB84}_\lambda(\mathcal{A})$.

Theorem 6.8 (Certified Deletion of BB84 States). No adversary $\mathcal{A}$ wins the certified deletion game of BB84 states (game 3) with non-negligible probability. Formally, for all adversary $\mathcal{A}$,

$$\Pr(\text{CD2-BB84}_\lambda(\mathcal{A}) = 1) = \text{negl}(\lambda)$$

6.1.2 Secret-Preserving Public Interaction from Coset States

In some circumstances, we want an unclonable object to allow that every party can interact with it in some non-trivial way without learning any

secret information (for instance, to verify that it has been correctly generated). Based on the hidden *subspace states* of Aaronson and Christiano [112], Coladangelo, Liu, Liu, and Zhandry [113] proposed a way to tackle this challenge by introducing *coset states*, which provide a way to perform public interaction while preserving security. Let $A \subset \mathbb{F}_2^n$ be a subspace of dimension $n/2$. The corresponding subspace state $|A\rangle$ is the superposition of all vectors in A , that is (up to a normalization factor) $\sum_{a \in A} |a\rangle$. Consider in addition two vectors s and s' of $\mathbb{F}_2^n$. The corresponding coset state is the one-time pad encryption (a quantum analogue of the classical one-time pad) of $|A\rangle$ using s and s' as keys, which results in the following state:

$$\begin{aligned} |A_{s,s'}\rangle &:= X^s Z^{s'} |A\rangle \\ &= \frac{1}{2^{n/4}} \sum_{a \in A} (-1)^{s' \cdot a} |a + s\rangle \end{aligned}$$

where $s' \cdot a$ denotes the scalar product of s' and a .

A coset state $|A_{s,s'}\rangle$ holds information on both the *primal coset* $A + s$ and the *dual one* $A^\perp + s'$ — where $A^\perp := \{x \in \mathbb{F}_2^n : \forall y \in A, x \cdot y = 0\}$ — in the sense that it is composed of the superposition of all vectors in $A + s$, and applying a Hadamard gate on it yields, up to a global phase, $H|A_{s,s'}\rangle = |A_{s',s}^\perp\rangle$, that is the superposition of all vectors in $A^\perp + s'$. To see why this holds, consider a subspace state $|A\rangle$. Applying a Hadamard gate to this state yields

$$\frac{1}{2^{n/4}} H \sum_{a \in A} |a\rangle = \frac{1}{2^{n/4}} \frac{1}{2^{n/2}} \sum_{a \in A} \sum_{y \in \mathbb{F}_2^n} (-1)^{a \cdot y} |y\rangle$$

Now, for all $y \in A^\perp$, $a \cdot y = 0$, meaning that the amplitude of $|y\rangle$ in the sum above is $2^{-n/4}$. As there are $2^{n/2}$ vectors in $A^\perp$, the sum of the squared amplitudes of these vectors is 1, meaning that the amplitude of all vectors not in $A^\perp$ is 0, hence yielding $|A^\perp\rangle$.

It remains to show that the same holds for coset states, this is immediate based on the observation that, up to a global phase, $HX^s Z^{s'} = X^{s'} Z^s H$.

Coset states have a strong form of unclonability, called *direct product hardness*, which states that it is hard, given a coset state, to output

both a vector in the primal coset and one in the dual coset. Interestingly, this hardness still holds when, in addition to the coset state, the *computationally bounded* adversary is given obfuscated membership programs for the primal coset and the dual coset. Concretely, this direct product hardness property is the following.

Game 4: Direct Product Hardness Game of Coset States

Parameters: a security parameter λ .

Parties: a challenger and a QPT adversary $\mathcal{A}$.

1. The challenger samples a subspace $A \subset \mathbb{F}_2^\lambda$ of dimension $\lambda/2$, and two vectors s and s' in $\mathbb{F}_2^\lambda$. Then, the challenger prepares the coset state $|A_{s,s'}\rangle$ and two obfuscated² membership programs P_{A+s} and $P_{A^\perp+s'}$ where $P_{A+s}(x)$ returns 1 if and only if $x \in A+s$, and $P_{A^\perp+s'}$ is defined similarly for the dual coset $A^\perp+s'$.
2. Finally, the challenger sends the coset state $|A_{s,s'}\rangle$ and the two programs P_{A+s} and $P_{A^\perp+s'}$ to $\mathcal{A}$.
3. $\mathcal{A}$ returns two vectors u and v , and wins the game if and only if $u \in A+s$ and $v \in A^\perp+s'$.

We denote the binary random variable indicating whether the adversary $\mathcal{A}$ wins the game by $\text{DPH-coset}_\lambda(\mathcal{A})$.

Theorem 6.9 (Direct Product Hardness of Coset States). No QPT adversary $\mathcal{A}$ wins the direct product hardness game (game 4) of coset states with non-negligible probability. Formally, for all QPT adversary $\mathcal{A}$,

$$\Pr(\text{DPH-coset}_\lambda(\mathcal{A}) = 1) = \text{negl}(\lambda)$$

We provide a figure describing the game above in figure 6.1.

We present below another property of coset states, namely *monogamy-of-entanglement* of coset states, conjectured by Coladangelo, Liu, Liu, and Zhandry [113], and later proved by Culf and Vidick [107]. Loosely speaking, this property ensures that if an adversary Alice splits a coset

²Precisely, the form of obfuscation used in the theorem above is the *indistinguishable obfuscation*. We refer the curious reader to the paper of Goldwasser and Rothblum [114] for more information on this notion.

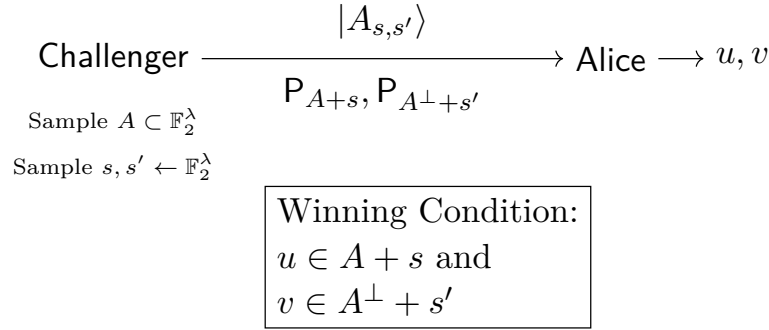


Figure 6.1: Direct Product Hardness of Coset States. No computationally bounded adversary can win this game with non-negligible probability.

state in two states and then gives them to two other adversaries Bob and Charlie, then Bob and Charlie cannot respectively output a vector in the primal coset and one in the dual coset, even given the description of the corresponding subspace. Similarly, as the direct product hardness, this hardness still holds if the adversaries are given membership programs for the primal and dual cosets.

Game 5: Monogamy-of-Entanglement Game of Coset States

Parameters: a security parameter λ .

Parties: a challenger and three QPT adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$, such that $\mathcal{B}$ and $\mathcal{C}$ cannot communicate.

1. The challenger samples a subspace $A \subset \mathbb{F}_2^\lambda$ of dimension $\lambda/2$, and two vectors s and s' in $\mathbb{F}_2^\lambda$. Then, the challenger prepares the coset state $|A_{s,s'}\rangle$ and two obfuscated membership programs P_{A+s} and $\text{P}_{A^\perp+s'}$ where $\text{P}_{A+s}(x)$ returns 1 if and only if $x \in A + s$, and $\text{P}_{A^\perp+s'}$ is defined similarly for the dual coset $A^\perp + s'$.
2. Finally, the challenger sends the coset state $|A_{s,s'}\rangle$ and the two programs P_{A+s} and $\text{P}_{A^\perp+s'}$ to $\mathcal{A}$.
3. $\mathcal{A}$ prepares two (possibly entangled) quantum states σ_B and σ_C , then sends σ_B to $\mathcal{B}$ and σ_C to $\mathcal{C}$.
4. The challenger sends the description of the subspace A to both $\mathcal{B}$ and $\mathcal{C}$.

5. $\mathcal{B}$ returns u and $\mathcal{C}$ returns v ; they win if and only if $u \in A + s$ and $v \in A^\perp + s'$.

We denote the binary random variable indicating whether the triple of adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ wins the game by $\text{MoE-coset}^\lambda(\mathcal{A}, \mathcal{B}, \mathcal{C})$.

Theorem 6.10 (Monogamy-of-Entanglement of Coset States). No triple of adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ wins the monogamy-of-entanglement game (game 5) of coset states with non-negligible probability. Formally, for all triples of adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$,

$$\Pr\left(\text{MoE-coset}^\lambda(\mathcal{A}, \mathcal{B}, \mathcal{C}) = 1\right) = \text{negl}(\lambda)$$

We provide an illustration of the game above in Figure 6.2. A proof of the theorem can be derived from the monogamy-of-entanglement game of BB84 states introduced in Theorem 6.1, by invoking a simple but useful connection between coset states and BB84 states. To see this, let A be a subspace spanned by canonical vectors, that is $A = \text{Span}\{e_i \mid i \in I\}$ for some set $I \subseteq \{1, \dots, \lambda\}$ with its complement $\bar{I}$, and $s, s' \in \mathbb{F}_2^\lambda$. Let $\theta \in \{0, 1\}^\lambda$ be such that $\theta_i = 1$ if and only if $i \in I$. Further, let $x \in \{0, 1\}^\lambda$ be such that $x_i = s_i$ if $i \in I$ and $x_i = s'_i$ if $i \in \bar{I}$. We then have the following:

$$|A_{s,s'}\rangle = |x_1^{\theta_1}\rangle \otimes \cdots \otimes |x_\lambda^{\theta_\lambda}\rangle,$$

where we write $|x_i^{\theta_i}\rangle = H^{\theta_i} |x_i\rangle$. A similar connection can be shown for general subspace A , allowing to reduce coset states to BB84 states, and thus the two monogamy-of-entanglement games to each other.

As mentioned above, coset states allow to construct several functionalities involving public interaction, as *public-key quantum money*, where any party can verify the validity of some quantum states; *single-decryptors*, where any party holding the quantum secret key can decrypt ciphertexts, but cannot extract a secret key from it; *copy-protection*, where any party can evaluate a given quantum program, but cannot extract its classical description; *etc.* As the constructions using coset states in the literature are more involved than the ones simply using BB84 states, we focus on two constructions, namely for a public-key quantum money scheme, and a single-decryptor one, that we present in Sections 6.2.2 and 6.4.3 respectively.

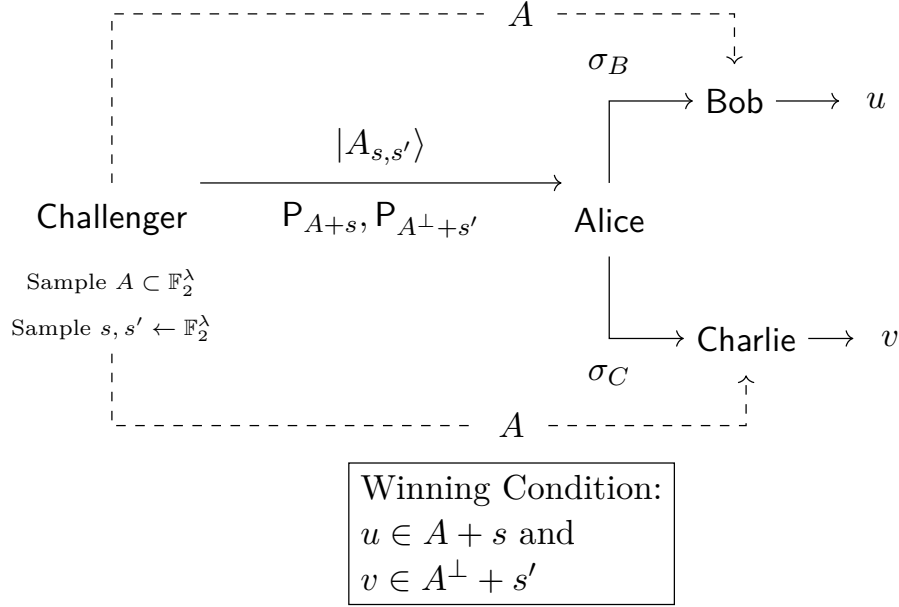


Figure 6.2: Monogamy-of-Entanglement of Coset States. No computationally bounded adversary can win this game with non-negligible probability.

6.2 Quantum Money

Quantum money is a functionality introduced by Wiesner [6] in the same paper in which he introduced conjugate coding. His idea was to take advantage of the no-cloning theorem to build a digital money scheme in which every unit of money (or *quantum coins*) is made of a quantum state and thus cannot be duplicated. In this part, we present the quantum money functionality, and we review some of the important protocols that have been proposed for it.

6.2.1 Private-Key Quantum Money

Quantum money exists in two main fashions: public-key quantum money and private-key quantum money. First, we present private-key quantum money. A private-key quantum money protocol is composed of three parties: a bank, a trusted entity who mints the money, that is, creates new quantum coins; a client, who receives the money from the bank; and a vendor who sells something to the client in exchange for money. The functionality is referred to as private as the bank is the only party

who can verify the validity of a coin. A vendor who wants to test the validity of a given coin must thus send it to the bank and wait for its answer.

Definition 6.1 (Private-Key Quantum Money scheme). A private-key quantum money protocol with a trusted bank and potentially malicious clients is made of a *Mint* algorithm used by the bank and an interactive protocol *Verify* between a vendor and the bank such that:

- *Mint* takes as input a security parameter 1^λ and returns a quantum state ρ as a “quantum coin”.
- *Verify* is a protocol in which a vendor sends an alleged quantum coin to the bank, and the bank either *accepts* or *rejects* it. In the following, we will write $\text{Verify}(\rho) = \top$ (resp. $\text{Verify}(\rho) = \perp$) to denote the fact that the bank accepts (resp. rejects) a coin ρ .

A private-key quantum money protocol must in addition fulfill the following properties:

- (*Correctness*) A coin that has been generated by the bank and that has not been modified must be accepted by the verification protocol, except with negligible probability. More precisely,

$$\Pr\left(\text{Verify}(\rho) = \top : \rho \leftarrow \text{Mint}(1^\lambda)\right) \geq 1 - \text{negl}(\lambda)$$

where λ is a security parameter for this quantum money protocol.

- (*k-unforgeability*) No malicious user $\mathcal{A}$, given k valid coins, must be able to produce $k+1$ valid coins. We abuse the notation and write $\text{Verify}(\rho_1, \dots, \rho_k)$ to denote a procedure that takes k quantum coins as input, and accepts if and only if all the coins are accepted by the *Verify* procedure. Similarly, we write $\text{Mint}(1^\lambda, k)$ to denote a procedure that runs *Mint* k times to produce and returns k coins. The *k-unforgeability* property then asks that for all algorithm $\mathcal{A}$,

$$\Pr\left(\text{Verify}(\widetilde{\rho}_1, \dots, \widetilde{\rho}_{k+1}) = \top : \begin{array}{l} (\widetilde{\rho}_1, \dots, \widetilde{\rho}_{k+1}) \leftarrow \mathcal{A}(\rho_1, \dots, \rho_k) \\ (\rho_1, \dots, \rho_k) \leftarrow \text{Mint}(1^\lambda, k) \end{array}\right) \leq \text{negl}(\lambda)$$

We provide an illustration of this unforgeability property in figure 6.3.

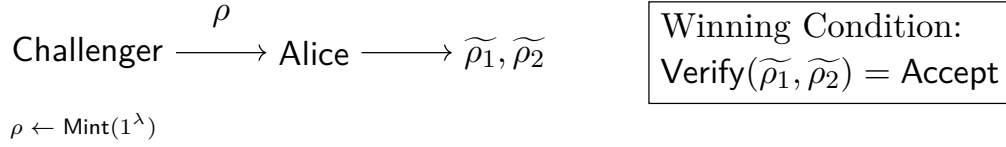


Figure 6.3: 1-unforgeability of a quantum money scheme presented as a game. In order for the scheme to be secure, no adversary $\mathcal{A}$ should be able to win this game with non-negligible probability.

Wiesner’s quantum money. In this part, we present the quantum money protocol introduced by Wiesner [6]. In this protocol, every pair of parties among the bank, the client and the vendor can communicate through a classical authenticated channel and a quantum channel. We start by giving a high-level description of the construction and then provide a formal description (construction 1).

The bank maintains a database, in which every quantum coin they have minted is recorded. In order to mint money, the bank samples three random bitstrings (s, x, θ) and stores this triple in the database. The bank then sends s as the serial number of the coin and the BB84 state $|x^\theta\rangle$ as the coin to the client.

A vendor who wants to verify the validity of a quantum coin sends the coin and its serial number to the bank. The bank then measures every qubit in the basis it has been prepared. If all the measurements output the expected value (the one stored in the database) — which happens with probability 1 if the coin is valid — then the bank accepts. If at least one of the values is not the expected one, the bank rejects.

Construction 1: Wiesner’s Quantum Money

- $\text{Mint}(1^\lambda)$
 1. Sample $s, x, \theta \leftarrow \{0, 1\}^\lambda$ such that $I = \{i : \theta_i = 0\}$ is of size $\lambda/2$;
 2. If s is already contained in the database:
 - rerun the Mint procedure from the start;
 - otherwise store s, x, θ in the database;
 3. Return $(s, |x^\theta\rangle)$.

- $\text{Verify}(s, \rho)$
 1. Search for a record (s, x, θ) for any $x, \theta \in \{0, 1\}^\lambda$ in the database;
 2. If there is no such record, return $\perp$;
 3. Otherwise, for all $i \in \{1, \dots, \lambda\}$:
 - (a) Measure the i -th qubit ρ_i of ρ in the basis θ_i ³; let x'_i denote the outcome;
 - (b) Test whether $x'_i = x_i$.
 4. If all tests pass, return $\top$, otherwise, return $\perp$.

By measuring in the basis θ_i , we mean measuring in the computational basis if $\theta_i = 0$, and in the Hadamard one otherwise.

Theorem 6.11. Construction 1 has 1-unforgeability.

Proof. We reduce the security of this construction to the monogamy-of-entanglement of BB84 states (Theorem 6.1). Assume an adversary $\mathcal{A}$ breaks the 1-unforgeability property of this construction with non-negligible probability. That is, given a pair $(s, |x^\theta\rangle)$, $\mathcal{A}$ returns two pairs (s_B, ρ_B) and (s_C, ρ_C) that both pass the verification test. In the first part of the proof, we show that $s_B = s_C = s$ with probability negligibly close to 1. Observe first that, as (s_B, ρ_B) and (s_C, ρ_C) are valid, it means that both s_B and s_C are contained in the database — otherwise the bank would abort as specified in item 2 of the verification procedure. Also, the only information $\mathcal{A}$ has on the bank's database is that it contains s . Then, producing pairs such that $s_B \neq s$ or $s_C \neq s$ would s_B or s_C not being contained in the database, and the bank aborting, with probability negligibly close to 1. We can therefore assume that $s_B = s_C = s$ with probability negligibly close to 1. We now present a triple of adversaries $(\mathcal{A}', \mathcal{B}, \mathcal{C})$ who win the monogamy-of-entanglement game of BB84 with non-negligible probability.

- $\mathcal{A}'$, given $|x^\theta\rangle$, runs $\mathcal{A}$ on this state to get two pairs (s, ρ_B) and (s, ρ_C) .
- $\mathcal{A}'$ sends ρ_B to $\mathcal{B}$, and ρ_C to $\mathcal{C}$.

- $\mathcal{B}$, when receiving ρ_B from $\mathcal{A}'$, and θ from the challenger, measures ρ_B in the basis θ , then returns the outcome x'_B .
- $\mathcal{C}$ does the same with ρ_C .

With non-negligible probability, as ρ_B and ρ_C are both valid for s , then $x'_B = x'_C = x$, and $\mathcal{B}$ and $\mathcal{C}$ win the game. This leads to a contradiction which completes the proof. $\square$

Remark 6.1. We wrote the proof above using a reduction from the monogamy-of-entanglement property of BB84 as it fits this section better. This proof is folklore and, to the best of our knowledge, does not formally appear in the literature. Note however that a work of Molina, Vidick, and Watrous [115] proves, using semi-definite programming techniques, that $(3/4)^\lambda$ is a tight upper-bound on the counterfeiting probability of an adversary in this scheme.

This quantum money protocol was the first one being presented and, while being information theoretically sound, it had some issues. Firstly, it is a private scheme and thus, requires the vendor to send qubits to the bank. Also, this scheme requires the bank to maintain a huge database in order to store all the bases and bits for every quantum banknote.

Solving the database issue. Bennett, Brassard, Breidbard, and Wiesner [116] show how avoid the use of such a huge database at the cost of losing the information theoretic security, their scheme being only computationally secure. Roughly, they use a pseudo-random function F ⁴, known only by the bank, then sample a serial number s as in the original scheme, but this time choose x and θ according to $F(s)$ — typically, set x and θ as the first and second halves of $F(s)$. The verification is done in the same way as in the original scheme, except that instead of recovering (x, θ) from the database, the bank computes them in the same way as in the minting process, using $F(s)$. We give an example of a coin in this protocol in Table 6.1.

⁴We recall that the security of a pseudo-random function states that its outcome is computationally indistinguishable from the one of a truly random function.

Although it removes the huge database problem, this scheme does not solve the first issue, namely the vendor still needs to ask the bank to verify banknotes. Note that revealing the pseudo-random function F to the clients would allow them to perform this verification on their own but if one of them is malicious, they could use this information to forge money. Solving this first issue requires to construct a public-key quantum money scheme.

s	01011001
$F(s)$	00111011
x	0011
θ	1011
$ \psi\rangle$	$ +0 - -\rangle$

Table 6.1: Example of a coin $|\psi\rangle$ minted using the quantum money protocol described above.

6.2.2 Public-Key Quantum Money

A public-key quantum money scheme features a bank, a client and a vendor as well, but the vendor can verify the validity of the money locally, meaning that they do not have to communicate with the bank. More formally, a public-key quantum money scheme has the same **Mint** and **Verify** algorithms as its private counterpart, and also has the same correctness and unforgeability properties. Intuitively however, a public-key quantum money scheme is harder to construct than a private one, as the adversary in the unforgeability property has now the ability to run locally the verification algorithm, meaning that they possibly have access to more information on the minting procedure than in the private version.

Several attempts had been made in order to construct such a public scheme since the definition of quantum money. Although significant progress has been done these last years, we still do not have a candidate for this scheme whose security is provably based on standard assumptions. As a consequence, we can still consider public-key quantum money an open problem. We make a non-exhaustive list of these candidates

in the following, and discuss where they failed or what they lack to be considered as a completely satisfying scheme.

In the very beginning of the quantum cryptography era, Bennett, Brassard, Breidbard, and Wiesner [116] proposed a candidate scheme whose security relies on the hardness of factoring the product of two primes. However, as it later turned out that factorizing Blum integers is easy for a quantum computer ([5]), the security of this scheme does not hold anymore. Almost thirty years later, several new candidates emerged: Farhi, Gosset, Hassidim, Lutomirski, and Shor [117], Kane [118] and Kane, Sharif, and Silverberg [119] proposed candidates based on knot theory and quaternion algebras respectively, but the problems based on these fields have not been extensively studied by cryptographers, hence their hardness is not considered as a standard assumption. Two works by Aaronson [16], and Aaronson and Christiano [112] defined a secure public-key quantum money scheme based on a quantum oracle for the former, and on a classical one for the latter. However, it was unclear at that time how to build such oracles — in fact, both papers propose a construction for the oracles that was subsequently broken (Lutomirski, Aaronson, Farhi, Gosset, Kelner, Hassidim, and Shor [120], Conde Pena, Duran Diaz, Faugere, Hernandez Encinas, and Perret [121]). Zhandry [12] described a way to construct Aaronson *et al.*'s oracles based on a strong form of obfuscation, which is not known to exist under standard assumptions. In the same paper, Zhandry proposed a scheme from *quantum lightning*⁵, a powerful primitive whose candidate given in the same paper was later broken (Roberts [122]). More recently, Khesin, Lu, and Shor [123] proposed a candidate based on random lattices, which was broken by Liu, Montgomery, and Zhandry [124].

Public-key quantum money from subspace states. As mentioned above, Zhandry [12] showed how to construct the oracles needed in the public-key quantum money construction of Aaronson and Christiano [112]. The idea of Aaronson *et al.*'s scheme is the following: the bank keeps as secret a random subspace description A (see Section 6.1.2), and prepares quantum coins of the form $|A\rangle$. Informally, the verification

⁵We give a description on quantum lightning in Section 6.2.3.

procedure then consists in checking whether the quantum state is a superposition of vectors in A , then applying a Hadamard gate, and checking that the resulting state is a superposition of vectors in $A^\perp$ through the help of membership oracles for A and $A^\perp$. Aaronson *et al.* showed that this verification procedure ensures that the state is $|A\rangle$, and proved that cloning a random subspace state is hard, even when given access to membership oracles. We describe this formally in construction 2.

Construction 2: Public-Key Quantum Money

Ingredients: an obfuscation scheme iO .

- $\text{Mint}(1^\lambda)$
 1. Sample a subspace $A \leftarrow \{0, 1\}^{\lambda \times \lambda/2}$ of dimension $\lambda/2$;
 2. Prepare the membership programs $P_A, P_{A^\perp}$ ⁶;
 3. Obfuscate these programs: $\widehat{P}_A \leftarrow \text{iO}(P_A)$, $\widehat{P}_{A^\perp} \leftarrow \text{iO}(P_{A^\perp})$;
 4. Return $|A\rangle$ as the quantum coin, and $(\widehat{P}_A, \widehat{P}_{A^\perp})$, as the corresponding verification key.
- $\text{Verify}(\rho, (\widehat{P}_A, \widehat{P}_{A^\perp}))$
 1. Run $\widehat{P}_A$ in superposition over ρ , let b_0 denote the outcome, and ρ' the resulting state;
 2. Apply a Hadamard gate on ρ' ; let ρ'' denote the resulting state;
 3. Run $\widehat{P}_{A^\perp}$ in superposition over ρ'' ; let b_1 denote the outcome;
 4. Return $\top$ if $b_0 = b_1 = 1$, otherwise return $\perp$.

On input a vector $u \in \{0, 1\}^\lambda$, P_A returns 1 if $u \in A$ and 0 otherwise, and $P_{A^\perp}$ returns 1 if $u \in A^\perp$ and 0 otherwise.

The construction of the membership oracles, based on non-standard assumptions, by Zhandry yields a secure public-key quantum money scheme, under these non-standard assumptions.

6.2.3 Quantum Lightning

In quantum money, even in the public settings, users have to trust the bank. Zhandry [12] had the idea of a new functionality, called quantum lightning, which can be seen as a quantum money where users do not have to trust the bank. More precisely, this functionality provides a way to prepare quantum states called *quantum bolts* associated with a serial number, and to verify that such a bolt has been prepared honestly. Unlike quantum money, this functionality ensures that no one, *even the one who prepared the bolt*, can produce two valid bolts (even identical) with the same serial number.

Definition 6.2 (Quantum Lightning). A quantum lightning scheme is composed of two algorithms (**Gen**, **Verify**) defined in the following way.

- **Gen** takes as input a security parameter λ and returns a quantum bolt $|\psi\rangle$, together with a serial number s for this bolt.
- **Verify** takes as input a quantum bolt $|\psi\rangle$ and a serial number s and either accepts (returns $\top$) or rejects it (returns $\perp$).

A quantum lightning scheme must in addition fulfil the correctness and security properties defined below.

- (*Correctness*) A quantum lightning scheme has correctness if a pair (*bolt*, *serial number*) generated by **Gen** is accepted with probability close to 1. More precisely,

$$\Pr \left(\text{Verify}(|\psi\rangle, s) = \top : \begin{array}{l} (|\psi\rangle, s) \leftarrow \text{Gen}(1^\lambda) \\ (\text{Gen}, \text{Verify}) \leftarrow \text{Setup}(1^\lambda) \end{array} \right) \geq 1 - \text{negl}(\lambda)$$

- (*Unicity of serial number*) There must be only one valid serial number associated to a quantum bolt. More precisely, for all s' ,

$$\Pr \left(\text{Verify}(|\psi\rangle, s') = \top \wedge s \neq s' : \begin{array}{l} (|\psi\rangle, s) \leftarrow \text{Gen}(1^\lambda) \\ (\text{Gen}, \text{Verify}) \leftarrow \text{Setup}(1^\lambda) \end{array} \right) \leq \text{negl}(\lambda)$$

- (*Security*) A quantum lightning scheme is secure if no adversary is able to produce two valid bolts with the same serial number. More formally,

$$\Pr \left(\text{Verify}(\widetilde{\rho}_1, \widetilde{\rho}_2) = \top : \widetilde{\rho}_1, \widetilde{\rho}_2 \leftarrow \mathcal{A}(1^\lambda) \right) \leq \text{negl}(\lambda)$$

6.2.4 Quantum Lightning Construction

Quantum lightning is a very powerful cryptographic primitive. However, we still do not know how to construct a protocol for it. Zhandry [12] proposed a construction based on an assumption on the hardness of a matrix problem, but this assumption was broken by Roberts [122]. While invalidating Zhandry’s construction, the latter does not prove that quantum lightning is impossible, so we can still hope to find a protocol for this functionality. In fact, Zhandry [125] proposed a new template for constructing quantum lightning, and gives a candidate scheme based on a non-standard assumption on isogenies.

Interestingly, there is also a close connection between the notion of collapsing functions from post-quantum cryptography (that we introduced in Section 4.2.2) and unclonable cryptography, in particular public-key quantum money and quantum lightning. Recall that a collapsing function f says that one should not be able to distinguish a superposition of pre-images $\frac{1}{\sqrt{k}} \sum_i^k |x_i\rangle$ from a measured pre-image $|x_i\rangle$ for some $i \in [k]$, for some image $y = f(x_i)$ for all $i \in [k]$. While collapsing functions give rise to secure post-quantum cryptography like commitment schemes as we have seen before, its precise opposite is necessary for public-key quantum money and quantum lightning: if no verification can distinguish a state in a superposition of many supports from its measured state, a simple forgery comes ahead. This intuition was formalized in a series of win-win results proven by Zhandry [12] and follow-up works (e.g., Dall’Agnol and Spooner [126]).

Theorem 6.12. If there exists a computationally binding (non-interactive) commitment scheme, then either the scheme is collapse-binding, or it can be transformed into a secure public-key quantum money (quantum lightning) scheme.

6.2.5 Applications of Quantum Lightning

Quantum lightning is a powerful functionality that can be used to build numerous cryptographic primitives such as public-key quantum money, verifiable randomness and decentralized currency.

Public-key quantum money. Roughly, for constructing a public-key quantum money scheme with quantum lightning and a public signature scheme, the bank generates new banknotes by generating a quantum bolt and by signing the serial number. The banknote is then the triple (*quantum bolt, serial number, signature*) and every user can verify it first by verifying that the signature is valid and then that the serial number is valid for the bolt.

Verifiable randomness. For producing verifiable randomness, one can generate a bolt and its serial number and sends the serial number as the random string together with the bolt as the proof. Any user who wants to verify that the string has been genuinely generated (using a random process) just has to verify that the serial number corresponds to the bolt.

Decentralized currency. Finally, Zhandry proposed an idea for a decentralized currency using quantum lightning. The idea is that a pair (*Gen, Verify*) is publicly known by all users and, in order to mint money, a user must generate a bolt whose serial number starts with a certain number of zeroes (fixed in advance). The only way for a user to generate such a bolt is to repeatedly generate bolts until one has a serial number starting with this number of zeroes. Then, the coin is the pair (*bolt, serial number*) and any other user can verify that it was genuinely minted by first verifying that the serial number starts with enough zeroes and then running *Verify* on the bolt and the serial number. This minting method is reminiscent of the *proof of work* in Bitcoin, which ensures that coins are hard enough to produce.

Such a decentralized currency protocol, while being interesting in theory, has several issues that prevents it to be used as it is. However, Coladangelo and Sattath [127] expanded this idea by proposing a hybrid

protocol for a blockchain based decentralized currency. Roughly, they use a classical blockchain and combine it with quantum lightning: users can perform transactions classically (as they would do with the classical blockchain only) but they also can exchange an amount of money for a quantum banknote “holding” this amount of money.

This banknote is actually composed of a quantum bolt and its serial number, and the user that creates the banknote publishes (classically) in the blockchain the serial number and the amount of coin she wants the banknote to hold. She is now able to perform transactions by sending this banknote to any other user, who can verify the banknote’s validity by looking for the serial number of the banknote on the blockchain (which gives him the amount of money associated to the banknote) and by using the quantum lightning verification procedure to ensure that the bolt is indeed associated with this serial number. The major advantage with this protocol is that this new way of doing transactions allows instantaneous verification, while with standard blockchain’s transactions, users have to wait some time before they accept a transaction.

6.3 Unclonable Encryption

In this part, we present a recent unclonable functionality, namely unclonable encryption.

An unclonable encryption scheme can be seen as a classical encryption scheme with quantum ciphertexts. These ciphertexts must be unclonable in the sense that, roughly, no adversary can, given a ciphertext of a message m , split this ciphertext into two ciphertexts that can be decrypted to m . This functionality has been defined by Broadbent and Lord [13] symmetrically. Inspired by Wiesner’s conjugate coding (see 3.2), the authors also proposed a one-time secure construction, that is, the security holds if the encryption key is used only once and then discarded. Ananth and Kaleoglu [14] extended this work to asymmetric encryption and proposed two constructions, one for private key and one for public key encryption. Note that these two constructions are secure even if the key is used more than one time. In the following, we provide a formal definition of unclonable encryption, and present the construction of Broadbent and Lord mentioned above. We also give an

informal presentation of the recent notion of encryption schemes with certified deletion.

6.3.1 Unclonable Encryption

Although we only define unclonable encryption in the symmetric fashion for ease of reading, this definition can easily be modified for public key encryption. We refer the interested reader to the work of Ananth and Kaleoglu [14] for more information.

Definition 6.3 (Unclonable Encryption). An unclonable encryption scheme is composed of three algorithms (**KeyGen**, **Enc**, **Dec**) defined in the following way:

- **KeyGen** takes as input a security parameter λ and returns a classical key k .
- **Enc** takes as input a key and a classical message and returns a quantum state as the ciphertext of this message.
- **Dec** takes as input a key and a quantum ciphertext and returns a classical message.

We define the correctness, indistinguishability and unclonability properties of an unclonable encryption scheme.

- (*Correctness*) An unclonable encryption scheme has correctness if the decryption algorithm **Dec**, run with a key k and a message m encrypted with k returns m with high probability. More precisely, for all message m ,

$$\Pr \left(\text{Dec}(k, c) = m : \begin{array}{l} c \leftarrow \text{Enc}(k, m) \\ k \leftarrow \text{KeyGen}(1^\lambda) \end{array} \right) = 1 - \text{negl}(\lambda)$$

where λ is the security parameter of the scheme.

- (*Indistinguishability*) An unclonable encryption scheme has indistinguishable security if no QPT adversary can distinguish between

the ciphertexts of two messages of the same length. More formally, for all m_0, m_1 such that $|m_0| = |m_1|$, and all QPT adversary $\mathcal{A}$,

$$|\Pr(\mathcal{A}(\text{Enc}(k, m_0)) = 0) - \Pr(\mathcal{A}(\text{Enc}(k, m_1)) = 0)| \leq \text{negl}(\lambda)$$

where both probability are taken over $k \leftarrow \text{KeyGen}(1^\lambda)$.

- (*Unclonability*) An unclonable encryption scheme is unclonable if, when given the encryption of a random message, no adversary Alice can split it and share it with two adversaries Bob and Charlie, in such a way that, when given the secret key, both Bob and Charlie are able to learn the message. Formally, for all triple of QPT adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$,

$$\Pr(\text{UE-Pir}_\lambda(\mathcal{A}, \mathcal{B}, \mathcal{C}) = 1) \leq \text{negl}(\lambda)$$

where $\text{UE-Pir}_\lambda(\mathcal{A}, \mathcal{B}, \mathcal{C})$ is defined in game 6 below.

Game 6: Unclonability Piracy Game of an Unclonable Encryption Scheme

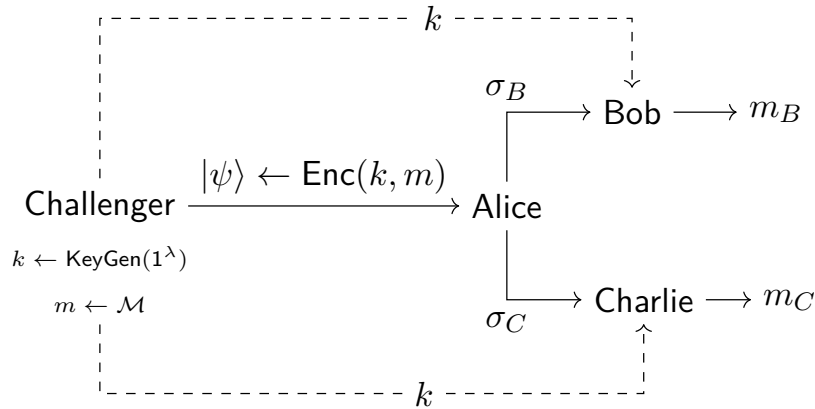
Parameters: a security parameter λ .

Parties: a challenger and a triple of QPT adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ such that $\mathcal{B}$ and $\mathcal{C}$ cannot communicate.

1. The challenger samples a key $k \leftarrow \text{KeyGen}(1^\lambda)$ and a message m , then computes $|\psi\rangle \leftarrow \text{Enc}(k, m)$ and sends it to $\mathcal{A}$.
2. $\mathcal{A}$ prepares two (possibly entangled) quantum states σ_B and σ_C and sends σ_B to $\mathcal{B}$ and σ_C to $\mathcal{C}$. From now on, $\mathcal{B}$ and $\mathcal{C}$ cannot communicate.
3. The challenger sends k to $\mathcal{B}$ and $\mathcal{C}$.
4. $\mathcal{B}$ returns m_B and $\mathcal{C}$ returns m_C . They win if and only if $m_B = m_C = m$.

We denote the binary random variable indicating whether the adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ win the game by $\text{UE-Pir}_\lambda(\mathcal{A}, \mathcal{B}, \mathcal{C})$.

We provide an illustration of this game in figure 6.4.



Winning Condition:

$$m_B = m_C = m$$

Figure 6.4: Unclonability game for unclonable encryption. An unclonable encryption scheme is unclonable if no adversaries can win this game with non-negligible probability.

Remark 6.2. Although we do not define it here, note that Broadbent and Lord [13] also define a stronger security property for unclonable encryption schemes, namely *unclonable indistinguishability*. Roughly speaking, this property mixes the unclonable and indistinguishable properties: when given the encryption of one of two chosen messages, no adversary Alice can split it and share it with two adversaries Bob and Charlie, in such a way that, when given the secret key, both Bob and Charlie are able to guess which message has been encrypted.

Construction. We now present the protocol of Broadbent and Lord [13] for λ -long messages. Note that this protocol has correctness, indistinguishability, and unclonability (following from the monogamy-of-entanglement of BB84 states (Theorem 6.1)); but it does not achieve unclonable indistinguishability. Actually, constructing an unclonable encryption scheme with unclonable indistinguishability remains an open problem at the time of writing.

Construction 3: Unclonable Encryption

- $\text{KeyGen}(1^\lambda)$
 1. Sample $r, \theta \leftarrow \{0, 1\}^\lambda$ such that $I = \{i : \theta_i = 0\}$ is of size $\lambda/2$;
 2. Return (r, θ) as the secret key.
- $\text{Enc}((r, \theta), m)$
 1. Prepare $|\psi\rangle = H^\theta |m \oplus r\rangle$ ^a;
 2. Return $|\psi\rangle$ as the quantum ciphertext.
- $\text{Dec}((r, \theta), \rho)$
 1. Apply H^θ on ρ ; let ρ' denotes the resulting state;
 2. Measure ρ' in the computational basis to get an outcome c ;
 3. Return $c \oplus r$ as the decrypted plaintext.

^athe conjugate coding of $m \oplus r$ with the basis θ

Theorem 6.13. Construction 3 has unclonability.

Proof. We reduce the security of this construction to the monogamy-of-entanglement of BB84 states (Theorem 6.1). Assume a triple of adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ breaks the unclonability property of this construction with non-negligible probability. That is, given a quantum ciphertext $|(m \oplus r)^\theta\rangle$, $\mathcal{A}$ returns two states σ_B and σ_C , and $\mathcal{B}$ and $\mathcal{C}$, respectively given σ_B and σ_C as well as the key (r, θ) , both return the message m with non-negligible probability. We present a triple of adversaries $(\mathcal{A}', \mathcal{B}', \mathcal{C}')$ who win the monogamy-of-entanglement game of BB84 with non-negligible probability.

- $\mathcal{A}'$, given $|x^\theta\rangle$, runs $\mathcal{A}$ on this state to get σ_B and σ_C .
- $\mathcal{A}'$ samples $r \leftarrow \{0, 1\}^\lambda$.
- $\mathcal{A}'$ sends (r, σ_B) to $\mathcal{B}'$, and (r, σ_C) to $\mathcal{C}'$.

- $\mathcal{B}'$, when receiving (r, ρ_B) from $\mathcal{A}'$, and θ from the challenger, runs $\mathcal{B}$ on $(\rho_B, (r, \theta))$; let m'_B denotes the outcome.
- $\mathcal{C}'$ does the same with ρ_C and $\mathcal{C}$; let m'_C denotes the outcome.
- $\mathcal{B}'$ returns $m'_B \oplus r$, and $\mathcal{C}'$ returns $m'_C \oplus r$.

Observe that, after the random r is sampled by $\mathcal{A}'$, there is a unique m such that $x = m \oplus r$. Furthermore, the state $|x^\theta\rangle$ follows the same distribution as the quantum ciphertext $|(m \oplus r)^\theta\rangle$ from the unclonability piracy game. Then, with non-negligible probability, as $m'_B = m'_C = m$, $\mathcal{B}'$ and $\mathcal{C}'$ win the game. This leads to a contradiction which completes the proof. $\square$

History of unclonable encryption. As mentioned above, based on a work of Gottesman [128], Broadbent and Lord [13] defined this primitive in the private settings and gave a construction of a one-time private unclonable encryption scheme based on BB84 states. Ananth and Kaleoglu [14] later extended the definition to the public settings, presenting in the same paper a generic transformation to construct both private and public unclonable encryption schemes from a one-time one. However, none of the constructions above achieve the stronger *unclonable-indistinguishability* security property (Broadbent and Lord [13]). Unclonable-indistinguishability is considerably more difficult to achieve than the simple unclonability as, while constructions with this property has been proposed in the QROM (see Section 4.2.1) by Ananth, Kaleoglu, and Liu [105], [129], and under unproven conjectures by Ananth and Behera [130] and Chevalier, Hermouet, and Vu [131], achieving a construction in the plain model (that is, without oracle) remains an open problem at the time of writing.

6.3.2 Certified Deletion

As we just saw above, using quantum states as ciphertexts in an encryption allows for some stronger properties than what a classical encryption scheme gives, namely unclonability. Recent works also focused on a different, yet also inherently quantum, property that an encryption

scheme with quantum ciphertexts can have: the notion of certified deletion of ciphertexts. We can illustrate how useful this property is with the following simple example. Consider Alice who wants to store some private information on a remote - *untrusted* - server, so that she can retrieve them later. As Alice does not trust the server, she encrypts the information before sending it. However, when she finally retrieves this information and/or asks the server to delete them, nothing proves her that the server really deleted them. A malicious server could have, indeed, kept the information, waiting for the moment where they will be powerful enough to break the encryption scheme (assuming its security is based on computational assumptions), or even for a possible leak of the key. If the encryption scheme used by Alice, on the other hand, has certified deletion, she can ask the server for a *certificate of deletion* of the ciphertext, ensuring that the server really destroyed the ciphertext.

In the rest of this subsection, we define formally what an encryption scheme with certified deletion is, present the protocol of Bartusek and Khurana [15] as an example, and finally discuss the recent advanced in the area.

Definition 6.4 (Encryption Scheme with Certified Deletion). For ease of reading, we only define the scheme for message space of one bit. A (public key) encryption scheme with certified deletion is composed of five algorithms (KeyGen, Enc, Dec, Del, Verify) defined in the following way.

- KeyGen takes as input a security parameter and returns a secret key sk , and a public key pk .
- Enc takes as input a public key and a classical message and returns a quantum state as the ciphertext of this message, and a verification key.
- Dec takes as input a secret key and a quantum ciphertext and returns a classical message.
- Del takes as input a quantum ciphertext and returns a classical certificate of deletion.

- **Verify** takes as input a verification key and a certificate of deletion, and returns either $\top$ (for acceptance) or $\perp$ (for rejection).

Correctness and indistinguishability properties of an encryption scheme with certified deletion are defined as for an unclonable encryption scheme (see Definition 6.3), and we define below the correctness and security of deletion properties.

- (*Correctness of deletion*) An encryption scheme with certified deletion has correctness of deletion, if the verification algorithm **Verify** accepts any honestly generated certificate and its corresponding verification key. More formally, for all messages m ,

$$\Pr \left(\text{Verify}(vk, cert) = \top : \begin{array}{l} cert \leftarrow \text{Del}(\rho_m) \\ \rho_m, vk \leftarrow \text{Enc}(pk, m) \\ sk, pk \leftarrow \text{KeyGen}(1^\lambda) \end{array} \right) = 1 - \text{negl}(\lambda)$$

- (*Security of deletion*) An encryption scheme with certified deletion has security of deletion if, after an adversary provides a valid certificate of deletion, this adversary, even given the decryption key, cannot learn any information on the encrypted message. Formally, for all QPT adversary $\mathcal{A}$,

$$\Pr(\text{CD-Pir}_\lambda(\mathcal{A}) = 1) \leq \frac{1}{2} + \text{negl}(\lambda)$$

where $\text{CD-Pir}_\lambda(\mathcal{A})$ is defined in game 7 below.

Game 7: Piracy Game of an Encryption Scheme With Certified Deletion

Parameters: a security parameter λ .

Parties: a challenger and a QPT adversary $\mathcal{A}$.

1. The challenger samples a key $sk, pk \leftarrow \text{KeyGen}(1^\lambda)$ and a message $m \leftarrow \{0, 1\}$. Then they compute $vk, \rho_m \leftarrow \text{Enc}(pk, m)$, and finally sends pk, ρ_m to the adversary.
2. The adversary sends a certificate of deletion $cert$ to the challenger.

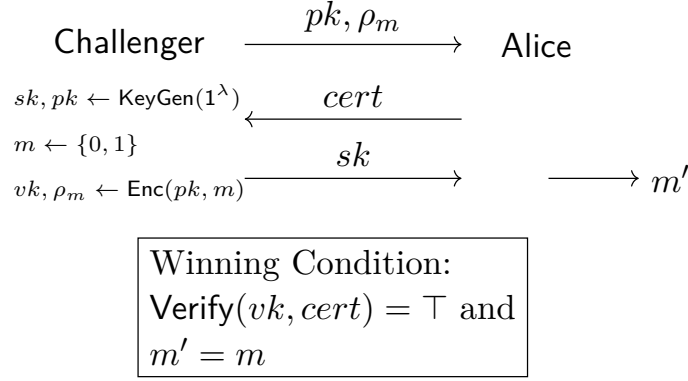


Figure 6.5: Security of deletion game. An encryption scheme with certified deletion has security of deletion if no adversary can win this game with non-negligible advantage over $1/2$.

3. The challenger sends sk to the adversary.
4. The adversary returns m' and wins if $\text{Verify}(vk, cert) = \top$ and $m' = m$.

We denote the binary random variable indicating whether the adversary $\mathcal{A}$ wins the game by $\text{CD-Pir}_\lambda(\mathcal{A})$.

We provide an illustration of this security property in figure 6.5.

Remark 6.3. Note that, for ease of presentation, the definition that we presented above slightly differs from the one used in the latest papers on the topic. We refer the interested reader to the recent work of Bartusek, Khurana, Malavolta, Poremba, and Walter [132].

Remark 6.4. Note also, that we can actually consider that, in the guessing step of the protocol — where the adversary is given the key and asked to guess b — we sometimes consider that the adversary becomes unbounded. That is, they can perform actions such as exhaustive search over an exponentially large space. When formally defining such a case, we simply consider two adversaries: the first one is computationally bounded and must return the certificate and send its “memory state” to the second one, which is unbounded and performs the guessing step.

Construction of encryption with certified deletion. We give a brief presentation of the scheme of Bartusek and Khurana [15]. This scheme uses a classical public key encryption scheme $(\text{KeyGen}, \text{Enc}, \text{Dec})$ as subroutines, and is for one bit-long message spaces. The security of this scheme follows from the stronger certified deletion property of BB84 states (Theorem 6.8).

Construction 4: Encryption With Certified Deletion

Ingredients: a classical public key encryption scheme $(\text{PKE.KeyGen}, \text{PKE.Enc}, \text{PKE.Dec})$ with semantic security.

- $\text{KeyGen}(1^\lambda)$
 1. Sample $(sk, pk) \leftarrow \text{PKE.KeyGen}(1^\lambda)$;
 2. Return (sk, pk) as the secret and public keys for this scheme.
- $\text{Enc}(pk, m \in \{0, 1\})$
 1. Sample $x, \theta \leftarrow \{0, 1\}^\lambda$ such that $I = \{i : \theta_i = 0\}$ is of size $\lambda/2$;
 2. Compute $b = m \oplus \bigoplus_{i \in I} x_i$;
 3. Compute $c = \text{PKE.Enc}(pk, \theta \| b)$ ⁷;
 4. Return $(|x^\theta\rangle, c)$ as the quantum ciphertext, and (x, θ) as the verification key.
- $\text{Dec}(sk, (\rho, c))$
 1. Compute $(\theta, b) \leftarrow \text{PKE.Dec}(sk, c)$;
 2. Measure ρ in basis θ ⁸; let x denotes the outcome;
 3. Return $b \oplus \bigoplus_{i \in I} x_i$ as the plaintext.
- $\text{Del}((\rho, c))$
 1. Measure ρ in the Hadamard basis; let x' denotes the outcome.
 2. Return x' as the certificate of deletion.

- $\text{Verify}((x, \theta), x')$

1. Return $\top$ if $x_i = x'_i$ for all $i \notin I$, otherwise return $\perp$.

The operator $\parallel$ is the concatenation operator.

By measuring in the basis θ , we mean measuring in the computational basis if $\theta = 0$, and in the Hadamard one otherwise.

Theorem 6.14. Construction 4 has security of deletion.

Proof. The goal of this proof is to reduce this new game to the certified deletion property of BB84 states (Theorem 6.8). First note that the only way sk gives information on x , is by allowing the adversary to decrypt c , and learn θ . We can then simply send θ instead of sk to the adversary in step 3. Then, to proceed with the reduction, we want to show that c does not help $\mathcal{A}$ to win the game, as the encryption is semantically secure. We do it by proving that $\mathcal{A}$ has the same winning probability in this piracy game, and in an “hybrid” game where $\text{PKE.Enc}(pk, \theta)$ is replaced by $\text{PKE.Enc}(pk, 0)$, which directly proves that c does not help $\mathcal{A}$ to win the game. This is done by the following reduction.

Assume that $\mathcal{A}$ wins the original game with probability p , and the hybrid one with probability $p + \varepsilon$, where ε is non-negligible in λ . Assume without loss of generality that ε is positive. We describe another adversary $\mathcal{A}'$ who breaks the semantic security of PKE, that is $\mathcal{A}'$ distinguishes between the encryption of two different messages with non-negligible advantage. $\mathcal{A}'$ starts by sampling a basis θ as in the game, then submits $(m_0, m_1) = (\theta, 0)$ as the two plaintexts to the challenger of the semantic security game. $\mathcal{A}'$ receives $c = \text{PKE.Enc}(pk, m_b)$ — where b is the random bit she needs to guess — and uses it to simulate the challenger in the piracy game of certified deletion above. That is, she plays the game with $\mathcal{A}$, and then checks whether $\mathcal{A}$ wins the game or not. If yes, she makes the guess 1 for b , otherwise she makes the guess 0. The probability that her guess is correct is then

$$\begin{aligned}
 & \frac{1}{2} \Pr(\mathcal{A} \text{ wins} \mid b = 0) + \frac{1}{2} \Pr(\mathcal{A} \text{ loses} \mid b = 1) \\
 &= \frac{1}{2} - \frac{1}{2} [\Pr(\mathcal{A} \text{ wins} \mid b = 0) - \Pr(\mathcal{A} \text{ wins} \mid b = 1)] \\
 &= \frac{1}{2} - \frac{1}{2} \varepsilon
 \end{aligned}$$

which is non-negligibly greater than $1/2$, hence breaking the semantic security of PKE, which completes the proof. $\square$

History of certified deletion. In this paragraph, we give a brief history of the recent advances in the certified deletion area. Certified deletion was introduced by Broadbent and Islam [110], in a paper in which they provided the first security definition and a one-time secure construction. Bartusek and Khurana [15] and Hiroka, Morimae, Nishimaki, and Yamakawa [133], [134] extended the idea of certified deletion to other cryptographic primitives, from commitments to more advanced encryption schemes as attribute-based or functional encryption. However, this last work considered private verification only. The following works focused on constructing certified deletion schemes for the same primitives, this time with public verification. Bartusek, Khurana, Malavolta, Poremba, and Walter [132], Hiroka, Morimae, Nishimaki, and Yamakawa [133], and Bartusek, Garg, Goyal, Khurana, Malavolta, Raizes, and Roberts [135], [136] showed how to construct such schemes with less and less computational assumptions, ending up with schemes that only require one-way functions.

6.4 Copy-Protection

Another interesting task that can be improved using quantum mechanics is the task of copy-protection. For this task, a software vendor wants to sell a copy of their software to a client, but they do not want the client to be able to copy it. Classically, although one may not easily be able to copy a software installed on their computer, it is in theory impossible for the vendor to completely prevent such copying. Indeed, as long as the program is completely stored on the client's computer, they have full information on it, hence they can copy it. Thus, the copy-protection task is classically impossible. However, we can think of a way to encode the program into a quantum state, such that being able to copy the program implies breaking the no-cloning theorem. As promising as it is, it turns out that constructing such encoding is not an easy task. Indeed, since the introduction of copy-protection by Aaronson [16], copy-protection constructions have been found for only a few families

of functions (in the rest of the section, we refer to program or software as a classical function).

In the following, we write a history of this functionality by presenting several interesting results in this field. We also present the *secure software leasing* (SSL) functionality, which is very close to copy-protection. The idea of this functionality, defined by Ananth and La Placa [137], is that the vendor does not *sell* the software to the client, but instead they *lease* it and expect the client to send them back after some time. secure software leasing is actually weaker than copy-protection, and we will see in the following that some copy-protection protocols can be slightly modified to construct secure software leasing protocols.

6.4.1 Definitions

Before presenting constructions and history of copy-protection and secure software leasing, we define these functionalities.

Definition 6.5 (Copy-Protection). (Quantum) copy-protection is the task of encoding a function into a quantum state such that anyone owning this state can compute the function on any input they want and cannot produce two copies of the function. More formally, we describe a copy-protection scheme for a certain family of functions $\mathcal{F} = \{\mathcal{F}_\lambda\}_{\lambda \in \mathbb{N}}$, where for all λ , all functions in λ have the same domain and codomain, as composed of two algorithms, **Protect** and **Eval**, such that:

- **Protect** takes as input a security parameter 1^λ , the classical description of a function $f \in \mathcal{F}_\lambda$ and outputs a quantum state ρ .
- **Eval** takes as input a security parameter 1^λ , a quantum state ρ and an input x from the domain of $\mathcal{F}_\lambda$ and outputs y , a classical value in the codomain of $\mathcal{F}_\lambda$.

In addition, a copy-protection scheme must have *correctness* and *anti-piracy* properties.

- (*Correctness*) The correctness property ensures that anyone with a quantum copy-protected function can run **Eval** on it with any

input and get the correct image with probability close to 1:

$$\forall f \in \mathcal{F}_\lambda, \forall x, \Pr[\text{Eval}(1^\lambda, \text{Protect}(1^\lambda, f), x) = f(x)] \geq 1 - \text{negl}(\lambda)$$

- (*Anti-piracy*) The anti-piracy property ensures that, when given a quantum encoding of a function, no adversary Alice can split it and share it with two adversaries Bob and Charlie, in such a way that, both Bob and Charlie are both able to compute the function. More precisely, no triple of malicious colluding adversaries $\mathcal{A}, \mathcal{B}$ and $\mathcal{C}$ can win the following game with non-negligible advantage over some *trivial winning probability* related to the distribution $\mathcal{D}_x(\lambda)$ define below. We give more information on this trivial probability in Remark 6.5.

Formally, a copy-protection scheme for a family $\mathcal{F}$ has anti-piracy security if, for all triple of QPT adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$,

$$\Pr(\text{CP-Pir}_\lambda^\mathcal{F}(\mathcal{A}, \mathcal{B}, \mathcal{C}) = 1) \leq p_{\text{triv}}^{\text{CP}}(\lambda, \mathcal{D}_x) + \text{negl}(\lambda)$$

where $\text{CP-Pir}_\lambda^\mathcal{F}(\mathcal{A}, \mathcal{B}, \mathcal{C})$ is defined in game 8 below, and $p_{\text{triv}}^{\text{CP}}(\lambda, \mathcal{D}_x)$ is defined in Remark 6.5.

Game 8: Piracy Game of a Copy-Protection Scheme for a family $\mathcal{F}$

Parameters: a security parameter λ ; an efficiently samplable distribution $\mathcal{D}_f(\lambda)$ on the family of functions $\mathcal{F}_\lambda$; and an efficiently samplable distribution $\mathcal{D}_x(\lambda)$ over pairs of inputs for $\mathcal{F}_\lambda$.

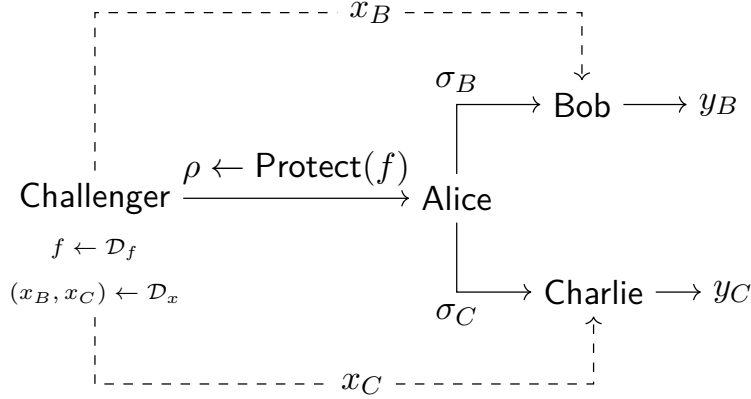
Parties: a challenger and a triple of QPT adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$.

1. The challenger samples a function f from $\mathcal{D}_f(\lambda)$, then computes $\rho \leftarrow \text{Protect}(1^\lambda, f)$ and sends it to $\mathcal{A}$.
2. $\mathcal{A}$ prepares two (possibly entangled) quantum states σ_B and σ_C , and sends σ_B to $\mathcal{B}$ and σ_C to $\mathcal{C}$. From now on, $\mathcal{B}$ and $\mathcal{C}$ cannot communicate.
3. The challenger samples (x_B, x_C) from $\mathcal{D}_x(\lambda)$ and sends x_B to $\mathcal{B}$ and x_C to $\mathcal{C}$.
4. $\mathcal{B}$ returns y_B and $\mathcal{C}$ returns y_C . They win if $y_B = f(x_B)$ and

$$y_C = f(x_C).$$

We denote the binary random variable indicating whether the adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ win the game as $\text{CP-Pir}_\lambda^{\mathcal{F}}(\mathcal{A}, \mathcal{B}, \mathcal{C})$.

We provide an illustration of this game in figure 6.6.



Winning Condition:
 $y_B = f(x_B)$ and
 $y_C = f(x_C)$

Figure 6.6: Anti-Piracy game for copy-protection. A copy-protection scheme is secure if no adversaries can win this game with a non-negligible advantage over the trivial winning probability.

Remark 6.5 (Trivial winning probability). Consider any copy-protection scheme for a family of functions $\mathcal{F}$ and distributions $\mathcal{D}_f$ and $\mathcal{D}_x$ for the corresponding anti-piracy game. Now, consider a triple of adversaries such that Alice simply forwards the quantum encoding of the function she receives to Bob, and nothing to Charlie. Bob then uses the quantum encoding to compute the image of its challenge x_B , and Charlie outputs a random image. We denote the probability of winning of such adversaries as the trivial winning probability.

Secure software leasing (SSL). As said before, secure software leasing is very close to copy-protection. It is the task of vendor encoding a

function into a quantum state such that any client owning this state (a) can compute the function on any input they want and (b) has to return the state to the vendor after some time. As in copy-protection, a secure software leasing scheme is described for a certain family of functions $\mathcal{F}$ with the same domains and codomains and is composed of three algorithms, **Lease**, **Eval**, and **Verify**.

Informally, an honest execution of a secure software leasing scheme is: a vendor runs **Lease** on a function, and leases the output to a client. The client can then compute the function on every input they want using **Eval**, and after some time, returns the state to the vendor. The latter finally runs **Verify** on this state to check whether the client behaved honestly or tried to cheat (for instance by trying to copy the state).

More formally, the **Lease** and **Eval** algorithms are defined in the same way as the **Protect** and **Eval** algorithm of copy-protection, and **Verify** takes as input a quantum state ρ and the description of a function f and either accepts (returns $\top$) or rejects it (returns $\perp$).

In addition, the *correctness* and *finite-term lessor security* properties must hold. The *correctness* property is defined in the same way as in copy-protection.

The *finite-term lessor security* property is similar to the *anti-piracy* property of copy-protection but incorporates the fact that the client must return the protected function to the vendor. This property ensures that no malicious adversary $\mathcal{A}$ can win game 9 with non-negligible probability.

Formally, a secure software leasing scheme has finite-term lessor security if, for all QPT adversary $\mathcal{A}$,

$$\Pr \left(\text{SSL-Pir}_{\lambda}^{\mathcal{F}}(\mathcal{A}) = 1 \right) \leq p_{triv}^{SSL}(\lambda, \mathcal{D}_x) + \text{negl}(\lambda)$$

where $\text{SSL-Pir}_{\lambda}^{\mathcal{F}}(\mathcal{A})$ is defined in game 9 below, and $p_{triv}^{SSL}(\lambda, \mathcal{D}_x)$ is defined similarly as $p_{triv}^{CP}(\lambda, \mathcal{D}_x)$.

Game 9: Piracy Game of a Secure Software Leasing Scheme

Parameters: a security parameter λ , an efficiently samplable distribution $\mathcal{D}_f(\lambda)$ on $\mathcal{F}_{\lambda}$ and an efficiently samplable distribution $\mathcal{D}_x(\lambda)$ on the inputs for $\mathcal{F}_{\lambda}$.

Parties: a challenger and a QPT adversary $\mathcal{A}$.

1. The challenger samples a function f from $\mathcal{D}_f(\lambda)$, then computes $\rho \leftarrow \text{Lease}(1^\lambda, f)$ and sends it to $\mathcal{A}$.
2. $\mathcal{A}$ prepares a quantum state σ_C and sends it to the challenger.
3. The challenger samples x from $\mathcal{D}_x(\lambda)$ and sends it to $\mathcal{A}$.
4. $\mathcal{A}$ returns y , and wins if $\text{Verify}(\sigma_C, f) = \top$ and $y = f(x)$.

We denote the binary random variable indicating whether the adversary $\mathcal{A}$ wins the game by $\text{SSL-Pir}_\lambda^{\mathcal{F}}(\mathcal{A})$.

Revocable cryptography. Based on the work of Ananth and La Placa [137] on secure software leasing, a recent line of work try to construct more primitive in the same way. Instead of asking the quantum object (such as quantum signing token, quantum program) to have unclonability, or certified deletion property (see Section 6.3.2), this new *revocable cryptography* field asks it to be *revocable*. What it means for a quantum object to be revocable is that, if a client Bob receives it from a vendor Alice, he can, after using it as much as he wants, return it to Alice, who can then verify it. Conditioned on acceptance, it ensures that Bob does not have the object anymore, and thus cannot perform the corresponding task (signing a message, evaluating the program, etc.). Except for secure software leasing, we do not tackle this field in details in this paper. However, we provide a brief description of important works in this field below.

Unruh [57] was the first to define this notion, presenting a revocable timed-released encryption. A timed-released encryption hides an information until a fixed amount of time has passed; its revocable version then means that one holding the encryption can return it before this amount of time has passed, therefore definitely losing access to the message. Ananth and La Placa [137], Kitagawa, Nishimaki, and Yamakawa [138], and Broadbent, Jeffery, Lord, Podder, and Sundaram [139] all proposed constructions for secure software leasing, with different assumptions. Three papers, by Chardouvelis, Goyal, Jain, and Liu [140] and Ananth, Poremba, and Vaikuntanathan [141], [142] focused on building public-key encryption, and variants (such as functional

encryption, fully-homomorphic encryption) with revocable decryption keys: once Bob returns its quantum decryption key to Alice, she knows with certainty that he cannot decrypt any more messages. The authors of the second and third papers also provide a construction for revocable pseudorandom functions. In such a construction, a “quantum program” allows Bob to evaluate a pseudorandom function on any input. At any time, Alice can request Bob to return this quantum state, ensuring that the latter cannot evaluate the pseudorandom function anymore. Morimae, Poremba, and Yamakawa [143] designed the first digital signature scheme with a revocable signature key. Finally, Ananth, Mutreja, and Poremba [144] focus on the case of multi-copy security.

6.4.2 Copy-Protection and Secure Software Leasing Protocols for Point Functions

While it is impossible to construct copy-protection for any family of functions (Aaronson [16]), this functionality exists for specific families. In this section, we introduce such families: point functions and compute-and-compare functions. Then, we present the copy-protection protocol defined by Coladangelo, Majenz, and Poremba [17]. Then we show how to extend it in order to construct a secure software leasing protocol. Although the copy-protection protocol unfortunately does not achieve standard security, it is interesting by its simplicity, and the resulting secure software leasing protocol achieves standard security⁹. Finally, we explain how a copy-protection protocol for point functions can be easily used to construct a copy-protection protocol for compute-and-compare functions.

Point functions and compute-and-compare functions. First, we introduce the point functions. A point function is parametrized by a point $y \in \{0, 1\}^n$, takes as input a classical bitstring $x \in \{0, 1\}^n$ and outputs 1 if and only if $x = y$. Otherwise, it outputs 0. Point functions can be extended to compute-and-compare functions, that are parametrized by a function f , a lock $y \in \{0, 1\}^n$ and a secret s . A compute-and-compare

⁹We write that a protocol has non-standard security if an adversary can win the corresponding security game with small, yet non-negligible probability.

function $CC[f, y, s]$ takes as input a bitstring $x \in \{0, 1\}^n$ and outputs s if and only if $f(x) = y$. Otherwise, it outputs a rejection symbol $\perp$.

Definition 6.6 (Point Function). A point function f_y with point $y \in \{0, 1\}^n$ is defined in the following way:

$$f_y(x) = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases}$$

Definition 6.7 (Compute-and-Compare Function). A compute-and-compare function $CC_{f,y,s}$ with function f , lock y and secret s is defined in the following way:

$$CC_{f,y,s}(x) = \begin{cases} s & \text{if } f(x) = y \\ 0 & \text{otherwise} \end{cases}$$

We are now ready to present copy-protection and secure-software-leasing protocols.

Copy-protection protocol for point functions. This protocol copy-protects a family of functions $\mathcal{F} = \{f_y\}_y$ indexed by their point $y \in \{0, 1\}^n$. Note that the protocol we describe is not exactly the one presented by Coladangelo, Majenz, and Poremba [17], as we make a few simplifications for ease of understanding.

Construction 5: Copy-Protection of Point Functions

- $\text{Protect}(1^\lambda, y)$
 1. Sample $r \leftarrow \{0, 1\}^n$;
 2. Return $(|r^y\rangle, r)$ as the quantum protection of y .
- $\text{Eval}(1^\lambda, (\rho, r), x)$
 1. Measure ρ in the basis x^{10} ; let r' denote the outcome;
 2. Return 1 if $r' = r$, otherwise return 0.

By measuring in the basis x , we mean measuring in the computational basis if $x = 0$, and in the Hadamard one otherwise.

Note that if $x = y$, then the measurement in the evaluation procedure is done in the same basis as the one used for encoding the state, hence

the outcome is r . Otherwise, if $x \neq y$, some (at least one) of the qubits of $|r^y\rangle$ will not be measured in the right basis, hence the outcome of the measurement will be different from r with significant probability. Remark that in the protocol presented by Coladangelo, Majenz, and Poremba [17], two hash functions are used, essentially to make sure that the probability of evaluating the protection of a point y on $x \neq y$ is negligible.

Secure software leasing protocol for point functions. We recall that the protocol for copy-protection of point functions presented above does not achieve standard security. Here, we present an extension of this protocol, defined in the same paper, for secure software leasing, which achieves standard security.

Construction 6: Secure Software Leasing of Point Functions

- $\text{Lease}(1^\lambda, y)$
 1. Sample $r \leftarrow \{0, 1\}^n$;
 2. Return $(|r^y\rangle, r)$ as the quantum protection of y .
- $\text{Eval}(1^\lambda, (\rho, r), x)$
 1. Measure ρ in the basis x ; let r' denote the outcome;
 2. Return 1 if $r' = r$, otherwise return 0.
- $\text{Verify}(1^\lambda, (\rho, r), y)$
 1. Run $\text{Eval}(1^\lambda, (\rho, r), y)$; let z denote the outcome;
 2. Return $\top$ if $z = 1$, else return $\perp$.

As for the copy-protection scheme, we refer the curious reader to Coladangelo, Majenz, and Poremba [17] for a proof of the security of this construction.

Copy-protection of compute-and-compare functions. As said above, a copy-protection scheme (PF.Protect, PF.Eval) for point functions can easily be extended to a copy-protection scheme for compute-and-compare

functions. We present how to do it in the following. The family of functions is now $\mathcal{F} = \{CC_{f,y,s}\}$, every function is parametrized by a function f , a lock y and a secret s .

Construction 7: Copy-Protection of Compute-And-Compare Functions

Ingredients: a copy-protection scheme for point-functions (PF.Protect, PF.Eval).

- Protect($1^\lambda, f, y, s$)
 1. Run PF.Protect($1^\lambda, y$) to get $|\psi_y\rangle$;
 2. Return $(|\psi_y\rangle, f, s)$ as the quantum encoding.
- Eval($1^\lambda, (\rho, f, s), x$)
 1. Run PF.Eval($1^\lambda, \rho, f(x)$);
 2. Return s if the outcome is 1, otherwise return $\perp$.

As we can see, it is enough to protect only the lock as a point function. Intuitively, the security comes from the fact that an adversary who wants to copy a protected function obtained from this protocol needs to copy the lock, which they cannot do by anti-piracy security of the copy-protection for point functions.

6.4.3 Single-Decryptors

Copy-protection does not exist only for point functions and compute-and-compare functions. In this subsection, we present *single-decryptors*, which can be seen as copy-protection schemes where the protected function is the decryption algorithm of some encryption scheme. In other words, a single-decryptor scheme is an encryption scheme in which the decryption key is a quantum state, and cannot be copied. This functionality has been defined recently by Georgiou and Zhandry [145], and they show how to construct a (private) single-decryptor scheme, based on unclonable encryption. Later on, Coladangelo, Liu, Liu, and Zhandry [113] showed how to construct such a scheme, based on *coset states*, a generalization of subspace states (see 6.1.2) where

the subspace state is *quantum one-time padded* (a quantum analogue of classical one-time pad) with secret values (see the latter paper and the work of Culf and Vidick [107] for more information on coset states). In the following, we define single-decryptor schemes more formally.

Definition 6.8 (Single-Decryptor). A *single-decryptor* scheme is made of four algorithms (Setup, QKeyGen, Enc, Dec) defined as follows:

- Setup takes as input a security parameter 1^λ and returns a secret key sk and a public key pk
- QKeyGen takes as input a secret key sk and returns a quantum decryption key $|\psi\rangle$.
- Enc is a classical probabilistic algorithm that takes as input a public key pk and a message m and returns a ciphertext c .
- Dec takes as input a quantum decryption key ρ and a ciphertext c and returns a message m .

A *single-decryptor* scheme must in addition be *correct*, *CPA-secure* and must satisfy *anti-piracy* security. This anti-piracy notion of security is defined through a game and combines both the ideas of unclonability and CPA security. Moreover, it can be shown that this *CPA-style* anti-piracy implies CPA security.

Definition 6.9 (Anti-Piracy Security (CPA-style)). We formalize here what *anti-piracy* security means. No polynomially bounded adversary must be able to win the following game with a probability greater than $1/2 + \text{negl}(\lambda)$. This game is parametrized by a security parameter λ , and is between a challenger and a triple of colluding adversaries $\mathcal{A}, \mathcal{B}$ and $\mathcal{C}$. Formally, a secure software leasing scheme has finite-term lessor security if, for all triple of QPT adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$,

$$\Pr(\text{SD-Pir}_\lambda(\mathcal{A}, \mathcal{B}, \mathcal{C}) = 1) \leq \frac{1}{2} + \text{negl}(\lambda)$$

where $\text{SD-Pir}_\lambda(\mathcal{A}, \mathcal{B}, \mathcal{C})$ is defined in game 10 below.

Game 10: Piracy Game of a Single-Decryptor Scheme

Parameters: a security parameter λ .

Parties: a challenger and a triple of QPT adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ such that $\mathcal{B}$ and $\mathcal{C}$ cannot communicate.

1. The challenger samples a pair $(sk, pk) \leftarrow \text{Setup}(1^\lambda)$. The challenger prepares $|\psi\rangle \leftarrow \text{QKeyGen}(sk)$ and sends it to $\mathcal{A}$.
2. $\mathcal{A}$ prepares two (possibly entangled) quantum states σ_B and σ_C and chooses two messages m_0 and m_1 . She sends σ_B to $\mathcal{B}$, σ_C to $\mathcal{C}$ and (m_0, m_1) to the challenger.
3. The challenger computes $c_0 \leftarrow \text{Enc}(pk, m_0)$ and $c_1 \leftarrow \text{Enc}(pk, m_1)$. The challenger samples two random bits b_B and b_C and sends c_{b_B} to $\mathcal{B}$ and c_{b_C} to $\mathcal{C}$.
4. $\mathcal{B}$ and $\mathcal{C}$ respectively return b'_B and b'_C .
5. $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ win if and only if $b'_B = b_B$ and $b'_C = b_C$.

We denote the binary random variable indicating whether the triple of adversaries $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ wins the game by $\text{SD-Pir}_\lambda(\mathcal{A}, \mathcal{B}, \mathcal{C})$.

We provide an illustration of this game in figure 6.7.

A single-decryptor scheme from coset states. In this paragraph, we give an informal presentation of the single-decryptor scheme introduced in Coladangelo, Liu, Liu, and Zhandry [113]. This scheme is based on the coset states, that we presented in Section 6.1.2.

Construction 8: Single-Decryptor

- $\text{Setup}(1^\lambda)$
 1. Sample a random subspace $A \leftarrow \{0, 1\}^{\lambda \times \lambda/2}$ of dimension $\lambda/2$, and two vectors $s, s' \leftarrow \{0, 1\}^\lambda$;
 2. Prepare the obfuscated membership programs P_{A+s} and $P_{A^\perp+s'}$;
 3. Return (A, s, s') as the secret key, and $(P_{A+s}, P_{A^\perp+s'})$ as

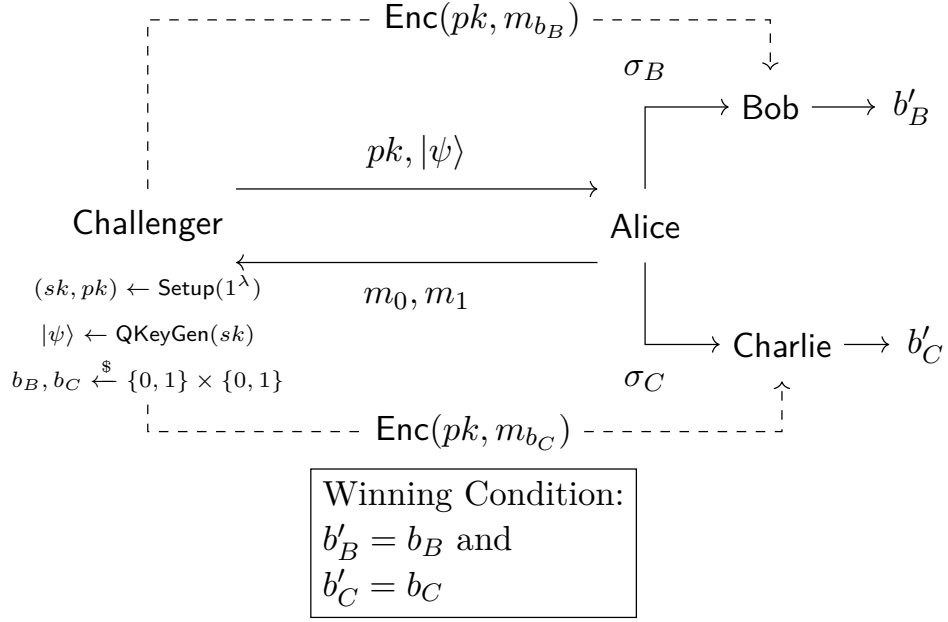


Figure 6.7: Anti-Piracy game for single decryptor. A single decryptor scheme has anti-piracy security if no adversaries can win this game with probability greater than $1/2$.

the public key.

- $\text{QKeyGen}((A, s, s'))$
 1. Return the coset state $|A, s, s'\rangle$.
- $\text{Enc}((P_{A+s}, P_{A^\perp+s'}), m)$
 1. Sample a bit $r \leftarrow \{0, 1\}$;
 2. Prepare and obfuscate the program Q_m described in program 1, using λ , m , r , P_{A+s} , and $P_{A^\perp+s'}$ as parameters;
 3. Return (Q_m, r) as the encryption.
- $\text{Dec}(\rho, (Q, r))$
 1. If $r = 1$, apply a Hadamard gate on ρ ; let ρ' denote the resulting state;
 2. Run the program Q on superposition over ρ' ;
 3. Abort if the program returns $\perp$, otherwise return the outcome.

Program 1: Program Q_m

Hardwired: a security parameter λ , a message m , a bitstring r , two membership programs P_0 and P_1 .

Input: a vector $u \in \{0, 1\}^\lambda$.

1. Run P_r on u .
2. If the outcome is 1, indicating that the program accepts, return m .
3. Otherwise, return $\perp$.

Anti-piracy security. The security of this protocol comes from the obfuscation of Q_m . Informally, applying obfuscation to this program states that, if an adversary gets any significant information on m given Q_m and an auxiliary information σ , then this adversary can also learn a vector u that passes the test. Thus, if adversaries Alice, Bob and Charlie break the anti-piracy security of this scheme, that is, if Bob and Charlie both distinguish correctly the encryption of m_0 from the encryption of m_1 , then it means that Bob gets some information on m_{b_B} given $Q_{m_{b_B}}$ and σ_1 . Hence, assuming that the random r used for Bob's encryption is 0 - which happens with probability $1/2$ - Bob can learn a vector u in the primal coset. Similarly and for the same reasons, assuming that the random r used for Charlie's encryption is 1 - which also happens with probability $1/2$ - Charlie can learn a vector v in the dual coset. Thus, they are able to return respectively a vector in the primal coset and a vector in the dual coset with non-negligible probability, which is known to be impossible according to the monogamy-of-entanglement property of coset states (Theorem 6.10), proving that this scheme is secure.

Remark 6.6. Note that, similarly to construction 2, this scheme uses a strong form of obfuscation that only exists under non-standard assumptions. As a result, the security of the scheme presented above also holds under these non-standard assumptions.

6.4.4 History of Copy-Protection

In this section, we present a history of different results and constructions regarding copy-protection and secure software leasing. We do not mention results on single-decryptor schemes in this subsection, as we already discussed it in Section 6.4.3.

Aaronson [16] defined copy-protection and presented a construction based on a quantum oracle. In this paper, he also proved that no *learnable functions* (roughly, functions whose behavior can be learned from a polynomial number of input-output pairs) can be copy-protected. This field did not get a lot of attention until 2020, where Ananth and La Placa [137] defined secure software leasing and proved two impossibility results: one on the impossibility of secure software leasing for learnable functions and the other one on the impossibility of copy-protection for the so-called *de-quantumizable functions*, a family of functions that are not learnable, thus extending the impossibility result of Aaronson. In the same paper, they constructed a secure software leasing protocol for a subclass of evasive functions, namely *searchable compute-and-compare functions*.

The same year, Coladangelo, Majenz, and Poremba [17] presented a first concrete copy-protection protocol for point functions in the quantum oracle model. However, this protocol does not achieve standard security (an adversary can win the anti-piracy game with a constant - yet small - probability). Nevertheless, they extended this protocol to a secure software leasing one, with standard security.

Aaronson, Liu, Liu, Zhandry, and Zhang [146] provided a new construction for copy-protection, this time based only on a classical oracle, improving his previous construction. They also provided a *copy-detection* construction for any watermarked functions. We will not detail the notions of *copy-detection* (a task close to copy-protection but weaker than it) and *watermarkable functions*, we refer the reader to the papers of Aaronson, Liu, Liu, Zhandry, and Zhang [146] and of Cohen, Holmgren, Nishimaki, Vaikuntanathan, and Wichs [147] for more information on these notions. Soon after, Kitagawa, Nishimaki, and Yamakawa [148] constructed a secure software leasing scheme for pseudorandom functions and searchable compute-and-compare functions

based on a relaxed version of watermarking.

Broadbent, Jeffery, Lord, Podder, and Sundaram [11] showed that copy-protection can be achieved without any assumptions (we recall that we always implicitly assumed that the adversary was computationally bounded in the previous constructions). To achieve it, the authors used a weaker definition for correctness, namely correctness with respect to a distribution, and a new adversary model: the *honest-malicious adversary*. In this model, in the anti-piracy security game, $\mathcal{B}$ can perform any computation on σ_B to return y_B as in the original definition, but $\mathcal{C}$ has to use the Eval algorithm on σ_C and x_C and returns the output as y_C . They also proved that copy-protection with honest-malicious adversary implies secure software leasing.

Coladangelo, Liu, Liu, and Zhandry [113] presented the first construction of a copy-protection scheme in the plain model by using coset states to copy-protect a subclass of pseudo-random functions. Some ideas of this last paper were exploited by Liu, Liu, Qian, and Zhandry [149] to construct a copy-protection scheme of digital signatures. In the same paper, the authors presented constructions of single-decryptors, copy-protection of digital signatures, and copy-protection of pseudo-random functions with (bounded) collusion resistance. This result was later extended by Çakan and Goyal [150] with unbounded collusion resistance.

Finally, two papers (Ananth, Kaleoglu, and Liu [105], [129]) improved the result of Coladangelo, Majenz, and Poremba [17] by constructing a copy-protection scheme for point functions in the quantum oracle model (the former paper leverages coset states, while the latter only BB84 states), with standard security.

Remark that, apart from the copy-protection of pseudorandom functions of Coladangelo, Liu, Liu, and Zhandry [113], all the other schemes are constructed in the quantum random oracle model. In particular, copy-protection for point functions in the plain model (that is without using the random oracle model) remains elusive. With this in mind, Kitagawa and Nishimaki [151] and Chevalier, Hermouet, and Vu [152] both proposed a construction for such a scheme. Although the security of these two constructions are indeed in the plain model, the former only achieves security based on a relaxed security notion

(named one-out-of-many security), and the latter achieves security for a specific — “less natural” — challenge distribution. The existence of a copy-protection scheme of point functions with a “natural” distribution is still an open problem at the time of writing. However, Ananth and Behera [130] and Chevalier, Hermouet, and Vu [131] made progress on this task by proposing constructions based on unproven assumptions.

6.5 Perspectives and Open Problems

In this section, we summarize the various open problems mentioned in this chapter, and discuss promising research directions related to them.

Public-key quantum money. The major open problem in public-key quantum money is to construct it from standard assumptions. As mentioned above, several works Farhi, Gosset, Hassidim, Lutomirski, and Shor [117], Kane [118], and Kane, Sharif, and Silverberg [119] proposed a construction relying on the hardness of not well-studied knots-based or quaternion-algebras-based problems. As these problems are not well-studied, and hence considered as non-standard, investigating how they relate to well-studied problems (*e.g.* code-based, or lattices-based problems) could improve our knowledge on the hardness of this task. The only secure public-key quantum money scheme known so far (Zhandry [12]) being based on the oracle-based template from Aaronson and Christiano [112], another natural research direction would be to try to construct these oracles without using obfuscation.

Quantum lightning. Similarly to public-key quantum money, we do not know any construction of quantum lightning based on standard assumptions. A promising way to construct this primitive seems to be the group-actions template defined by Zhandry [125] and Mutreja and Zhandry [153]. The only candidate (proposed in the former paper) is also based on a non-standard assumption related to isogenies, thus studying this new problem, or trying to propose a candidate based on well-studied assumptions is a promising direction in this field.

Unclonable encryption. While unclonable encryption has been studied extensively, there is still no construction achieving unclonable-indistinguishability property in the plain model — that is, without oracles — even based on obfuscation. Constructing a scheme with such security level is a natural, yet challenging goal for anyone working in this field, and the concurrent works of Ananth and Behera [130] and Chevalier, Hermouet, and Vu [131] both reduce this task to much simpler ones: proving that the *simultaneous Golreich-Levin*, and *simultaneous compute-and-compare obfuscation* conjectures respectively introduced in these two papers are true.

Copy-protection. We finally turn to copy-protection. Although this primitive can in theory be defined for any family of functions, little is known on which families can be protected in practice, and which cannot. Indeed, there could be families for which the definition of copy-protection makes sense, but it is provably impossible to construct a secure scheme for them. The works of Aaronson [16] and Ananth and La Placa [137] actually provide examples of such families, and recent works Ananth and Behera [130], [154] makes progress on this question. However, we still do not have a definite answer, and continuing this work is crucial to improve our understanding of copy-protection.

Another important and unresolved question on this primitive is the existence of a copy-protection scheme of point functions in the plain model. The difficulties when trying to construct such a scheme are quite similar as the ones when trying to construct unclonable encryption with unclonable-indistinguishability property, thus focusing on proving the conjectures of Ananth and Behera [130] and Chevalier, Hermouet, and Vu [131] seems to be a promising direction.

References

- [1] R. P. Feynman, “Simulating physics with computers,” in *Feynman and computation*, CRC Press, 2018, pp. 133–153.
- [2] D. Deutsch, “Quantum theory, the church–turing principle and the universal quantum computer,” *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences*, vol. 400, no. 1818, 1985, pp. 97–117.
- [3] D. Deutsch and R. Jozsa, “Rapid solution of problems by quantum computation,” *Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences*, vol. 439, no. 1907, 1992, pp. 553–558.
- [4] E. Bernstein and U. Vazirani, “Quantum complexity theory,” *SIAM Journal on computing*, vol. 26, no. 5, 1997, pp. 1411–1473.
- [5] P. W. Shor, “Algorithms for quantum computation: Discrete logarithms and factoring,” in *Proceedings 35th annual symposium on foundations of computer science*, Ieee, pp. 124–134, 1994.
- [6] S. Wiesner, “Conjugate coding,” *ACM Sigact News*, vol. 15, no. 1, 1983, pp. 78–88.
- [7] M. Rabin, “How to exchange secrets by oblivious transfer,” Harvard University, Tech. Rep. TR-81, 1981.

- [8] C. H. Bennett and G. Brassard, “Quantum cryptography: Public key distribution and coin tossing,” *Theoretical Computer Science*, vol. 560, 2014, pp. 7–11. DOI: <https://doi.org/10.1016/j.tcs.2014.05.025>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0304397514004241>.
- [9] M. Bellare and P. Rogaway, “Random oracles are practical: A paradigm for designing efficient protocols,” in *ACM CCS 93*, D. E. Denning, R. Pyle, R. Ganesan, R. S. Sandhu, and V. Ashby, Eds., pp. 62–73, ACM Press, Nov. 1993. DOI: [10.1145/168588.168596](https://doi.org/10.1145/168588.168596).
- [10] J. v. d. Graaf, “Towards a formal definition of security for quantum protocols,” 1998.
- [11] A. Broadbent, S. Jeffery, S. Lord, S. Podder, and A. Sundaram, “Secure software leasing without assumptions,” in *Theory of Cryptography Conference*, Springer, pp. 90–120, 2021.
- [12] M. Zhandry, “Quantum lightning never strikes the same state twice. or: Quantum money from cryptographic assumptions,” *Journal of Cryptology*, vol. 34, 2021, pp. 1–56.
- [13] A. Broadbent and S. Lord, *Uncloneable quantum encryption via oracles*, Cryptology ePrint Archive, Report 2019/257, 2019.
- [14] P. Ananth and F. Kaleoglu, “Unclonable encryption, revisited,” in *Theory of Cryptography Conference*, Springer, pp. 299–329, 2021.
- [15] J. Bartusek and D. Khurana, “Cryptography with certified deletion,” in *CRYPTO 2023, Part V*, H. Handschuh and A. Lysyanskaya, Eds., ser. LNCS, vol. 14085, pp. 192–223, Springer, Cham, Aug. 2023. DOI: [10.1007/978-3-031-38554-4_7](https://doi.org/10.1007/978-3-031-38554-4_7).
- [16] S. Aaronson, “Quantum copy-protection and quantum money,” *2009 24th Annual IEEE Conference on Computational Complexity*, Jul. 2009. DOI: [10.1109/ccc.2009.42](https://doi.org/10.1109/ccc.2009.42). [Online]. Available: <http://dx.doi.org/10.1109/CCC.2009.42>.
- [17] A. Coladangelo, C. Majenz, and A. Poremba, “Quantum copy-protection of compute-and-compare programs in the quantum random oracle model,” *Quantum*, vol. 8, 2024, p. 1330.
- [18] A. Broadbent and C. Schaffner, “Quantum cryptography beyond quantum key distribution,” *Designs, Codes and Cryptography*, vol. 78, no. 1, 2016, pp. 351–382.

- [19] M. Bozzio, C. Crépeau, P. Wallden, and P. Walther, *Quantum cryptography beyond key distribution: Theory and experiment*, 2024. arXiv: [2411.08877](https://arxiv.org/abs/2411.08877) [quant-ph]. [Online]. Available: <https://arxiv.org/abs/2411.08877>.
- [20] O. Sattath, “Uncloneable cryptography,” *Communications of the ACM*, vol. 66, no. 11, 2023, pp. 78–86.
- [21] U. Mahadev, “Classical verification of quantum computations,” in *59th FOCS*, M. Thorup, Ed., pp. 259–267, IEEE Computer Society Press, Oct. 2018. DOI: [10.1109/FOCS.2018.00033](https://doi.org/10.1109/FOCS.2018.00033).
- [22] J. Zhang, “Classical verification of quantum computations in linear time,” in *63rd FOCS*, pp. 46–57, IEEE Computer Society Press, Oct. 2022. DOI: [10.1109/FOCS54457.2022.00012](https://doi.org/10.1109/FOCS54457.2022.00012).
- [23] J. Bartusek, Y. T. Kalai, A. Lombardi, F. Ma, G. Malavolta, V. Vaikuntanathan, T. Vidick, and L. Yang, “Succinct classical verification of quantum computation,” in *CRYPTO 2022, Part II*, Y. Dodis and T. Shrimpton, Eds., ser. LNCS, vol. 13508, pp. 195–211, Springer, Cham, Aug. 2022. DOI: [10.1007/978-3-031-15979-4_7](https://doi.org/10.1007/978-3-031-15979-4_7).
- [24] A. Natarajan and T. Zhang, “Quantum free games,” in *55th ACM STOC*, B. Saha and R. A. Servedio, Eds., pp. 1603–1616, ACM Press, Jun. 2023. DOI: [10.1145/3564246.3585208](https://doi.org/10.1145/3564246.3585208).
- [25] U. Mahadev, “Classical homomorphic encryption for quantum circuits,” in *59th FOCS*, M. Thorup, Ed., pp. 332–338, IEEE Computer Society Press, Oct. 2018. DOI: [10.1109/FOCS.2018.00039](https://doi.org/10.1109/FOCS.2018.00039).
- [26] A. Gheorghiu, T. Kapourniotis, and E. Kashefi, “Verification of quantum computation: An overview of existing approaches,” *Theory Comput. Syst.*, vol. 63, no. 4, 2019, pp. 715–808. DOI: [10.1007/S00224-018-9872-3](https://doi.org/10.1007/S00224-018-9872-3). [Online]. Available: <https://doi.org/10.1007/s00224-018-9872-3>.
- [27] Z. Ji, Y.-K. Liu, and F. Song, “Pseudorandom quantum states,” in *CRYPTO 2018, Part III*, H. Shacham and A. Boldyreva, Eds., ser. LNCS, vol. 10993, pp. 126–152, Springer, Cham, Aug. 2018. DOI: [10.1007/978-3-319-96878-0_5](https://doi.org/10.1007/978-3-319-96878-0_5).

- [28] W. Kretschmer, “Quantum pseudorandomness and classical complexity,” in *16th Conference on the Theory of Quantum Computation, Communication and Cryptography*, 2021.
- [29] S. Aaronson, A. Bouland, B. Fefferman, S. Ghosh, U. V. Vazirani, C. Zhang, and Z. Zhou, “Quantum pseudoentanglement,” in *ITCS 2024*, V. Guruswami, Ed., vol. 287, 2:1–2:21, LIPIcs, Jan. 2024. DOI: [10.4230/LIPIcs.ITCS.2024.2](https://doi.org/10.4230/LIPIcs.ITCS.2024.2).
- [30] A. Bouland, B. Fefferman, S. Ghosh, T. Metger, U. Vazirani, C. Zhang, and Z. Zhou, “Public-Key Pseudoentanglement and the Hardness of Learning Ground State Entanglement Structure,” in *39th Computational Complexity Conference (CCC 2024)*, R. Santhanam, Ed., ser. Leibniz International Proceedings in Informatics (LIPIcs), vol. 300, 21:1–21:23, Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024. DOI: [10.4230/LIPIcs.CCC.2024.21](https://doi.org/10.4230/LIPIcs.CCC.2024.21). [Online]. Available: <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.CCC.2024.21>.
- [31] H. Zhao, L. Lewis, I. Kannan, Y. Quek, H.-Y. Huang, and M. C. Caro, “Learning quantum states and unitaries of bounded gate complexity,” *PRX Quantum*, vol. 5, no. 4, 2024, p. 040 306.
- [32] T. Morimae and T. Yamakawa, “Quantum commitments and signatures without one-way functions,” in *CRYPTO 2022, Part I*, Y. Dodis and T. Shrimpton, Eds., ser. LNCS, vol. 13507, pp. 269–295, Springer, Cham, Aug. 2022. DOI: [10.1007/978-3-031-15802-5_10](https://doi.org/10.1007/978-3-031-15802-5_10).
- [33] P. Ananth, L. Qian, and H. Yuen, “Cryptography from pseudorandom quantum states,” in *CRYPTO 2022, Part I*, Y. Dodis and T. Shrimpton, Eds., ser. LNCS, vol. 13507, pp. 208–236, Springer, Cham, Aug. 2022. DOI: [10.1007/978-3-031-15802-5_8](https://doi.org/10.1007/978-3-031-15802-5_8).
- [34] M. A. Nielsen and I. Chuang, *Quantum computation and quantum information*, 2002.
- [35] W. K. Wootters and W. H. Zurek, “A single quantum cannot be cloned,” *Nature*, vol. 299, no. 5886, 1982, pp. 802–803.
- [36] J. S. Bell, “On the einstein podolsky rosen paradox,” *Physics Physique Fizika*, vol. 1, no. 3, 1964, p. 195.

- [37] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, “Proposed experiment to test local hidden-variable theories,” *Physical review letters*, vol. 23, no. 15, 1969, p. 880.
- [38] R. L. Rivest, A. Shamir, and L. M. Adleman, “A method for obtaining digital signatures and public-key cryptosystems,” *Communications of the Association for Computing Machinery*, vol. 21, no. 2, Feb. 1978, pp. 120–126. DOI: [10.1145/359340.359342](https://doi.org/10.1145/359340.359342).
- [39] V. S. Miller, “Use of elliptic curves in cryptography,” in *CRYPTO’85*, H. C. Williams, Ed., ser. LNCS, vol. 218, pp. 417–426, Springer, Berlin, Heidelberg, Aug. 1986. DOI: [10.1007/3-540-39799-X_31](https://doi.org/10.1007/3-540-39799-X_31).
- [40] N. Koblitz, “Elliptic curve cryptosystems,” *Mathematics of computation*, vol. 48, no. 177, 1987, pp. 203–209.
- [41] D. J. Bernstein, “Introduction to post-quantum cryptography,” in *Post-quantum cryptography*, Springer, 2009, pp. 1–14.
- [42] T. Gagliardoni, “Quantum security of cryptographic primitives,” Ph.D. dissertation, TU Darmstadt, 2017.
- [43] M. Kaplan, G. Leurent, A. Leverrier, and M. Naya-Plasencia, “Quantum differential and linear cryptanalysis,” *IACR Trans. Symm. Cryptol.*, vol. 2016, no. 1, 2016, pp. 71–94. DOI: [10.13154/tosc.v2016.i1.71-94](https://doi.org/10.13154/tosc.v2016.i1.71-94). [Online]. Available: <https://tosc.iacr.org/index.php/ToSC/article/view/536>.
- [44] I. Damgård, J. Funder, J. B. Nielsen, and L. Salvail, “Superposition attacks on cryptographic protocols,” in *ICITS 13*, C. Padró, Ed., ser. LNCS, vol. 8317, pp. 142–161, Springer, Cham, 2014. DOI: [10.1007/978-3-319-04268-8_9](https://doi.org/10.1007/978-3-319-04268-8_9).
- [45] A. Sahai and B. Waters, “How to use indistinguishability obfuscation: Deniable encryption, and more,” in *46th ACM STOC*, D. B. Shmoys, Ed., pp. 475–484, ACM Press, May 2014. DOI: [10.1145/2591796.2591825](https://doi.org/10.1145/2591796.2591825).
- [46] T. Gagliardoni, A. Hülsing, and C. Schaffner, “Semantic security and indistinguishability in the quantum world,” in *CRYPTO 2016, Part III*, M. Robshaw and J. Katz, Eds., ser. LNCS, vol. 9816, pp. 60–89, Springer, Berlin, Heidelberg, Aug. 2016. DOI: [10.1007/978-3-662-53015-3_3](https://doi.org/10.1007/978-3-662-53015-3_3).

- [47] M. Kaplan, G. Leurent, A. Leverrier, and M. Naya-Plasencia, “Breaking symmetric cryptosystems using quantum period finding,” in *CRYPTO 2016, Part II*, M. Robshaw and J. Katz, Eds., ser. LNCS, vol. 9815, pp. 207–237, Springer, Berlin, Heidelberg, Aug. 2016. DOI: [10.1007/978-3-662-53008-5_8](https://doi.org/10.1007/978-3-662-53008-5_8).
- [48] H. Kuwakado and M. Morii, “Security on the quantum-type even-mansour cipher,” in *2012 International Symposium on Information Theory and its Applications*, IEEE, pp. 312–316, 2012.
- [49] H. Kuwakado and M. Morii, “Quantum distinguisher between the 3-round feistel cipher and the random permutation,” in *2010 IEEE International Symposium on Information Theory*, IEEE, pp. 2682–2685, 2010.
- [50] G. Leander and A. May, “Grover meets simon - quantumly attacking the FX-construction,” in *ASIACRYPT 2017, Part II*, T. Takagi and T. Peyrin, Eds., ser. LNCS, vol. 10625, pp. 161–178, Springer, Cham, Dec. 2017. DOI: [10.1007/978-3-319-70697-9_6](https://doi.org/10.1007/978-3-319-70697-9_6).
- [51] A. Schrottenloher, “Quantum algorithms for cryptanalysis and quantum-safe symmetric cryptography,” Ph.D. dissertation, Sorbonne Université, 2021.
- [52] O. Goldreich, S. Micali, and A. Wigderson, “Proofs that yield nothing but their validity or all languages in NP have zero-knowledge proof systems,” *Journal of the ACM*, vol. 38, no. 3, Jul. 1991, pp. 691–729. DOI: [10.1145/116825.116852](https://doi.org/10.1145/116825.116852).
- [53] M. Zhandry, “Secure identity-based encryption in the quantum random oracle model,” in *CRYPTO 2012*, R. Safavi-Naini and R. Canetti, Eds., ser. LNCS, vol. 7417, pp. 758–775, Springer, Berlin, Heidelberg, Aug. 2012. DOI: [10.1007/978-3-642-32009-5_44](https://doi.org/10.1007/978-3-642-32009-5_44).
- [54] M. Zhandry, “How to record quantum queries, and applications to quantum indistinguishability,” in *CRYPTO 2019, Part II*, A. Boldyreva and D. Micciancio, Eds., ser. LNCS, vol. 11693, pp. 239–268, Springer, Cham, Aug. 2019. DOI: [10.1007/978-3-030-26951-7_9](https://doi.org/10.1007/978-3-030-26951-7_9).
- [55] Q. Liu and M. Zhandry, “Revisiting post-quantum Fiat-Shamir,” in *CRYPTO 2019, Part II*, A. Boldyreva and D. Micciancio, Eds., ser. LNCS, vol. 11693, pp. 326–355, Springer, Cham, Aug. 2019. DOI: [10.1007/978-3-030-26951-7_12](https://doi.org/10.1007/978-3-030-26951-7_12).

- [56] J. Don, S. Fehr, C. Majenz, and C. Schaffner, “Security of the Fiat-Shamir transformation in the quantum random-oracle model,” in *CRYPTO 2019, Part II*, A. Boldyreva and D. Micciancio, Eds., ser. LNCS, vol. 11693, pp. 356–383, Springer, Cham, Aug. 2019. DOI: [10.1007/978-3-030-26951-7_13](https://doi.org/10.1007/978-3-030-26951-7_13).
- [57] D. Unruh, “Revocable quantum timed-release encryption,” in *EUROCRYPT 2014*, P. Q. Nguyen and E. Oswald, Eds., ser. LNCS, vol. 8441, pp. 129–146, Springer, Berlin, Heidelberg, May 2014. DOI: [10.1007/978-3-642-55220-5_8](https://doi.org/10.1007/978-3-642-55220-5_8).
- [58] A. Ambainis, M. Hamburg, and D. Unruh, “Quantum security proofs using semi-classical oracles,” in *CRYPTO 2019, Part II*, A. Boldyreva and D. Micciancio, Eds., ser. LNCS, vol. 11693, pp. 269–295, Springer, Cham, Aug. 2019. DOI: [10.1007/978-3-030-26951-7_10](https://doi.org/10.1007/978-3-030-26951-7_10).
- [59] D. Unruh, “Computationally binding quantum commitments,” in *EUROCRYPT 2016, Part II*, M. Fischlin and J.-S. Coron, Eds., ser. LNCS, vol. 9666, pp. 497–527, Springer, Berlin, Heidelberg, May 2016. DOI: [10.1007/978-3-662-49896-5_18](https://doi.org/10.1007/978-3-662-49896-5_18).
- [60] A. Ambainis, A. Rosmanis, and D. Unruh, “Quantum attacks on classical proof systems: The hardness of quantum rewinding,” in *55th FOCS*, pp. 474–483, IEEE Computer Society Press, Oct. 2014. DOI: [10.1109/FOCS.2014.57](https://doi.org/10.1109/FOCS.2014.57).
- [61] J. Watrous, “Zero-knowledge against quantum attacks,” in *38th ACM STOC*, J. M. Kleinberg, Ed., pp. 296–305, ACM Press, May 2006. DOI: [10.1145/1132516.1132560](https://doi.org/10.1145/1132516.1132560).
- [62] N. Bitansky and O. Shmueli, “Post-quantum zero knowledge in constant rounds,” in *52nd ACM STOC*, K. Makarychev, Y. Makarychev, M. Tulsiani, G. Kamath, and J. Chuzhoy, Eds., pp. 269–279, ACM Press, Jun. 2020. DOI: [10.1145/3357713.3384324](https://doi.org/10.1145/3357713.3384324).
- [63] T. Vidick and J. Watrous, “Quantum proofs,” *Foundations and Trends® in Theoretical Computer Science*, vol. 11, no. 1-2, 2016, pp. 1–215.

- [64] D. Unruh, “Quantum proofs of knowledge,” in *EUROCRYPT 2012*, D. Pointcheval and T. Johansson, Eds., ser. LNCS, vol. 7237, pp. 135–152, Springer, Berlin, Heidelberg, Apr. 2012. DOI: [10.1007/978-3-642-29011-4_10](https://doi.org/10.1007/978-3-642-29011-4_10).
- [65] A. Chiesa, F. Ma, N. Spooner, and M. Zhandry, “Post-quantum succinct arguments: Breaking the quantum rewinding barrier,” in *62nd FOCS*, pp. 49–58, IEEE Computer Society Press, Feb. 2022. DOI: [10.1109/FOCS52979.2021.00014](https://doi.org/10.1109/FOCS52979.2021.00014).
- [66] A. Lombardi, F. Ma, and N. Spooner, “Post-quantum zero knowledge, revisited or: How to do quantum rewinding undetectably,” in *63rd FOCS*, pp. 851–859, IEEE Computer Society Press, Oct. 2022. DOI: [10.1109/FOCS54457.2022.00086](https://doi.org/10.1109/FOCS54457.2022.00086).
- [67] R. W. F. Lai, G. Malavolta, and N. Spooner, “Quantum rewinding for many-round protocols,” in *TCC 2022, Part I*, E. Kiltz and V. Vaikuntanathan, Eds., ser. LNCS, vol. 13747, pp. 80–109, Springer, Cham, Nov. 2022. DOI: [10.1007/978-3-031-22318-1_4](https://doi.org/10.1007/978-3-031-22318-1_4).
- [68] U. Vazirani and T. Vidick, “Fully device independent quantum key distribution,” *Communications of the ACM*, vol. 62, no. 4, 2019, pp. 133–133.
- [69] D. Boneh and M. Zhandry, “Secure signatures and chosen ciphertext security in a quantum computing world,” in *CRYPTO 2013, Part II*, R. Canetti and J. A. Garay, Eds., ser. LNCS, vol. 8043, pp. 361–379, Springer, Berlin, Heidelberg, Aug. 2013. DOI: [10.1007/978-3-642-40084-1_21](https://doi.org/10.1007/978-3-642-40084-1_21).
- [70] M. Zhandry, “How to construct quantum random functions,” in *53rd FOCS*, pp. 679–687, IEEE Computer Society Press, Oct. 2012. DOI: [10.1109/FOCS.2012.37](https://doi.org/10.1109/FOCS.2012.37).
- [71] M. V. Anand, E. E. Targhi, G. N. Tabia, and D. Unruh, “Post-quantum security of the CBC, CFB, OFB, CTR, and XTS modes of operation,” in *Post-Quantum Cryptography - 7th International Workshop, PQCrypto 2016*, T. Takagi, Ed., pp. 44–63, Springer, Cham, 2016. DOI: [10.1007/978-3-319-29360-8_4](https://doi.org/10.1007/978-3-319-29360-8_4).

- [72] A. Broadbent and S. Jeffery, “Quantum homomorphic encryption for circuits of low T-gate complexity,” in *CRYPTO 2015, Part II*, R. Gennaro and M. J. B. Robshaw, Eds., ser. LNCS, vol. 9216, pp. 609–629, Springer, Berlin, Heidelberg, Aug. 2015. DOI: [10.1007/978-3-662-48000-7_30](https://doi.org/10.1007/978-3-662-48000-7_30).
- [73] G. Alagic, T. Gagliardoni, and C. Majenz, “Unforgeable quantum encryption,” in *EUROCRYPT 2018, Part III*, J. B. Nielsen and V. Rijmen, Eds., ser. LNCS, vol. 10822, pp. 489–519, Springer, Cham, Apr. 2018. DOI: [10.1007/978-3-319-78372-7_16](https://doi.org/10.1007/978-3-319-78372-7_16).
- [74] C. Chevalier, E. Ebrahimi, and Q.-H. Vu, “On security notions for encryption in a quantum world,” in *INDOCRYPT 2022*, T. Isobe and S. Sarkar, Eds., ser. LNCS, vol. 13774, pp. 592–613, Springer, Cham, Dec. 2022. DOI: [10.1007/978-3-031-22912-1_26](https://doi.org/10.1007/978-3-031-22912-1_26).
- [75] T. V. Carstens, E. Ebrahimi, G. Tabia, and D. Unruh, *On quantum indistinguishability under chosen plaintext attack*, Cryptology ePrint Archive, Report 2020/596, 2020. [Online]. Available: <https://eprint.iacr.org/2020/596>.
- [76] D. Boneh and M. Zhandry, “Quantum-secure message authentication codes,” in *EUROCRYPT 2013*, T. Johansson and P. Q. Nguyen, Eds., ser. LNCS, vol. 7881, pp. 592–608, Springer, Berlin, Heidelberg, May 2013. DOI: [10.1007/978-3-642-38348-9_35](https://doi.org/10.1007/978-3-642-38348-9_35).
- [77] G. Alagic, C. Majenz, A. Russell, and F. Song, “Quantum-access-secure message authentication via blind-unforgeability,” in *EUROCRYPT 2020, Part III*, A. Canteaut and Y. Ishai, Eds., ser. LNCS, vol. 12107, pp. 788–817, Springer, Cham, May 2020. DOI: [10.1007/978-3-030-45727-3_27](https://doi.org/10.1007/978-3-030-45727-3_27).
- [78] S. Garg, H. Yuen, and M. Zhandry, “New security notions and feasibility results for authentication of quantum data,” in *CRYPTO 2017, Part II*, J. Katz and H. Shacham, Eds., ser. LNCS, vol. 10402, pp. 342–371, Springer, Cham, Aug. 2017. DOI: [10.1007/978-3-319-63715-0_12](https://doi.org/10.1007/978-3-319-63715-0_12).
- [79] P. W. Shor and J. Preskill, “Simple proof of security of the bb84 quantum key distribution protocol,” *Physical Review Letters*, vol. 85, no. 2, Jul. 2000, pp. 441–444. DOI: [10.1103/physrevlett.85.441](https://doi.org/10.1103/physrevlett.85.441). [Online]. Available: <http://dx.doi.org/10.1103/PhysRevLett.85.441>.

- [80] R. Renner, “Security of quantum key distribution,” *International Journal of Quantum Information*, vol. 6, no. 01, 2008, pp. 1–127.
- [81] G. Brassard and L. Salvail, “Secret-key reconciliation by public discussion,” in *Advances in Cryptology — EUROCRYPT ’93*, T. Hellesest, Ed., pp. 410–423, Berlin, Heidelberg: Springer Berlin Heidelberg, 1994.
- [82] J. Müller-Quade and D. Unruh, “Long-term security and universal composability,” in *Theory of Cryptography: 4th Theory of Cryptography Conference, TCC 2007, Amsterdam, The Netherlands, February 21–24, 2007. Proceedings 4*, Springer, pp. 41–60, 2007.
- [83] D. Unruh, “Everlasting multi-party computation,” in *Annual Cryptology Conference*, Springer, pp. 380–397, 2013.
- [84] W. Diffie and M. E. Hellman, “New directions in cryptography,” *IEEE Transactions on Information Theory*, vol. 22, no. 6, 1976, pp. 644–654. DOI: [10.1109/TIT.1976.1055638](https://doi.org/10.1109/TIT.1976.1055638).
- [85] A. K. Ekert, “Quantum cryptography based on bell’s theorem,” *Phys. Rev. Lett.*, vol. 67, 6 Aug. 1991, pp. 661–663. DOI: [10.1103/PhysRevLett.67.661](https://doi.org/10.1103/PhysRevLett.67.661). [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.67.661>.
- [86] C. H. Bennett, G. Brassard, and N. D. Mermin, “Quantum cryptography without bell’s theorem,” *Phys. Rev. Lett.*, vol. 68, 5 Feb. 1992, pp. 557–559. DOI: [10.1103/PhysRevLett.68.557](https://doi.org/10.1103/PhysRevLett.68.557). [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.68.557>.
- [87] D. Mayers and A. Yao, “Quantum cryptography with imperfect apparatus,” in *Proceedings 39th Annual Symposium on Foundations of Computer Science (Cat. No.98CB36280)*, pp. 503–509, 1998. DOI: [10.1109/SFCS.1998.743501](https://doi.org/10.1109/SFCS.1998.743501).
- [88] J. Barrett, L. Hardy, and A. Kent, “No signaling and quantum key distribution,” *Physical Review Letters*, vol. 95, no. 1, Jun. 2005. DOI: [10.1103/physrevlett.95.010503](https://doi.org/10.1103/physrevlett.95.010503). [Online]. Available: <http://dx.doi.org/10.1103/PhysRevLett.95.010503>.

- [89] S. Pironio, A. Acín, N. Brunner, N. Gisin, S. Massar, and V. Scarani, “Device-independent quantum key distribution secure against collective attacks,” *New Journal of Physics*, vol. 11, no. 4, Apr. 2009, p. 045 021. DOI: [10.1088/1367-2630/11/4/045021](https://doi.org/10.1088/1367-2630/11/4/045021). [Online]. Available: <http://dx.doi.org/10.1088/1367-2630/11/4/045021>.
- [90] E. Hänggi, R. Renner, and S. Wolf, “Efficient device-independent quantum key distribution,” in *Advances in Cryptology – EURO-CRYPT 2010*, H. Gilbert, Ed., pp. 216–234, Berlin, Heidelberg: Springer Berlin Heidelberg, 2010.
- [91] U. Vazirani and T. Vidick, “Robust device independent quantum key distribution,” in *Proceedings of the 5th Conference on Innovations in Theoretical Computer Science*, ser. ITCS ’14, pp. 35–36, Princeton, New Jersey, USA: Association for Computing Machinery, 2014. DOI: [10.1145/2554797.2554802](https://doi.org/10.1145/2554797.2554802). [Online]. Available: <https://doi.org/10.1145/2554797.2554802>.
- [92] T. Metger, Y. Dulek, A. Coladangelo, and R. Arnon-Friedman, “Device-independent quantum key distribution from computational assumptions,” *New Journal of Physics*, vol. 23, no. 12, 2021, p. 123 021.
- [93] B. W. Reichardt, F. Unger, and U. Vazirani, “Classical command of quantum systems,” *Nature*, vol. 496, no. 7446, 2013, pp. 456–460.
- [94] A. Broadbent and C. Schaffner, “Quantum cryptography beyond quantum key distribution,” *Designs, Codes and Cryptography*, vol. 78, no. 1, Dec. 2015, pp. 351–382. DOI: [10.1007/s10623-015-0157-4](https://doi.org/10.1007/s10623-015-0157-4). [Online]. Available: <http://dx.doi.org/10.1007/s10623-015-0157-4>.
- [95] E. Kashefi and S. Singh, *Quantum protocol zoo*, <https://wiki.veriqloud.fr/>.
- [96] A. C.-C. Yao, “How to generate and exchange secrets,” *27th Annual Symposium on Foundations of Computer Science (sfcs 1986)*, 1986, pp. 162–167.
- [97] O. Goldreich, S. Micali, and A. Wigderson, “How to play any mental game,” Jan. 1987, pp. 218–229. DOI: [10.1145/28395.28420](https://doi.org/10.1145/28395.28420).

- [98] C. Crépeau, D. Gottesman, and A. Smith, “Secure multi-party quantum computation,” ser. STOC ’02, pp. 643–652, Montreal, Quebec, Canada: Association for Computing Machinery, 2002. DOI: [10.1145/509907.510000](https://doi.org/10.1145/509907.510000). [Online]. Available: <https://doi.org/10.1145/509907.510000>.
- [99] Child, *Quantum Information and Computation*, vol. 5, no. 6, Sep. 2005. DOI: [10.26421/qic5.6](http://dx.doi.org/10.26421/qic5.6). [Online]. Available: <http://dx.doi.org/10.26421/QIC5.6>.
- [100] A. Broadbent, J. Fitzsimons, and E. Kashefi, “Universal blind quantum computation,” *2009 50th Annual IEEE Symposium on Foundations of Computer Science*, Oct. 2009. DOI: [10.1109/focs.2009.36](http://dx.doi.org/10.1109/FOCS.2009.36). [Online]. Available: <http://dx.doi.org/10.1109/FOCS.2009.36>.
- [101] M. Blum, “Coin flipping by telephone,” in *CRYPTO’81*, A. Gersho, Ed., vol. ECE Report 82-04, pp. 11–15, U.C. Santa Barbara, Dept. of Elec. and Computer Eng., 1981.
- [102] C. H. Bennett, G. Brassard, C. Crépeau, and M.-H. Skubiszewska, “Practical quantum oblivious transfer,” in *CRYPTO’91*, J. Feigenbaum, Ed., ser. LNCS, vol. 576, pp. 351–366, Springer, Berlin, Heidelberg, Aug. 1992. DOI: [10.1007/3-540-46766-1_29](https://doi.org/10.1007/3-540-46766-1_29).
- [103] A. B. Grilo, H. Lin, F. Song, and V. Vaikuntanathan, “Oblivious transfer is in MiniQCrypt,” in *EUROCRYPT 2021, Part II*, A. Canteaut and F.-X. Standaert, Eds., ser. LNCS, vol. 12697, pp. 531–561, Springer, Cham, Oct. 2021. DOI: [10.1007/978-3-030-77886-6_18](https://doi.org/10.1007/978-3-030-77886-6_18).
- [104] J. Bartusek, A. Coladangelo, D. Khurana, and F. Ma, “One-way functions imply secure computation in a quantum world,” in *CRYPTO 2021, Part I*, T. Malkin and C. Peikert, Eds., ser. LNCS, vol. 12825, pp. 467–496, Virtual Event: Springer, Cham, Aug. 2021. DOI: [10.1007/978-3-030-84242-0_17](https://doi.org/10.1007/978-3-030-84242-0_17).
- [105] P. Ananth, F. Kaleoglu, and Q. Liu, “Cloning games: A general framework for unclonable primitives,” in *CRYPTO 2023, Part V*, H. Handschuh and A. Lysyanskaya, Eds., ser. LNCS, vol. 14085, pp. 66–98, Springer, Cham, Aug. 2023. DOI: [10.1007/978-3-031-38554-4_3](https://doi.org/10.1007/978-3-031-38554-4_3).

- [106] M. Tomamichel, S. Fehr, J. Kaniewski, and S. Wehner, “A monogamy-of-entanglement game with applications to device-independent quantum cryptography,” *New Journal of Physics*, vol. 15, no. 10, Oct. 2013, p. 103 002. DOI: [10.1088/1367-2630/15/10/103002](https://doi.org/10.1088/1367-2630/15/10/103002). [Online]. Available: <https://dx.doi.org/10.1088/1367-2630/15/10/103002>.
- [107] E. Culf and T. Vidick, “A monogamy-of-entanglement game for subspace coset states,” *Quantum*, vol. 6, Sep. 2022, p. 791. DOI: [10.22331/q-2022-09-01-791](https://doi.org/10.22331/q-2022-09-01-791). [Online]. Available: <https://doi.org/10.22331/q-2022-09-01-791>.
- [108] N. Johnston, R. Mittal, V. Russo, and J. Watrous, “Extended non-local games and monogamy-of-entanglement games,” *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 472, no. 2189, 2016, p. 20 160 003.
- [109] A. Behera, O. Sattath, and U. Shinar, *Noise-tolerant quantum tokens for MAC*, Cryptology ePrint Archive, Report 2021/1353, 2021. [Online]. Available: <https://eprint.iacr.org/2021/1353>.
- [110] A. Broadbent and R. Islam, “Quantum encryption with certified deletion,” in *TCC 2020, Part III*, R. Pass and K. Pietrzak, Eds., ser. LNCS, vol. 12552, pp. 92–122, Springer, Cham, Nov. 2020. DOI: [10.1007/978-3-030-64381-2_4](https://doi.org/10.1007/978-3-030-64381-2_4).
- [111] S. Ben-David and O. Sattath, “Quantum tokens for digital signatures,” *Quantum*, vol. 7, 2023, p. 901.
- [112] S. Aaronson and P. Christiano, “Quantum money from hidden subspaces,” in *Proceedings of the forty-fourth annual ACM symposium on Theory of computing*, pp. 41–60, 2012.
- [113] A. Coladangelo, J. Liu, Q. Liu, and M. Zhandry, “Hidden cosets and applications to unclonable cryptography,” in *CRYPTO 2021, Part I*, T. Malkin and C. Peikert, Eds., vol. 12172, pp. 556–584, Aug. 2021.
- [114] S. Goldwasser and G. N. Rothblum, “On best-possible obfuscation,” in *TCC 2007*, S. P. Vadhan, Ed., ser. LNCS, vol. 4392, pp. 194–213, Springer, Berlin, Heidelberg, Feb. 2007. DOI: [10.1007/978-3-540-70936-7_11](https://doi.org/10.1007/978-3-540-70936-7_11).

- [115] A. Molina, T. Vidick, and J. Watrous, “Optimal counterfeiting attacks and generalizations for wiesner’s quantum money,” in *Conference on Quantum Computation, Communication, and Cryptography*, Springer, pp. 45–64, 2012.
- [116] C. H. Bennett, G. Brassard, S. Breidbard, and S. Wiesner, “Quantum cryptography, or unforgeable subway tokens,” in *Advances in Cryptology: Proceedings of CRYPTO ’82*, pp. 267–275, Plenum, 1982.
- [117] E. Farhi, D. Gosset, A. Hassidim, A. Lutomirski, and P. Shor, “Quantum money from knots,” in *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference*, pp. 276–289, 2012.
- [118] D. M. Kane, *Quantum money from modular forms*, 2018. arXiv: [1809.05925 \[quant-ph\]](#).
- [119] D. M. Kane, S. Sharif, and A. Silverberg, “Quantum money from quaternion algebras,” *Mathematical Cryptology*, vol. 2, no. 1, 2022, pp. 60–83.
- [120] A. Lutomirski, S. Aaronson, E. Farhi, D. Gosset, J. A. Kelner, A. Hassidim, and P. W. Shor, “Breaking and making quantum money: Toward a new quantum cryptographic protocol,” in *ICS 2010*, A. C.-C. Yao, Ed., pp. 20–31, Tsinghua University Press, Jan. 2010.
- [121] M. Conde Pena, R. Duran Diaz, J.-C. Faugere, L. Hernandez Encinas, and L. Perret, “Non-quantum cryptanalysis of the noisy version of aaronson–christiano’s quantum money scheme,” *IET Information Security*, vol. 13, no. 4, 2019, pp. 362–366.
- [122] B. Roberts, “Security analysis of quantum lightning,” in *EURO-CRYPT 2021, Part II*, A. Canteaut and F.-X. Standaert, Eds., ser. LNCS, vol. 12697, pp. 562–567, Springer, Cham, Oct. 2021. DOI: [10.1007/978-3-030-77886-6_19](#).
- [123] A. B. Khesin, J. Z. Lu, and P. W. Shor, *Publicly verifiable quantum money from random lattices*, 2022. arXiv: [2207.13135 \[quant-ph\]](#).

- [124] J. Liu, H. Montgomery, and M. Zhandry, “Another round of breaking and making quantum money: How to not build it from lattices, and more,” in *EUROCRYPT 2023, Part I*, C. Hazay and M. Stam, Eds., ser. LNCS, vol. 14004, pp. 611–638, Springer, Cham, Apr. 2023. DOI: [10.1007/978-3-031-30545-0_21](https://doi.org/10.1007/978-3-031-30545-0_21).
- [125] M. Zhandry, “Quantum money from abelian group actions,” *TheoretiCS*, vol. 4, 2025.
- [126] M. Dall’Agnol and N. Spooner, “On the necessity of collapsing for post-quantum and quantum commitments,” in *18th Conference on the Theory of Quantum Computation, Communication and Cryptography*, 2023.
- [127] A. Coladangelo and O. Sattath, “A quantum money solution to the blockchain scalability problem,” *Quantum*, vol. 4, Jul. 2020, p. 297. DOI: [10.22331/q-2020-07-16-297](https://doi.org/10.22331/q-2020-07-16-297). [Online]. Available: <https://dx.doi.org/10.22331/q-2020-07-16-297>.
- [128] D. Gottesman, “Unclonable encryption,” *arXiv preprint quant-ph/0210062*, 2002.
- [129] P. Ananth, F. Kaleoglu, X. Li, Q. Liu, and M. Zhandry, “On the feasibility of unclonable encryption, and more,” in *Annual International Cryptology Conference*, Springer, pp. 212–241, 2022.
- [130] P. Ananth and A. Behera, “A modular approach to unclonable cryptography,” in *Annual International Cryptology Conference*, Springer, pp. 3–37, 2024.
- [131] C. Chevalier, P. Hermouet, and Q.-H. Vu, *Towards unclonable cryptography in the plain model*, Cryptology ePrint Archive, Paper 2023/1825, 2023. [Online]. Available: <https://eprint.iacr.org/2023/1825>.
- [132] J. Bartusek, D. Khurana, G. Malavolta, A. Poremba, and M. Walter, *Weakening assumptions for publicly-verifiable deletion*, TCC 2023, 2023. [Online]. Available: <https://eprint.iacr.org/2023/559>.

- [133] T. Hiroka, T. Morimae, R. Nishimaki, and T. Yamakawa, “Quantum encryption with certified deletion, revisited: Public key, attribute-based, and classical communication,” in *ASIACRYPT 2021, Part I*, M. Tibouchi and H. Wang, Eds., ser. LNCS, vol. 13090, pp. 606–636, Springer, Cham, Dec. 2021. DOI: [10.1007/978-3-030-92062-3_21](https://doi.org/10.1007/978-3-030-92062-3_21).
- [134] T. Hiroka, T. Morimae, R. Nishimaki, and T. Yamakawa, *Certified everlasting functional encryption*, Cryptology ePrint Archive, Report 2022/969, 2022. [Online]. Available: <https://eprint.iacr.org/2022/969>.
- [135] J. Bartusek, S. Garg, V. Goyal, D. Khurana, G. Malavolta, J. Raizes, and B. Roberts, “Obfuscation and outsourced computation with certified deletion,” *Cryptology ePrint Archive*, 2023.
- [136] J. Bartusek, D. Khurana, and A. Poremba, “Publicly-verifiable deletion via target-collapsing functions,” in *Annual International Cryptology Conference*, Springer, pp. 99–128, 2023.
- [137] P. Ananth and R. L. La Placa, “Secure software leasing,” in *EUROCRYPT 2021, Part II*, A. Canteaut and F.-X. Standaert, Eds., ser. LNCS, vol. 12697, pp. 501–530, Springer, Cham, Oct. 2021. DOI: [10.1007/978-3-030-77886-6_17](https://doi.org/10.1007/978-3-030-77886-6_17).
- [138] F. Kitagawa, R. Nishimaki, and T. Yamakawa, “Secure software leasing from standard assumptions,” in *TCC 2021, Part I*, K. Nissim and B. Waters, Eds., ser. LNCS, vol. 13042, pp. 31–61, Springer, Cham, Nov. 2021. DOI: [10.1007/978-3-030-90459-3_2](https://doi.org/10.1007/978-3-030-90459-3_2).
- [139] A. Broadbent, S. Jeffery, S. Lord, S. Podder, and A. Sundaram, “Secure software leasing without assumptions,” in *TCC 2021, Part I*, K. Nissim and B. Waters, Eds., ser. LNCS, vol. 13042, pp. 90–120, Springer, Cham, Nov. 2021. DOI: [10.1007/978-3-030-90459-3_4](https://doi.org/10.1007/978-3-030-90459-3_4).
- [140] O. Chardouvelis, V. Goyal, A. Jain, and J. Liu, “Quantum key leasing for pke and fhe with a classical lessor,” in *Annual International Conference on the Theory and Applications of Cryptographic Techniques*, Springer, pp. 248–277, 2025.

- [141] P. Ananth, A. Poremba, and V. Vaikuntanathan, “Revocable cryptography from learning with errors,” in *TCC 2023, Part IV*, G. N. Rothblum and H. Wee, Eds., ser. LNCS, vol. 14372, pp. 93–122, Springer, Cham, Nov. 2023. DOI: [10.1007/978-3-031-48624-1_4](https://doi.org/10.1007/978-3-031-48624-1_4).
- [142] P. Ananth, Z. Hu, and Z. Huang, “Quantum key-revocable dual-regev encryption, revisited,” in *Theory of Cryptography Conference*, Springer, pp. 257–288, 2024.
- [143] T. Morimae, A. Poremba, and T. Yamakawa, *Revocable quantum digital signatures*, Cryptology ePrint Archive, Report 2023/1937, 2023. [Online]. Available: <https://eprint.iacr.org/2023/1937>.
- [144] P. Ananth, S. Mutreja, and A. Poremba, “Revocable encryption, programs, and more: The case of multi-copy security,” *arXiv preprint arXiv:2410.13163*, 2024.
- [145] M. Georgiou and M. Zhandry, *Unclonable decryption keys*, Cryptology ePrint Archive, Report 2020/877, 2020.
- [146] S. Aaronson, J. Liu, Q. Liu, M. Zhandry, and R. Zhang, “New approaches for quantum copy-protection,” in *Advances in Cryptology—CRYPTO 2021: 41st Annual International Cryptology Conference, CRYPTO 2021, Virtual Event, August 16–20, 2021, Proceedings, Part I 41*, Springer, pp. 526–555, 2021.
- [147] A. Cohen, J. Holmgren, R. Nishimaki, V. Vaikuntanathan, and D. Wichs, “Watermarking cryptographic capabilities,” in *Proceedings of the forty-eighth annual ACM symposium on Theory of Computing*, pp. 1115–1127, 2016.
- [148] F. Kitagawa, R. Nishimaki, and T. Yamakawa, “Secure software leasing from standard assumptions,” in *Theory of Cryptography Conference*, Springer, pp. 31–61, 2021.
- [149] J. Liu, Q. Liu, L. Qian, and M. Zhandry, “Collusion resistant copy-protection for watermarkable functionalities,” in *Theory of Cryptography Conference*, Springer, pp. 294–323, 2022.
- [150] A. Çakan and V. Goyal, “Unclonable cryptography with unbounded collusions and impossibility of hyperefficient shadow tomography,” in *Theory of Cryptography Conference*, Springer, pp. 225–256, 2024.

- [151] F. Kitagawa and R. Nishimaki, “One-out-of-many unclonable cryptography: Definitions, constructions, and more,” in *Theory of Cryptography Conference*, Springer, pp. 246–275, 2023.
- [152] C. Chevalier, P. Hermouet, and Q.-H. Vu, “Semi-quantum copy-protection and more,” in *Theory of Cryptography Conference*, Springer, pp. 155–182, 2023.
- [153] S. Mutreja and M. Zhandry, *Quantum state group actions*, 2024. arXiv: [2410.08547](https://arxiv.org/abs/2410.08547) [quant-ph]. [Online]. Available: <https://arxiv.org/abs/2410.08547>.
- [154] P. Ananth, A. Behera, and Z. Huang, *Copy-protection from UPO, revisited*, Cryptology ePrint Archive, Paper 2025/1207, 2025. [Online]. Available: <https://eprint.iacr.org/2025/1207>.